TFPT — Architecture and the E_8 Compiler

The two axioms $\{c_3, g_{\text{car}}\}$, the derivation map, and the Coxeter–cyclotomic $D_5 \times A_3 \to E_8$ construction

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What this document is about

The architecture layer: how the two axioms $c_3 = \frac{1}{8\pi}$ and $g_{\text{car}} = 5$ build the Coxeter–cyclotomic compiler — the carrier $C^+{=}D_5$, the family geometry $\mathbb{P}^1 \setminus \mu_4{=}A_3$, the μ_4 glue $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, the EM fixed point α^{-1} (with its ablation), and the whole number alphabet $16, 40, 41, 48, 240, 248$ as carrier traces.

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and five companions**, best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team); **P.** `tfpt_prime_front` — research documentation of the prime / zeta line: the seven load-bearing modules on the E_8 glue census (Hecke from geometry through the transport ledger), the sandbox localization program that follows them, and the kill list. Documentation, not a result: it makes no claim about the Riemann Hypothesis.

You are reading document 1 (Architecture).

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\mathrm{car}} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale	primitive [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1,1)\rangle \cong (E_8)_1$	[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

Gravity is parameter-free. The classical metric-sector field equation is no longer only an $R+R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2| 2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, `v358`) and Λ from α ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in `v473` without changing the typing) and the one unit v_{geo}; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem G_{net} (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route `SEAM.EQUIV.MMST.01` is *closed modulo cited theorems*; the twistor route `SEAM.EQUIV.TWISTOR.01` (Costello–Li, prepared by `CELEST.SEAM.01`) and the unconditional parent stay [O]. On the MMST route: the explicit lattice model (`v367`/`v368`) and the S_3 closure stack pin the target at every computable level — central charge $c = 8$ (`v376`), the $(E_8)_1$ character with 248 currents and one primary (`v377`), genus-one torus GSD = 1 (`v378`) and reflection positivity (`v379`) — and it is Lean-formalised as `FORM.SEAM.MMST.01`: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the `#print axioms` check is clean. The *only* residual of that route that stays [O] is the abstract continuum existence of the scaling limit (`v336`) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package (h_s=16/16=1∈$\mathbb{Z}$, the Longo–Rehren locality criterion; Böckenhauer; KLM μ=4/2^2=1; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route `seamResidualClosed′` (`v469`); so the MMST route is *closed modulo cited theorems*, not solved. `QGEO.SYM.01` (the μ_4 deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (`v335`, Lean `FORM.QGEO.BW.01`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin 1.648 > 0, `v76`/`v330`). So the honest residual is the one unit v_{geo} plus the typed F_{transfer} interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\mathrm{wall}}, G_{\mathrm{metric}}, F_{\mathrm{frontier}}$) are retained only for ledger continuity — the live residual is $v_{\mathrm{geo}} \oplus F_{\mathrm{transfer}}$, with the seam G_{net} closed modulo cited theorems on its MMST route (parent [O]) and `QG.AMB.01` a [C] redundancy.

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

[E] exact / machine-proven **[C]** conditional **[O]** open / axiom **[X]** kill test

P1 $c_3 = \frac{1}{8\pi}$ **P2** $g_{\text{car}} = 5$ *(two inputs, [O] axioms)*

↓

Anchor $a = (1,1,2)$: $e_1, e_2, e_3 = 4, 5, 2$ *([E] exact)*

↓

Discrete compiler: $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, $h = 30$, φ_0 *([E] exact)*

↓

SM skeleton: three families, hypercharges, flavor matrix R *([E])*

↓

Physical bridges: v_{geo}, G_{net}, F_{transfer} *([C] bridges)*

↓

Frontier readouts: Λ, inflation, η_B, dark matter, QG *([C]/[O])*

Contents

Part I
Architecture and the derivation map

Abstract

This document develops the *architecture layer* of TFPT: the two axioms, the derivation map, and the explicit E_8 construction. The thesis is that the bridge layer of TFPT is governed by *three nested grammars*: a **carrier grammar** (from $3+2$ come charges, families, hypercharge, $\Omega_{\rm adm}=48$, $b_1=41/10$), a **seed grammar** (one seed $u=\varphi_0^{\rm ret}$ generates the Cabibbo angle, birefringence, the baryon fraction and the reactor angle), and an **action grammar** (the large scale ratios are exponential actions of $\alpha_\star^{-1}$ with rungs $1/g_{\rm car}$, 1, 2). Two new exact results sharpen the action grammar: the *decuple bridge* $\rho_\Lambda/\bar{M}_{\rm Pl}^4=\frac{3}{4\pi^2}e^{2\delta_{\rm ph}}\,[\,v_{\rm geo}/(5\beta_{\rm rad}^2\bar{M}_{\rm Pl})\,]^{10}$ (the cosmological constant is, in essence, the tenth power of the stripped electroweak scale, with $10=2g_{\rm car}$), and the Hubble corollary $H_0/\bar{M}_{\rm Pl}=e^{-\alpha_\star^{-1}}/(2\pi\sqrt{\Omega_\Lambda})$. We also give the exact corrected tree form of the Einstein normaliser $\xi=\frac{3}{4}/(1+9/128\pi^3)$, the dual-root dictionary that generates all hypercharges and transport cusps from $y_\pm=\{1/2,-1/3\}$, the Λ-anchored metrology closure, the overdetermination of $g_{\rm car}=5$, and the E_8 carrier atlas in three clearly separated levels. Section 3 adds the strongest structural result: the carrier code is the $D_5=\mathfrak{so}_{10}$ half-spinor ($\mathrm{Aut}=W(D_5)$, order 1920; $R_4=$ Clebsch graph), the family geometry $\mathbb{P}^1\setminus\mu_4$ is an A_3, and E_8 is the explicit unimodular glue closure of $D_5\oplus A_3$ — with the discriminant-form norms $q(D_5)=5/4$ and $q(A_3)=3/4$ summing to the E_8 root norm 2 and coinciding with the TFPT constants $\delta_2/\delta_{\rm top}^2$ and $\xi_{\rm tree}$. Status markers used throughout: **[E]** exact identity, **[E]** Lie/lattice theorem, **[E]** formalised, **[E]** numerical fixed point, **[C]** physical/conditional, **[O]** axiom/open. (All markers are explicit; the legacy `[M]`/`[C]` compatibility aliases have been removed.)

1 How to read this document

Notation convention. We write $D_5 \oplus A_3$ for the *root-lattice* direct sum (used in the glue/lattice statements: discriminant forms, μ_4 glue, E_8 as an even unimodular lattice), and $D_5 \times A_3$ for the *group/representation* branching (used for subalgebra embeddings and E_8-rep decompositions). The objects are the same atoms; the symbol marks whether the statement is lattice-theoretic ($\oplus$) or branching/representation-theoretic ($\times$).

Nothing here introduces a new primitive assumption; it reorganises existing outputs and proves the cross-links. The whole bridge layer rests on two inputs,

$$\boxed{c_3 = \frac{1}{8\pi} \quad \text{(axiom P1)}, \qquad g_{\rm car} = 5 \quad \text{(axiom P2)}.} \tag{1}$$

Foundational postulates — the two genuine axioms (hardenable, not derivable)

Everything downstream is a theorem *given* two structural inputs that are not reduced further and are honestly flagged as the theory's axioms:

(P1) Boundary kernel.
A single oriented seam Σ carries a primitive, reflection-positive boundary kernel with unit winding $[u_\Sigma] = 1$; its self-linking normalisation fixes $c_3 = 1/(8\pi)$. *Status:* primitive postulate. *Hardenable by:* the Calderón projector / determinant-character chain (the P1 hardening) and the Lean carrier-rigidity development (`TfptCarrier.*`, boundary-polarization and Calderón interfaces).

(P2) Carrier interface.
This splits into an *axiom* and a *theorem*, which must not be conflated:

- **P2**$_{\rm interface}$ **[O] (axiom):** the seam is read out by a five-slot carrier $g_{\rm car} = 5$ (3 colour $+\,2$ weak) — a postulate (the readout interface).
- **P2**$_{\rm algebra}$ **[E] (theorem):** *given* the carrier, hypercharge ($y_- = -\frac{1}{3}$, $y_+ = \frac{1}{2}$), rigidity and the family count follow from the idempotent/lattice axioms — machine-verified in Lean (`Hypercharge.lean`, `LatticeRigidityGeneral.lean`, `Rigidity.lean`).

So P2 is an *interface axiom* whose *algebraic consequences are a formalised theorem* — "axiom or theorem?" has the precise answer: interface axiom, algebra theorem.

These two are *inputs*: they can be formalised and stress-tested (more Lean, more interface theorems) but not computed away. Every other number in this note — $N_{\rm fam} = 3$, $\Omega_{\rm adm} = 48$, $b_1 = 41/10$, $\alpha_\star^{-1}$, the flavor matrix, the E_8 glue, A_s — is a consequence of (P1),(P2).

Anchor-first refinement. The two inputs are themselves the elementary symmetric polynomials of the single parabolic anchor $a = (1, 1, 2)$ (the exponents-at-infinity of the splitting type $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$): $e_1(a) = 4 = |\mu_4|$, $e_2(a) = 5 = g_{\rm car}$, $e_3(a) = 2 = |\mathbb{Z}_2|$, with $c_3 = 1/(2e_1(a)\pi) = 1/(8\pi)$. The power sums $p_n(a) = 2 + 2^n$ then give $|R(E_8)| = p_1p_2p_3 = 240$, $\dim E_8 = 240 + (p_4 - p_3) = 248$ and the binary ladder $p_{n+1} - p_n = 2^n$. So the input set reduces to $\{a = (1, 1, 2),\, \pi\}$. [`verification/v23_anchor_generator.py`]

The current reduction state of P2 (2026-07-22) — text consolidation, markers unchanged

The axiom P2 stays *declared* (`AX.P2.01`); what has changed is how much of it is now carried by theorems. The honest ledger state: P1 ($c_3 = 1/(8\pi)$) and the *four marks* are **[E]**-anchored (Gauss–Bonnet $n = 2\chi = 4$, [`verification/v216_marks_gauss_bonnet.py`]); given the four marks, $g_{\rm car} = 5$ follows from the partition $4 = 2{+}1{+}1$ plus the Deligne/Birkhoff–Grothendieck/stability typing ([`verification/v491_p2_partition_corollary.py`], [`verification/v499_p2_weights_deligne_bg.py`]; residual R1, the module identification, **[C]**); and the clock's *order*, the $\mathbb{Z}_8$ tower, the deck's *class* (a half-period translation) and the *position* half of the alignment bit are theorems ([`verification/v506_seam_clock_rigidity.py`], [`verification/v507_seam_bit_origin.py`], [`verification/v510_seam_bit_freedom.py`]). What remains as carrier input is *one discrete symmetry-lift bit*: flag transitivity of the four marks (marks *and* their two sides indistinguishable, $V_4 \to D_4$) $\iff$ $\tau = i$ (equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle) — on the free seam circle bare mark-transitivity is automatic for *every* configuration (the pair-exchanging V_4) and no local jet sees the side bit, so the flag form is the sharp discrete statement of the datum and its negation is concretely measurable (odd-doublet split $(2/\pi)\ln\cot(\delta/2) \neq 0$)

([`verification/v512_seam_tau_flag.py`], `SEAM.TAU.FLAG.01`). The bit has since survived *ten* side-blind derivation attacks on that scoreboard — free RP/Θ existence (the eighth, v521), the exhausted Wick-computable free-plus-twist state class (the ninth, v525) and the first genuinely interacting seam toy (the tenth, v529) — and is now physically *defined* as the *twist-class choice*: facets #14/#15 extend the web to 15 exact equivalences (both directions each), the flip-axis fraction is a gauge-robust order parameter ($O = \frac{1}{2}$ at $m = 4$, 0 elsewhere), and its η holonomy is in principle interferometrically readable (v528, `SEAM.BIT.TWISTCLASS.01`; the bit remains formal input — a definition replaces no derivation, no marker moves); and it now carries the first *positive* selection datum: reflection positivity, imposed on the interacting mark family, keeps exactly *one* member alive — $\delta = \pi/2$ with positive coupling, the carrier of that order parameter ([`verification/v534_seam_straddle_cone.py`], `SEAM.STRADDLE.CONE.01`; toy-level evidence for a dynamical origin, not a derivation); plus the [C] identifications R1/R2 of v499. In one sentence: **the compiler's dimensionless inputs reduce to four marks (derived from P1-side topology), one discrete symmetry-lift bit — the twist-class choice (flag transitivity $\iff \tau = i$) — and π.** This is a reduction *state*, not an abolition of the axiom: `AX.P2.01` remains the declared interface input, and its dynamical realisation (`QGEO.REALIZE.01`) stays [C]/[O].

Seed normal form, the factorial spine, and the degree-2 pattern [E]/[C] ([`verification/v106_review_validation.py`])

Three review-validated sharpenings of the anchor picture. **(1) Seed normal form [E].** The seed formula is exactly

$$\boxed{\varphi_0^{\mathrm{ret}} = \frac{|\mu_4|}{N_{\mathrm{fam}}}\, c_3 \;+\; \Omega_{\mathrm{adm}}\, c_3^{|\mu_4|}} \qquad \left(\tfrac{4}{3} = \tfrac{|\mu_4|}{N_{\mathrm{fam}}},\quad 48 = \Omega_{\mathrm{adm}} = N_{\mathrm{fam}} \dim S^+\right),$$

a *linear seam term* plus a *topological fourth-order correction* over all admissible fermion states — a structured anchor readout, not a fitted expression. **(2) Factorial spine [E].** With $X = 6Y$ on one 16-state generation, $\operatorname{Tr} X = 0$ and $\operatorname{Tr} X^2 = 120 = 5!$ (computed from the explicit charge table); the *same* factorial moment is $|R^+(E_8)| = \sum(E_8 \text{ exponents}) = 120$, doubles to $|R(E_8)| = 240$, and scales to $|W(D_5)| = 2^4 \cdot 5! = 1920$ (the Hawking-power fingerprint) — one moment, four projections (hypercharge, roots, Weyl group, horizon). **(3) The degree-2 inventory.** The compiler repeatedly selects *degree two*: the Pascal closure $2^{g-1} = \sum_{k\le 2}\binom{g}{k}$ truncates at $K{=}2$ with the *unique* solution $g = 5$ (scan 1..40); the glue norms add to the root norm, $\frac{5}{4} + \frac{3}{4} = 2$; $c_3 = \frac{1}{2} \times \frac{1}{4\pi}$ (variational half × two-dimensional-slice Gauss–Bonnet); the pair sector $\binom{5}{2} = 10 = \mathcal{A}_\Lambda$; the hypercharge moment is a *second* moment; the seam is codimension *two*. *Named hypothesis [C] (recorded, not claimed):* **Quadratic Boundary Locality** — the seam supports only bilinear data, so the truncation $K = 2$ is *forced* rather than chosen; if proven, the Pascal *selection* (red-team Target B residual, typed [O]/[C]) upgrades to [E] and the chain locality $\Rightarrow K{=}2 \Rightarrow g{=}5 \Rightarrow 16 \Rightarrow D_5 \Rightarrow E_8$ closes the carrier choice. *Audit:* the even-code transition reading $240 = 16 \times 15$ is exact arithmetic but non-unique ($240 = 16 \times 5 \times 3$ is the established glue factorisation) and stays recorded, not promoted.

The Sheet-Pairing Lemma ([`verification/v109_sheet_pairing.py`]) — the first exact anchor of the hypothesis. Character-exact weight-multiset combinatorics of the D_5 spinor weights $(\pm\frac{1}{2})^5$ gives: *(i)* **no scalar within a sheet** — the zero-weight multiplicity of $S^+ \otimes S^+$ is 0, because the odd slot count $g = 5$ flips the sign parity of $-w$ (a parity theorem of the five-slot code); *(ii)* **the scalar lives across the sheets**:

$$\boxed{S^+ \otimes S^- = \Lambda^0 \oplus \Lambda^2 \oplus \Lambda^4} \qquad (256 = 1 + 45 + 210),$$

exact as weight multisets, with zero-mode grading $(1, 5, 10)$ = the carrier code; *(iii)* **sheet-diagonal = odd forms**: $(S^+ \otimes S^+) \cup (S^- \otimes S^-) = 2\Lambda^1 + 2\Lambda^3 + \Lambda^5$ exactly — within-sheet pairings are odd-form-valued, never scalar; *(iv)* the scalar-bearing tower **tops at Λ^{2K}** with $K = 2$ — bilinear seam data reach exactly the Pascal-ladder degree (the source-layer transport theorem; document 3). *Consequence [C]:* the seam's single scalar two-point kernel is necessarily a **sheet pairing** (chirality-off-diagonal — the same $\mathbb{Z}_2$ as the glue chiralities and the branch-kernel selection); the remaining QBL step is to show the Calderón kernel supplies exactly this pairing and nothing beyond.

The Calderón-sheet selection theorem ([`verification/v110_calderon_sheet.py`]) — which involutions certify a scalar kernel. On $H = S^+ \oplus S^-$ the certified bilinear data of a Calderón involution ε ($\varepsilon^2 = 1$) are the matrix elements $\langle f, \varepsilon g\rangle$. The lemma's singlet counts per sheet block, $(0, 0, 1, 1)$, give the exact selection statement **[E]**:

$$\boxed{\text{a scalar two-point datum exists} \iff \varepsilon \text{ is sheet-odd}}$$

— a sheet-odd involution (swapping $S^+ \leftrightarrow S^-$) certifies exactly $1 + 1 = 2 = |\mathbb{Z}_2|$ invariant scalar kernels (one per orientation), a sheet-even involution certifies *none*. The kernel count $2 = |\mathbb{Z}_2|$ matches exactly the *two Lagrangian glues* (swapped by single spinor conjugation): the Calderón scalar channel carries precisely the glue ambiguity, resolved by the same sheet choice. *Anti-overclaim control **[E]**:* the parity theorem is generic over the odd ladder ($g = 3, 5, 7$: within-sheet count 0, cross count 2^{g-1}) — sheet-oddness forces the *half-spinor relation* $K = (g-1)/2$, **not** the value $g = 5$; the g-selection stays with the independent rank-8/integrality closure. *Consequence:* the QBL residue splits into two named parts [C] (recorded, not claimed): **(a) interface** — "the seam Calderón involution is sheet-odd" (structurally natural: the one-sided collar/double-cover deck is the *same* $|\mathbb{Z}_2|$ that halves $c_3 = \frac{1}{2}(4\pi)^{-1}$, and exchanging cover sides is sheet-odd by construction); **(b) analytic core** — "the two-point kernel certifies only slot-degrees ≤ 2". Proving both closes QBL $\Rightarrow K{=}2 \Leftrightarrow g{=}5$.

The Quadratic-Transport Theorem ([`verification/v111_quadratic_transport.py`] **) — the transport half of the analytic core *is* a theorem.** Grade seam transport by Clifford degree in the ten field operators on the five-slot Fock space ($S = \Lambda^\bullet(\mathbb{C}^5)$, S^+ the 16-state code; integer Jordan–Wigner model, spanning claims by full-rank certificates mod p, valid over $\mathbb{Q}$). *Parity* **[E]**: all linear words are sheet-odd, all 45 quadratic words sheet-even — code-preserving transport has *even* Clifford degree (the same $\mathbb{Z}_2$ again). *Minimality* **[E]**: sheet-even words of degree ≤ 1 are the scalars alone. *Generation and completeness* **[E]**: the 45 quadratics ($= \mathfrak{so}(10)$, exactly the Λ^2 term of the certified tower $1 + 45 + 210$) generate the whole code from the vacuum *and* span the full operator algebra $\mathrm{End}(S^+)$ ($\dim = 256$) at word length two — every code operation whatsoever is a word in pair transport, and all higher even-degree elements (e.g. the 210 quartics) are redundant. Hence

$$\boxed{\text{the transport degree is exactly } 2}$$

— degree ≤ 1 generates nothing, degree 2 generates everything: a theorem, not a hypothesis. *Anti-overclaim control **[E]**:* the $g = 3$ world behaves identically (15 quadratics generate $\mathrm{End}(S^+) = 16$) — the theorem selects the *degree*, not the rank; the g-selection stays with Pascal closure + rank-8. *Residue after v109–v111* [C]: QBL's remaining content is two *interface* statements — (a) the seam Calderón involution is sheet-odd; (b′) the certified *state* inventory is the ≤ 2-slot tower. Nothing about transport remains hypothetical.

The Self-Counting Channel ([`verification/v112_selfcounting_channel.py`] **) — the Pascal closure is an *identity* of the certified channel.** For odd g, negation $w \mapsto -w$ maps the S^+ weights bijectively into the opposite sheet, so the Cartan-neutral kernels of the certified channel are exactly the pairs $(w, -w)$ — *one per code state*: $\#\{\text{neutral kernels}\} = \dim S^+ = 2^{g-1}$ exactly **[E]**. Grading the *same* neutral set by pair-degree m gives sizes $\binom{g}{m}$, $m = 0..K$ — for $g = 5$ the triple $(1, 5, 10)$ **[E]**. Two countings of one set:

$$\boxed{2^{g-1} = \sum_{m \leq K} \binom{g}{m} \quad \text{is the channel counting itself}}$$

(symbolically verified for all odd $g = 3..13$) — the Pascal *closure* (`v2`/`v108`) is not an extra requirement but an identity of the certified channel. *Hodge fold* **[E]**: the code's minus-sign grading $\binom{g}{2j}$ equals the folded binomial $\binom{g}{\min(2j,\, g-2j)}$ with $\min(2j, g-2j) \leq K$ — every code sector appears at pair-degree $\leq K$. *Anti-overclaim:* the identity holds for every odd g and carries no g-selection; the selection stays with the rank-8 closure $g + N_{\mathrm{fam}} = 8$ and ladder integrality. *Residue re-type [C] (recorded, not claimed):* clause (b′) loses its closure burden — given the sheet-odd kernel, the channel *automatically* contains one neutral kernel per code state at pair-degree $\leq K$; what remains physical is the single input "the seam certifies through one scalar two-point kernel" (for the established free $c = 8$ seam net, Wick extends the one kernel to the entire even tower by determinants). QBL is thereby the structural consistency frame of the carrier choice; the *selection* is carried by rank-8 + integrality.

One kernel is the whole net ([`verification/v113_quasifree_kernel.py`] **) — the QBL input merges with the holomorphy premise.** The remaining input is now anchored in the established Gate-A objects. *Bookkeeping* **[E]**: $c(D_5)_1 = \frac{45}{9} = 5 = g_{\mathrm{car}}$, $c(A_3)_1 = \frac{15}{5} = 3 = N_{\mathrm{fam}}$, and the carrier net $(D_5)_1 \times (A_3)_1 = SO(10)_1 \times SO(6)_1$ is $10 + 6 = 16$ *free Majorana fermions* ($c = \frac{16}{2} = 8$); the whole extension tower carrier (μ=16) $\to SO(16)_1(\mu$=4$) \to (E_8)_1(\mu$=1) carries the *same* sixteen fermions — only the certified glue grows (index $2 \times 2 = 4 = |\mu_4|$). *Wick/Pfaffian* **[E]**: in the exact ten-Majorana model *all* 210 vacuum four-point and all 210 six-point functions equal the Pfaffian of the single two-point kernel — one kernel determines all correlations. *The kernel is a Calderón involution of rank g* **[E]**: $M = \mathbf{1} + iA$ with $A^2 = -\mathbf{1}$, so $P = M/2$ is a projection with rank $P = 5 = g_{\text{car}}$ and $\varepsilon = iA$ an involution; at seam level (sixteen Majoranas) the rank is $8 = \operatorname{rank} E_8$:

$$\boxed{\text{the central charge is the rank of the one kernel}}$$

($c = 5$ carrier block, $c = 8$ seam hull); and the polarization fixes the vacuum *uniquely* (joint annihilator kernel one-dimensional). *Premise merge [C] (recorded, not claimed):* "single scalar two-point kernel" = "the seam state is quasi-free" = the defining property of the free-fermion net the holomorphy gate already posits — QBL adds *no* assumption beyond Gate A; when the gate closes, the carrier choice closes **[E]** along free net ⇒ one kernel ⇒ sheet-odd ⇒ self-counting tower ⇒ complete transport ⇒ carrier. Gate typing **[C]**/**[O]** unchanged.

Hardening (P1): the Gauss–Bonnet origin of $c_3 = \frac{1}{8\pi}$

$c_3 = 1/(8\pi)$ is not an arbitrary number: it is the *one-sided Gauss–Bonnet normalisation of the seam.* The seam construction compactifies the oriented two-dimensional normal slice N_Σ to a sphere S^2, so Gauss–Bonnet gives the seam curvature integral

$$\oint_{S^2} K\, dA = 2\pi\, \chi(S^2) = 4\pi \qquad (\chi(S^2) = 2),$$

and the boundary datum is *one-sided* (the Calderón/double-cover construction uses a single collar), contributing a factor $1/|\mathbb{Z}_2| = \frac{1}{2}$. Hence

$$\boxed{c_3 = \frac{1}{|\mathbb{Z}_2|\ \oint_{S^2} K\, dA} = \frac{1}{2 \cdot 4\pi} = \frac{1}{8\pi}}$$

equivalently $c_3 = \frac{1}{2}(4\pi)^{-1}$, the one-sided 2D heat-kernel coefficient. This locates both factors precisely:

- the *discrete* $8 = |\mathbb{Z}_2| \cdot 2 \cdot \chi(S^2) = 2 \cdot 2 \cdot 2$ *coincides with* $h(D_5) = \operatorname{rank} E_8 = \varphi(30)$ — the geometric and the bootstrap readings of the "8" agree;
- the *continuous* π is the irreducible Gauss–Bonnet / 2D-Gaussian primitive ($\int e^{-x^2} = \sqrt{\pi}$), the single primitive the self-consistency loop also isolates.

Status: this *hardens* (P1) — it gives c_3 a named geometric origin and reconciles the "8" with E_8 — but does *not* eliminate π; π remains the one continuous primitive. **[C]**

The one-paragraph version. There is a boundary (the seam). It yields $c_3 = 1/(8\pi)$ and a five-slot carrier $g_{\text{car}} = 5$, split 3 colour + 2 weak. Two base charges $y_- = -1/3$ and $y_+ = 1/2$ generate the whole fermion family; the 16 even-occupation states are one generation, three families give $3 \times 16 = 48$. One seed $u = \varphi_0^{\text{ret}}$ drives four low-energy observables. Finally $\alpha_\star^{-1} \approx 137$ is the exponential engine: the electroweak scale takes one fifth of it ($e^{-\alpha_\star^{-1}/5}$), the cosmological constant takes it twice ($e^{-2\alpha_\star^{-1}}$), and the Hubble scale is the square root of Λ. A small grammar generates many sectors.

2 Headline: the Pascal compiler on five carrier slots

The strongest new statement is that three of the theory's separate structures are the *same* truncated row of Pascal's triangle on the five-slot carrier.

Family = action ladder = transition count (one Pascal row)

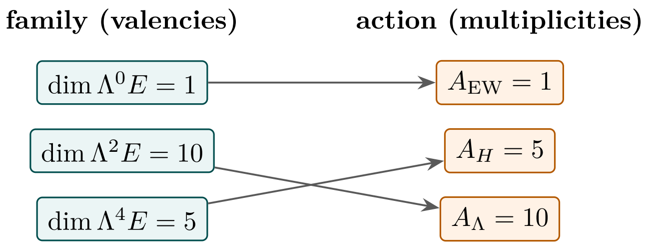
The even exterior algebra of E ($\dim E = g_{\mathrm{car}} = 5$), the action ladder, and the E_8 root count are one Pascal row:

$$S^+ = \Lambda^0 E \oplus \Lambda^2 E \oplus \Lambda^4 E, \qquad (\dim \Lambda^0, \dim \Lambda^2, \dim \Lambda^4) = (1, 10, 5), \qquad \dim S^+ = 1 + 10 + 5 = 16,$$

$$\mathcal{A}_{\mathrm{EW}} : \mathcal{A}_H : \mathcal{A}_\Lambda = \binom{5}{0} : \binom{5}{1} : \binom{5}{2} = 1 : 5 : 10 = (\dim \Lambda^0 E,\ \dim \Lambda^4 E,\ \dim \Lambda^2 E),$$

$$|R(E_8)| = \dim S^+(\dim S^+ - 1) = 16 \cdot 15 = 240.$$

Family (exterior degree), cosmology (action charge) and E_8 (root transitions) read the same $\{1, 5, 10\}$. [**E**]



even Hamming scheme on C^+: valencies $(1, 10, 5)$ $\leftrightarrow$ spectral multiplicities $(1, 5, 10)$

Figure 1: Exterior–Action duality. The Standard-Model family is the valency side of the even Hamming scheme on the five-slot code; the cosmological action ladder is its spectral (multiplicity) side. The Λ (pair) sector $\Lambda^2 E$ has dimension $\binom{5}{2} = 10$, the origin of the decuple exponent.

Natural sector assignment:

$$\boxed{EW \leftrightarrow \Lambda^0 E \text{ (singlet)}, \quad H \leftrightarrow \Lambda^4 E \cong E^\vee \ (\dim 5), \quad \Lambda \leftrightarrow \Lambda^2 E \ (\dim 10).}$$

The cosmological constant lives on the *pair* sector $\Lambda^2 E$, $\dim = \binom{5}{2} = 10$; this is why the decuple bridge has exponent 10 — Λ is a product over all $\binom{5}{2}$ carrier pairs (K_5 edges), not over the full 16-state family.

2.1 $g_{\mathrm{car}} = 5$ as the Pascal closure

A family is exactly the complete Pascal truncation to degree two:

$$\boxed{\dim S^+ = 2^{g_{\mathrm{car}}-1} = \binom{g_{\mathrm{car}}}{0} + \binom{g_{\mathrm{car}}}{1} + \binom{g_{\mathrm{car}}}{2} \iff g_{\mathrm{car}} = 5} \qquad (16 = 1 + 5 + 10), \quad [\mathbf{E}], \quad (2)$$

the unique positive-integer solution of $f(g) := 2^{g-1} - 1 - \frac{g(g+1)}{2} = 0$. *Proof of uniqueness.* Direct evaluation gives $f(1){=}{-1}$, $f(2){=}{-2}$, $f(3){=}{-3}$, $f(4){=}{-3}$, $f(5){=}0$, $f(6){=}{+10}$; so $g{=}5$ is the only root in $\{1, \dots, 6\}$. For $g \geq 6$ the forward difference is $f(g{+}1){-}f(g) = 2^{g-1}{-}(g{+}1) \geq 2^5{-}7 = 25 > 0$ and increasing, so f is strictly increasing on $[6, \infty)$ from $f(6){=}10 > 0$; hence $f(g) > 0$ for all $g \geq 6$ and there is *no* further root. Thus $g_{\mathrm{car}} = 5$ is the unique solution (no finite-range caveat). [`verification/v2_carrier_pascal.py`] This is sharper than $2g_{\mathrm{car}} = \binom{g_{\mathrm{car}}}{2}$: the family code (even states, 2^{g-1}) equals the action code (sum of the first three Pascal entries) only at $g_{\mathrm{car}} = 5$.

Carrier uniqueness theorem: $g_{\rm car} = 5$ and the 3+2 split are forced [E]

The carrier rank and its colour/weak split are not chosen — they are the unique solutions of two integer conditions tied to the E_8 closure and the SM gauge algebra:

$$\boxed{g_{\rm car} = 5 \text{ is the unique } g \text{ with } N_{\rm fam} = \tfrac{2^{g-1}-1}{g} \in \mathbb{Z}^+ \text{ and } \operatorname{rank} E_8 = g + N_{\rm fam} = 8} \tag{3}$$

$$\boxed{(b,s) = (3,2) \text{ is the unique split of } \; b+s = g_{\rm car} = 5, \;\; b^2+s^2 = 13 = |R(A_3)|+1} \tag{4}$$

The integer-family carriers are $g \in \{3,5,7,\dots\}$ (rank $4,8,16$), so rank $E_8 = 8$ picks $g_{\rm car} = 5$ uniquely; and $b^2+s^2 = 13$ with $b+s = 5$ forces $(3,2)$, reproducing $\dim \mathfrak{g}_{\rm SM} = (b^2-1)+(s^2-1)+1 = 12 = |R(A_3)|$. The induced hypercharge is anomaly-free ($\operatorname{Tr} Y = \operatorname{Tr} Y^3 = 0$ over the 16 states). This hardens P2: the *rank* and the *split* are forced; only the seam→carrier interface itself remains a declared input. [`verification/v14_carrier_uniqueness.py`]

The Pascal grammar: one $\Lambda^\bullet$ mechanism in four places (anti-numerology) [E]. The recurring small integers are *not* coincidences — they are the *same* exterior-algebra (Pascal) mechanism read in four spaces, which is why the binomials of rows 4 and 5 keep reappearing:

space	Pascal reading
carrier (one generation)	$\dim S^+ = \Lambda^{\rm even}(5) = \binom{5}{0}+\binom{5}{2}+\binom{5}{4} = 1+10+5 = 16$
scale ladder (cosmology)	$\mathcal{A}_{\rm EW} : \mathcal{A}_H : \mathcal{A}_\Lambda = \binom{5}{0} : \binom{5}{1} : \binom{5}{2} = 1:5:10$
flavor u/d readout	$\lVert \operatorname{Pl}(K) \rVert_1 = \binom{4}{0}+\binom{4}{1}+\binom{4}{2} = 1+4+6 = 11 = 16-g_{\rm car}$
mass volume	$\det M_{\rm SM} \sim (\varphi_0^{\rm ret})^{6+9+10} = (\varphi_0^{\rm ret})^{25} = (\varphi_0^{\rm ret})^{g_{\rm car}^2}$

So the carrier is the even row 5, the scale ladder is the lower row 5, the u/d leg is the cumulative row 4 up to $\deg u^c = 2$, and the mass volume closes on $g_{\rm car}^2$. *One* grammar (Λ^k of the 5-slot carrier and the $4 = |\mu_4|$ family fundamental), not four lucky hits. [`verification/v44_carrier_exterior.py`] [`verification/v45_family_exterior.py`] [`verification/v46_grand_mass_volume.py`]

2.2 One master moment generates 40, 41, 240, δ_2

With $X = 6Y$, the quadratic hypercharge trace on one family is $\operatorname{Tr}_{S^+} X^2 = 120 = 5!$, and it generates the whole integer chain:

$$\sum_{f,j} L_{f,j} = \frac{\operatorname{Tr}_{S^+} X^2}{3} = 40, \qquad 10 b_1 = \frac{\operatorname{Tr}_{S^+} X^2}{3} + 1 = 41,$$
$$|R(E_8)| = 2 \operatorname{Tr}_{S^+} X^2 = 240, \qquad \delta_2 = 4! \, \operatorname{Tr}_{S^+} X^2 \, c_3^8. \tag{5}$$

The flavor budget, the $U(1)$ budget, the E_8 count and the second seam defect all hang on the second hypercharge moment $5!$. [E]

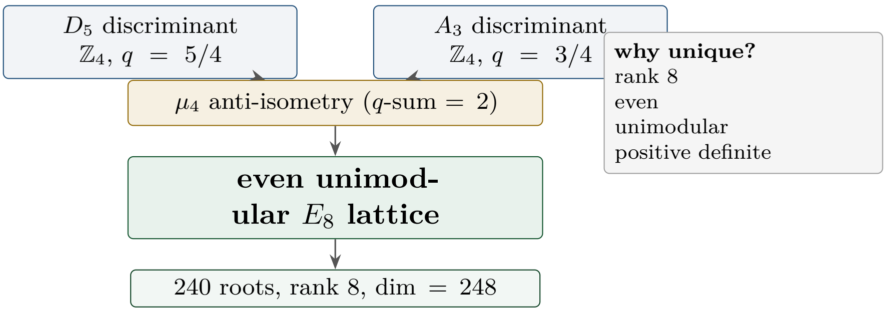
2.3 Compact E_8 compression and the role of ν^c

$$\boxed{\begin{aligned} |R(E_8)| &= \dim S^+ \cdot g_{\rm car} \cdot N_{\rm fam} = 16 \cdot 5 \cdot 3 = 240, \\ \dim E_8 &= \dim S^+ (\mathcal{A}_H + \mathcal{A}_\Lambda) + (g_{\rm car} + N_{\rm fam}) = 240 + 8 = 248 \end{aligned}} \tag{6}$$

with $N_{\rm fam} = (2^{g_{\rm car}-1} - 1)/g_{\rm car} = 15/5 = 3$ and rank $E_8 = g_{\rm car} + N_{\rm fam} = 8$. [E]. The right-handed neutrino ν^c is the Pascal singleton $\Lambda^0 E = (1,1)_0$ that closes $16 = 1 + 5 + 10$; removing it breaks the truncation and shifts α^{-1} by 2.16×10^{-3}.

3 The $D_5 \times A_3$ compiler: E_8 as a closure, not an input

This section upgrades the E_8 atlas (Section 10) from a counting coincidence to an explicit lattice construction. The whole content is consolidated and verified in the standalone companion note *"The Even Carrier Code and the $\mathbb{Z}_{30}$ Coxeter Compiler."* The thesis: the carrier code is the D_5 half-spinor, the family geometry is an A_3, and E_8 is the standard unimodular glue closure of $D_5 \oplus A_3$ — with the four punctures μ_4 as the glue code.



$D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ — the unique even unimodular rank-8 glue [E] exact (lattice).

The compiler in one line:

> five carrier slots $\Rightarrow D_5$, four family corners $\mu_4 \Rightarrow A_3$, $(D_5 \oplus A_3) + \mu_4$-glue $\Rightarrow E_8$.

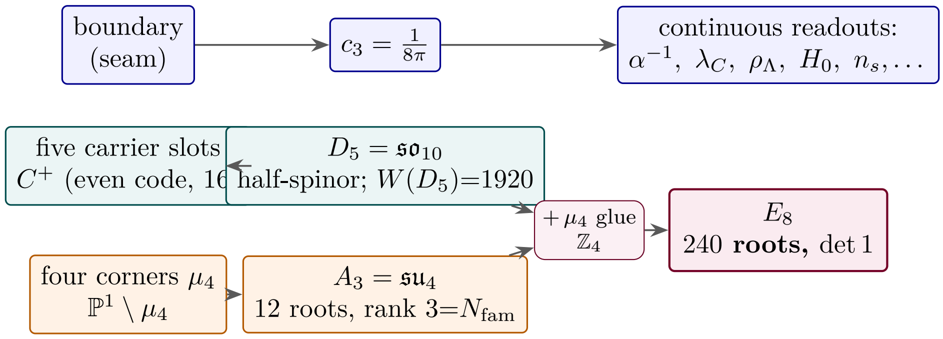
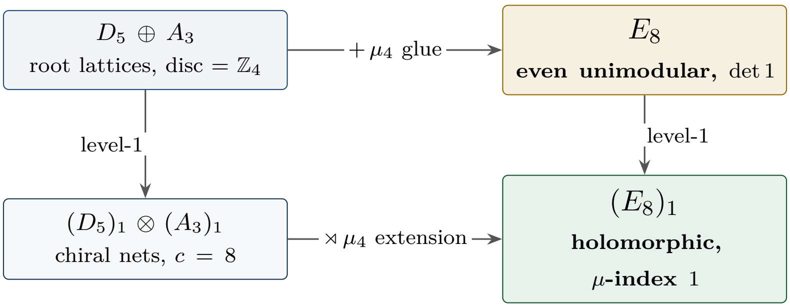


Figure 2: The explanation hierarchy. The boundary fixes the seam seed c_3; the five-slot even carrier code is the D_5 half-spinor; the four punctures μ_4 give A_3; their common $\mathbb{Z}_4$ discriminant (μ_4) glues $D_5 \oplus A_3$ into the unimodular E_8. E_8 is the *closure*, not the input; c_3 dresses the continuous observables.

The glue is *one* operation in two categories. The same order-$|\mu_4|$=4 discriminant glue closes $D_5 \oplus A_3$ into E_8 *twice over*: once on *root lattices* (the unimodular pushout $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, v1) and once on *chiral conformal nets* (the simple-current / anyon-condensation extension $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$, v92/v281). These are not two coincidences but the two functorial faces of a single commuting square (v234/v277/v260); the boundary realisation on which the strict-TOE residual turns (`SEAM.EQUIV.01`) is exactly the statement that the raw seam realises this square — now *closed modulo cited theorems on its MMST route* `SEAM.EQUIV.MMST.01` (the lattice/S_3 stack pins it at every computable level, Lean `FORM.SEAM.MMST.01`; only the cited MMST/OS continuum-existence theorems stay [O], v367/v368/v376–v379/v336; the twistor route `SEAM.EQUIV.TWISTOR.01` and the unconditional parent stay [O]). The extension leg of that citation package is itself re-founded at net level ([`verification/v469_seam_crossedproduct_route.py`]): the 128-spinor step is the *local* $\mathbb{Z}_2$ simple-current crossed product ($h_s(SO(16)_1) = 16/16 = 1 \in \mathbb{Z}$, the Longo–Rehren locality criterion; KLM $\mu = 4/2^2 = 1$), on 1995–2001 peer-reviewed subfactor theory, with the lattice-VOA preprint route demoted to an independent second witness, and the realisation input reduced to invariant level (R1′). Premise (i) of that package (c_-=8) now also carries a *ground-state* witness ([`verification/v489_seam_modular_commutator.py`]): the Kim–Shehab–Kim modular commutator $J(A,B,C) = (\pi/3)c_-$, evaluated on the exact BdG ground state of the collar model, gives $c_- = 8 = g_{\mathrm{car}} + N_{\mathrm{fam}}$ for the 16-layer carrier (per layer 0.500000 at L=48; trivial and orientation-flip controls pass) — independent of both the Chern route (v367) and the KLM count (v378). `SEAM.EQUIV.01` stays [O].



One operation, two categories: the order-$|\mu_4|$=4 glue is the lattice pushout $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and the simple-current / anyon-condensation extension $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$; the square commutes. [E] exact.

3.1 Carrier $= D_5 = \mathfrak{so}_{10}$ (proved)

The 16 even-occupation states $C^+ = \{x \in \mathbb{F}_2^{g_{\mathrm{car}}} : |x| \text{ even}\}$ are exactly the weights of the D_5 half-spinor $\frac{1}{2}(\pm1,\ldots,\pm1)$ (even number of minus signs). The flip-weight-4 relation is the **Clebsch graph** $\mathrm{SRG}(16,5,0,2)$ and the flip-weight-2 relation is its complement $\mathrm{SRG}(16,10,6,6)$; the *affine* automorphism group of the scheme (translations $\rtimes$ coordinate permutations) is the D_5 Weyl group,

$$\boxed{\mathrm{Aut}_{\mathrm{aff}}(C^+) = C^+ \rtimes S_5, \quad |\mathrm{Aut}_{\mathrm{aff}}(C^+)| = 2^{g_{\mathrm{car}}-1} g_{\mathrm{car}}! = 1920 = |W(D_5)| = r\,|R(E_8)|} \quad \text{[E]}, \tag{7}$$

(the linear code automorphism group alone is only S_5; the order-1920 group is the affine/scheme one) and the Clebsch edge count is the flavor budget $|E(R_4)| = \frac{16\cdot5}{2} = 40 = |R(D_5)| = \sum_{f,j} L_{f,j}$.

3.2 Family $= A_3 = \mathfrak{su}_4$ from $\mathbb{P}^1 \setminus \mu_4$ (proved, with the metric settled)

The carrier geometry fixes $X_f^\circ \cong \mathbb{P}^1 \setminus \mu_4$, $H_1 = \mathbb{Z}^3$, rigid D_4 generated by $z \mapsto iz$, $z \mapsto 1/z$. The twelve puncture-difference cycles $\gamma_i - \gamma_j$ carry the residue pairing $\mathrm{Res}(\omega_i, \gamma_j) = \delta_{ij}$ and form the A_3 root system:

$$\boxed{\#\{\gamma_i - \gamma_j\} = 12 = |R(A_3)| = \dim \mathfrak{g}_{\mathrm{SM}}, \quad \mathrm{rank} = 3 = N_{\mathrm{fam}}, \quad \mathrm{Gram} = \mathrm{Cartan}(A_3),\ \det = 4} \quad \text{[E]}. \tag{8}$$

The corner rotation $z \mapsto iz$ acts on $H^1 = \mathbb{C}^3$ as the A_3 Coxeter element (order $4 = h(A_3) = |\mu_4|$, spectrum $\{i, -1, -i\}$, $\chi = \Phi_4\Phi_2$), an isometry of the Cartan form — the exact analogue of the E_8 Coxeter element reading Φ_{30}.

The μ_4 clock is the order-4 character of the E_8 cycle, and $(2,3,5)$ is the icosahedron ([`verification/v223_coxeter_totative_clock.py`], [`verification/v219_icosahedral_mckay.py`]). The two “Coxeter” clocks are one object read at two scales. The E_8 exponents are the totatives $(\mathbb{Z}/30)^\times = \{1,7,11,13,17,19,23,29\}$, and the multiplicative order-4 element 7 (7^2=19, 7^4=1 mod 30) generates a literal μ_4 clock $\langle 7\rangle = \{1,7,13,19\}$ that permutes the four conjugate Coxeter 2-planes — the same μ_4 as the seam corner rotation. And the period $30 = 2\cdot3\cdot5$ itself is not a free factorisation: by the McKay correspondence E_8 is the exceptional *top* of the tower of finite $SU(2)$ subgroups ($2T \to \hat{E}_6$, $2O \to \hat{E}_7$, $2I \to \hat{E}_8$), so its branch data is the *icosahedron* and the three atoms $(|\mathbb{Z}_2|, N_{\mathrm{fam}}, g_{\mathrm{car}}) = (2,3,5)$ are its rotation-axis orders. The binary icosahedral group $2I$ (order $120 = |R^+(E_8)|$) has nine irreducible-representation degrees $\{1,2,2,3,3,4,4,5,6\}$ equal to the affine-E_8 Kac marks, with $\sum d_i = 30 = h(E_8)$ and $\sum d_i^2 = 120$. This is a *backward certificate* of the closed E_8 [E] (it explains the $(2,3,5)$ choice [C]; it does *not* replace the P2 input).

The metric, settled (former [C]). The transcendental geometric (Green) metric $G_{ij} = -\log|p_i - p_j|$ on μ_4 is only D_4-symmetric, with a regularisation-independent anisotropy $-\log 2$ on the root space, so it is *not* proportional to the A_3 Cartan form. But it does not need to be: the E_8-relevant A_3 is the canonical *residue* pairing (integral, S_4-symmetric, $=$ Cartan(A_3)), and the geometric metric is a D_4-equivariant deformation preserving the root system, Weyl group and Coxeter element. So the lattice structure is exact [E]; the analytic metric is a characterised deformation, not an open gap.

3.3 The μ_4 glue and the $5/4 + 3/4 = 2$ discriminant theorem (proved)

The two factors have equal discriminant groups and E_8 is unimodular, so the glue index is exactly $|\mu_4|$:

$$\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4, \qquad [E_8 : D_5 \oplus A_3] = \sqrt{\frac{\det D_5 \det A_3}{\det E_8}} = \sqrt{\frac{4{\cdot}4}{1}} = 4 = |\mu_4|. \tag{9}$$

The discriminant-form norms sum to the E_8 root norm, and — strikingly — they are pre-existing TFPT constants:

$$\boxed{\begin{aligned} &q(D_5) = \tfrac{5}{4} = \frac{\delta_2}{\delta_{\mathrm{top}}^2}, \qquad q(A_3) = \tfrac{3}{4} = \xi_{\mathrm{tree}}, \\ &q(D_5) + q(A_3) = 2 = |E_8\ \mathrm{root}|^2 \end{aligned}} \qquad \textbf{[E]}\ [\texttt{verification/v1\_e8\_glue.py}] \tag{10}$$

This is literally the E_8 spinor-root split $\frac{1}{2}(\pm)^8 = \frac{1}{2}(\pm)^5 \oplus \frac{1}{2}(\pm)^3$, $5 \cdot \frac{1}{4} + 3 \cdot \frac{1}{4} = 2$.

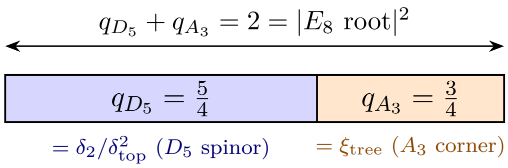


Figure 3: The discriminant-norm glue. The E_8 spinor root $\frac{1}{2}(\pm)^8$ (norm 2) splits across $D_5 \times A_3$ as $5 \cdot \frac{1}{4} + 3 \cdot \frac{1}{4}$; the two halves are the pre-existing TFPT constants $\delta_2/\delta_{\mathrm{top}}^2 = \frac{5}{4}$ and $\xi_{\mathrm{tree}} = \frac{3}{4}$.

The standard branching is realised explicitly:

$$\mathfrak{e}_8 = (45,1)\oplus(1,15)\oplus(10,6)\oplus(16,4)\oplus(\overline{16},\bar{4}), \qquad 248 = 45{+}15{+}60{+}64{+}64, \qquad 240 = 40{+}12{+}60{+}64{+}64, \tag{11}$$

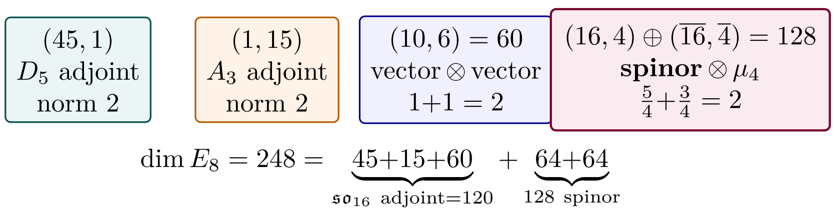


Figure 4: The $E_8 \supset D_5 \times A_3$ branching as norm-2 blocks. The adjoint half is the hypercharge moment $\mathrm{Tr}_{S^+} X^2 = 120 = \dim \mathfrak{so}_{16}$; the spinor half is two sheets over the four corners μ_4.

and assembling $D_5 \oplus D_3 \subset D_8$ with the even spinor coset produces all 240 roots (norm 2, Cartan det $= 1$, every root an integer combination of one simple-root basis): *the* 240*-metrisation is no longer blind — E_8 is built.* **[E]**

Hecke structure of the glue census is lattice-native (HECKE.GEOM.01). The μ_4-glue class thetas that refine the 240-root branching ($240 = 52{+}64{+}60{+}64$ over $\mathbb{Z}/4$) now carry a machine-checked Hecke package ([`verification/v535_hecke_from_geometry.py`]): Kneser p-neighbours realise

#iso_lines $= \sigma_3(p)\,\#\mathbb{P}^3(\mathbb{F}_p)$ at $p = 2, 3, 5, 7$; the marked neighbour-sum is the affine element $\nu_p = a\,\mathrm{Id} + b\,T_p$ with cusp output $a_3 = -4$, $a_5 = -2$; and the seven census q-series span exactly the 2-adic oldform hull of E_4 and f_8 (levels $\{1, 2, 4, 8, 16\}$ and $\{8, 16\}$). Classical scaffolding named classical; weight-4 →GL(1) stays [O]; no RH claim. [E] (in-suite mechanics)

Eichler trace layer at the E_8 lattice (`HECKE.GEOM.EICHLER.01`). The same census channels now carry a machine-checked Eichler/Witt package ([`verification/v536_eichler_trace_layer.py`]; 23 checks, ~ 30 s; builds on `HECKE.GEOM.01` without duplicating Kneser/ν_p/oldform checks). *(A)* The elementary Witt shadow $\lambda_{\mathrm{Eis}} = \sigma_3(\#\mathbb{P}^3 - \sigma_3) = L - \sigma_3^2$ is closed in p (identically and for all odd $p \le 100$). *(B)* The Eichler identity $\lambda_{\mathrm{geom}} = \lambda_{\mathrm{Eis}} + a_p^2$ is two-sided at $p = 3$ (live type-(i)/(ii) split: $352 = 364 - 12$) and at $p = 5$ via a T36-validated freeze ($3784 = 3906 - 122$); the assembler $R(p) = \sigma_3 - 1 - c(p^2)/8$ equals a_p^2 with Kneser filter = identity. *(C)* Closed Type-A/B local densities $N_A = \min\bigl(240(1+p^3),\ p^7 + p^4 - p^3 - 1\bigr)$ (B empty iff $p^4 < 241$; $p = 7$ live 82560/743040; injectivity at $p = 11, 13$). *(D)* Signed extraction $a_p = -c(p)/8$ from geometric Shell(p) recovers $(-4, -2, +24)$ and $b \in \{24, 124, 368\}$. *(E)* $O(1)$ assembler for all odd $p \le 31$ against the eta product f_8. Classical Eichler/Siegel/Witt named classical; the claim is in-suite two-sidedness and the closed compiler densities. Honest fences: Shell(p^2) for $p \ge 7$ out of scope; fibre refinement scoped to $\{5, 7, 11\}$; weight-4 →GL(1) stays [O]; no RH claim; no marker moves. [E] (mechanics) / [O] (weight drop)

Half-integral bridge (`HECKE.GEOM.HALFINT.01`). The weight-4 cusp f_8 now has a machine-checked half-integral companion in the compiler theta-monoid ([`verification/v537_halfintegral_bridge.py`]; 20 checks, ~ 90 s; companion to `HECKE.GEOM.01`/`HECKE.GEOM.EICHLER.01`). *(A)* In the 70-element weight-5/2 theta-monoid (plain and q^2-scaled $\vartheta_2, \vartheta_3, \vartheta_4$) there is exactly one object with the signed Shimura identity $\mathrm{Sh}_{t=2,\psi=1}(g) = -8\,f_8$ (witness $g = \vartheta_2(q^2)^2\,\vartheta_3(q^2)\,\vartheta_4\,\vartheta_4(q^2)$, verified coefficientwise through $n \le 120$; three further monoid elements lift to scales $+8, -16, +16$). *(B)* g is an exact $T(p^2)$-eigenform of weight 5/2, trivial nebentypus, with eigenvalues $a_p(f_8)$ for $p = 3, 5, 7, 11, 13$ $(-4, -2, 24, -44, 22)$; trial FE of the 5/2-side around centre 5/4 with $N = 512$, $\varepsilon = +1$. *(C)* $U_4(g) = 0$ exactly (mass mod 4 supported on $\{1, 2\}$); level $32 = 4 \cdot 8$ with $M = 8$ even lies outside Kohnen 1982 — the plus-route is structurally closed (negative result part of the claim). *(D)* With AFE-based twist L-values ($\varepsilon_d = \chi_d(8)$; $\varepsilon = -1 \Rightarrow L(2) = 0$), the Waldspurger quotient $R(d) = b(|d|)^2/\bigl(|d|^{3/2} L(f_8 \times \chi_d, 2)\bigr)$ is constant on ten fundamental discriminants $d \equiv 1 \pmod 8$ ($R = 23.1873585645\ldots$, relative spread $\le 10^{-12}$, including $d = 89$); the class $d \equiv 5 \pmod 8$ vanishes structurally. Classical Shimura/Kohnen/Waldspurger/Baruch–Mao named classical; the claim is the unique signed-scale monoid preimage, the in-suite chain, and the *measured* constant R. Honest fences: GL(2) centre $s = 2$, not the ξ-line; no RH claim; `ZETA.HP.CARRIER` untouched; no marker moves. [E] (Shimura/Hecke/Kohnen fence) / [C] (measured R)

Compiler relative-trace identity (`HECKE.GEOM.RTF.01`). The three preceding packages are machine-checked as three *projections* of one finite relative-trace identity [E_8/glue orbital counting] = [Newform/Waldspurger spectral sum] ([`verification/v538_relative_trace_identity.py`]; 18 checks, ~ 14 s; promotion of discovery T53; companion to `HECKE.GEOM.01`/`HECKE.GEOM.EICHLER.01`/ `HECKE.GEOM.HALFINT.01`). *(R1)* $\mathrm{Tr}_V(\nu_p \circ \pi)$ is computed *both sides independently*: geometric $a_p = -c(p)/8$ from the glue census (no modular input) equals spectral a_p from $\eta(2\tau)^4\eta(4\tau)^4$ (no geometry) for all $(p, \pi) \in \{3, 5, 7\} \times \{\mathrm{Id}, \pi_{\mathrm{Eis}}, \pi_{\mathrm{cusp}}, \pi_{E_4}, \pi_{f_8}\}$ (anchors e.g. p=7: full trace 727680, f_8-channel 19840). *(R2)* The Eichler split $\lambda_{\mathrm{geom}} = \lambda_{\mathrm{Eis}} + a_p^2$ *is* the RTF orbit decomposition (principal orbit $\to \lambda_{\mathrm{Eis}} = L - \sigma_3^2$; elliptic $\to a_p^2$; cross terms cancel; anchors $\lambda_{\mathrm{Eis}} = 336/3780/19264$, $a_p^2 = 16/4/576$). *(R3)* The period side is lattice counting: g is the quaternary form $n = (x^2 + y^2)/2 + 2z^2 + u^2 + 2w^2$ (x, y odd; sign $(-1)^{|u|+|w|}$), with $b_{\mathrm{rep}} = b$ through $n \le 200$, and [signed lattice count]$^2 = R \cdot |d|^{3/2} \cdot L(f_8 \times \chi_d, 2)$ at $R = 23.1873585645$ (rel $\sim 10^{-12}$). Verdict `ONE-FORMULA`. Classical Jacquet-type RTF / Eichler / Witt / Waldspurger / theta correspondence named classical; NEW is the compiler-native realisation and bilateral independence. Honest fences: the identity is *finite* ($p \in \{3, 5, 7\}$, finite d-windows); the *infinite* RTF (sum over all p and d + archimedean term) remains the named open step; no RH claim; not `ZETA.HP.CARRIER`-relevant; no marker moves. [E] (R1/R2 exact) / [C] (R3 measured R)

Weil structure of the compiler family (`RTF.GNS.WEIL.01`). The compiler family carries a fully identified Weil structure up to *two explicitly isolated obstructions* ([`verification/v539_weil_structure_family.py`]; 25 checks, $\sim$ 6 s; promotion of discovery T55/T61/T62/T63; companion to `HECKE.GEOM.RTF.01`). *(A)* The canonical GNS pairing is the direct integral over the Gelfand spectrum of the character subalgebra: metric $\ell^2(d, b^2/|d|)$ diagonal; fibres $\sigma(d) = (\chi_d(3), \ldots, \chi_d(13))$ partition the live family; Λ_{fam} constant on each fibre; twist-mixing certificate per fibre $\leq 5.037 \times 10^{-16} < 10^{-12}$. *(B)* The trivial Sato–Tate isotype is algebraically the GL(1) core $G_0 = (1+Y)/(1-Y) = \zeta_p(w-3)^2/\zeta_p(2w-6)$, globally $\zeta(u)^2/\zeta(2u) = \sum 2^{\omega(n)} n^{-u}$ (Dirichlet coefficients exact through $n \leq 2000$; character expansions $\Phi_1 = \chi_0 + \chi_2$, $\Phi_2 = \chi_4 - \chi_2$, $\Phi_3 = 2\chi_0 - \chi_4 + \chi_6$, $\Phi_4 = \chi_8 - \chi_6$; core series $c_{k,0} = [1, 0, 2, 0, 2, 0]$). *(C)* The family form satisfies the exact linear relation $Q_{\text{fam}} = 2Q_\zeta(g) - 2Q_\zeta(g_\flat) + \text{Arch} + \text{Corr}$ ($\text{Prime}_F = 2P_\zeta(g) - 2P_\zeta(g_\flat)$ with max rel 9×10^{-16}; $Q_{\text{fam}} = Q_F + \text{Corr}$ with rel 2×10^{-16}). **Obstruction 1:** the doubling term enters with a *minus* (family positivity does *not* imply $Q_\zeta \geq 0$). **Obstruction 2:** the Plancherel correction is demonstrably non-automorphic (no polar/arch absorption; no finite Euler factor; identified as $e^{-\sum p^{-u}}$-type). Finite-class positivity $Q_{\text{fam}} \in [4.369, 11.486]$ is measured [C]; dense-class positivity stays open / RH-adjacent and is *not* claimed. Classical Weil 1952 / Gelfand / SU(2)–Chebyshev / Sato–Tate / 2^ω-series / GNS named classical; NEW is the compiler-native identification and the machine-checked isolation of the two obstructions. Honest fences: *not* "almost RH" — the claim structurally shows why family positivity does not imply RH; `ZETA.HP.CARRIER` untouched; no marker moves. [E] (A/B/C/obstructions exact) / [C] (finite-class Q_{fam})

Amplitude route and positive linear carrier (`RTF.GNS.AMP.01`). The route out of the square plane behind the two `RTF.GNS.WEIL.01` obstructions is now machine-checked ([`verification/v540_amplitude_linear_carrier.py`]; 34 checks, $\sim$ 3 s; promotion of discovery T67–T72; companion to `HECKE.GEOM.RTF.01/RTF.GNS.WEIL.01`). *(A)* The amplitude Dirac $D = \begin{pmatrix} 0 & V \\ V^\mathsf{T} & 0 \end{pmatrix}$ with $V[d,m] = \sqrt{w_d}\, b(d)\, \chi_d(m)$, $w_d = |d|^{-1}$, exists exactly and is Hecke-equivariant: $D = D^\mathsf{T}$, $D^2 = \operatorname{diag}(VV^\mathsf{T}, K)$ with $V^\mathsf{T} V = K$ (rel 2.7×10^{-16}), spectral $\pm$-symmetry; m-side intertwining $VA_p^\mathsf{T} = \hat{A}_p V$ exact on the p-free locus ($p = 3, 5, 7$) and d-side Shimura $b(dp^2) = b(d)\,(a_p - \chi_d(p)p)$ integer-exact on 644 pairs. *(B)* The geometric polarisation $b = N_+ - N_-$ is exact (parity-split lattice count of the quinary form, plus the Θ-monomial identity coefficientwise); $\Theta = N_+ + N_-$ is a *pure* Siegel–Weil Eisenstein $T(p^2)$-eigenform with eigenvalues $\sigma_3(p) = 1 + p^3$ ($p \leq 13$ exact; cusp component zero), carrying the Cohen seed identity $\Theta(d) = -48\, L(-1, \chi_d)$ exact-rational on 159 live d (Cohen 1975: $H(2,d) = L(-1,\chi_d)$; generalised Bernoulli $B_{2,\chi}$). *(C)* Every coefficient bilinear form inherits the even-k deletion: the Cauchy–Littlewood window lemma with *free* Satake symbols has numerator $1 - stuv\, X^2$, hence $1 - p^6 X^2$ pair-independently at $st = uv = p^3$ (five channels checked); simultaneously the deletion *is* the square-class double counting of the towers, fam $[1, 0, 2, 0] + 2\flat\, [0, 2, 0, 2] + \delta_{k1} = [2, 2, 2, 2]$ (generating-function exact; Möbius square sieve). *(D)* The positive linear carrier $\ell^2(d, \mu)$ with $\mu = 48\, |L(-1, \chi_d)|\, |d|^{-a}$ carries *full* weights $\lambda_k = 1 + p^{3k} - (\chi p)^k$ (layer $[1, 1, 1, 1]$; no even-k kill) and the plus balance $Q = Q_\zeta(g_-) + Q_\zeta(g_+)$, $g_\pm = e^{\pm 3u/2} g$ (rel 2×10^{-15}); the $\flat$/doubling kernel enters *with plus* — the sign of the v539 minus inverted. *(E)* The completed functional equation $\Lambda_\Theta(s) = 8^{1-s}\, \Lambda_{\Theta^\dagger}(5/2 - s)$ holds exactly (Fricke closed via the Jacobi inversions; rel $\sim 10^{-40}$ modular and in the strip; poles $5/2$ and 0 swapped; centre $5/4$, plus-balance locus off-centre by exactly $1/4$); the guaranteed cone and the gap functional are FE-self-dual/-covariant. **Open boundary as content:** the positivity is *Euler-region* positivity (edge L-values, not the central line); the residual distance to the Weil cone is the FE-covariant gap functional λ^* on $n \equiv 6 \pmod 8$ — named, measured, Farkas-certified, and not removable by any finite signed theta library. Classical Cohen 1975 / Siegel–Weil / Jacobi–Fricke / Landen / Hecke split-Mellin / Cauchy–Littlewood / Shintani / Weil 1952 / Farkas named classical; NEW is the compiler-native consolidation. Honest fences: *not* "almost RH"; `ZETA.HP.CARRIER` untouched; no marker moves. [E] (A–E exact) / [C] (sampled cone facts; λ^* named open)

Matching lemma and transport ledger package (`RTF.GNS.LEDGER.01`). The discovery proof package T78–T85 is now load-bearing as one module in which every core check is *re-*

computed, not cited ([verification/v541_matching_lemma_ledger.py]; 33 checks, ~ 10 s; companion to HECKE.GEOM.RTF.01/RTF.GNS.WEIL.01/RTF.GNS.AMP.01). *(A) Window proof.* The matching lemma of the hybrid certificate recipe is *proved exact-integer on* $[4, 10^6]$: the certificate $7S(j) < 40A(j)$ (i.e. $\rho_{\text{design}}F(j) < 1$ with $\rho = 21/20$) holds at *every* atom $j \le 10^6$ (0 violations over 939 870 clash atoms; full enumeration, proven no-overflow, big-integer rechecks), with the exact Fraction margin $X = 22242575693/270725400000 = 0.082159$ at the exact argmax $j^* = 554400$ and $\rho_{\text{crit}} = 1.1440 > 21/20$; the maximal-greedy identity makes the certificate *uniform* in the target set M and the greedy weights. Structure laws at 0 tolerance: the 8-law $\Theta(4n) = 8\Theta(n)$ ($n \le 250000$); the seed-tower divisor formula $\Theta(4^a s f^2) = 8^a\Theta(s)\alpha_s(f)$ and the ψ w-table (all $n \le 50000$; Euler factor and w-recursion sympy-exact); the Cohen seeds $\Theta(s) = c\,L(-1, \chi_\Delta)$ with *one* exact-rational constant per residue class $\{-48, -80, -8, -8\}$ on all squarefree $s \le 2000$; the bracket $\Theta \le 2|\psi| \le 3\Theta$ on the full window. *(B) Transport ledger.* The ledger $Q_{\text{Weil}} = Q_{\text{cert}} + \Delta_{\text{arch}} + \Delta_2$ closes *exactly*: $\Delta_{\text{pole}} \equiv 0$ and $\Delta_{\text{conv}} \equiv 0$ are *proven identities* (sympy kernel collapse; quadrature 2.8×10^{-16}; exact scale invariance), the residual is 4.4×10^{-16} on 24 exact-autocorrelation battery rows, and the odd-prime side of the full Weil functional agrees *exactly* with the certified plus combination $P_\zeta^{\text{odd}}(h) = P_\zeta(g_-) + P_\zeta(g_+)$ (rel 3.9×10^{-16}): the prime side was never the wall. *(C) Character-exact signed envelope.* $-\psi = (\chi_{-4} + \frac{1}{4}\chi_8 + \frac{1}{4}\chi_{-8}) \cdot \Theta$ coefficient-exact on all odd $n \le 50000$ on both independent builds (character system solved sympy-exact); the signed certificate $7S_{\text{net}} < 40A$ holds with 0 violations and margin factor 8.86×; the confinement is a *set equality* on 10^6: {zero-credit clash atoms} = {χ_{-4}-coherent odd composites}. *(D) The archimedean term is internal.* Via Legendre duplication $(2\pi)^{-s}\Gamma(s) = \frac{1}{2}\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1)$ (sympy-exact + 1.8×10^{-41}) the ledger kernel obeys $k_\zeta = K_{\text{fam}} - K_{\text{shift}}$ *pointwise* (7.9×10^{-31} on 51 points; Gauss digamma anchor at $t = 0$), and $\Delta_{\text{arch}}(h) = A_{\text{fam}}(h) - A_{\text{shift}}(h)$ on the battery subset at rel 9.9×10^{-16} — I5 becomes the self-consistency of *one* heat family. *(E) λ-equivariant channel.* The CM carrier $g_\lambda = \sum_{\mathfrak{a}} \lambda_1(\mathfrak{a})\, q^{N\mathfrak{a}}$ exists exactly in two independent routes ($\mathbb{Z}[i]$ lattice sum of z^4 vs Cornacchia reconstruction, 0 mismatches $n \le 10^4$; CM laws exact; $\text{supp}(c_1) = \{\mathbb{Z}[i]\text{-norms}\}$ as a set equality with $c_1 \ne 0$ at every coherent atom; $\vartheta_3^2 \equiv r_2$ on 10^6); the canonical phase mean $\mu_1 = c_1/(c_0 d^2) \in [-1, 1]$ is exact, and the λ-window certificate over all $j \le 10^6$ gives $\max 7|S_\lambda|/(40A) = 0.0653$ strictly below the unlifted 0.2365 (factor 3.6×; exact Fraction rechecks) — the sign-mixing of the Grossencharacter phases *is* the cancellation channel; coherent targets are reachable only by coherent rescalings (T81 reachability). *(F) Frontier facts [C]:* unlifted coherent-chain crossing $k^* = 14$ (exact Fractions, $\log_{10} N^* = 23.1$); absolute Robin tail factor 6.16 at $J = 10^6$ and divergent — the constant route closes nothing beyond the window. **Named limits as content:** (i) *one* classically-shaped open lemma remains — the correlated-cancellation lemma on credit-rich non-coherent tail atoms (uniform form); provably-shaped $\ne$ formal proof; (ii) I5 in one-family form ($Q_{\text{cert}} + \Delta_2 + A_{\text{fam}} - A_{\text{shift}} \ge 0$) is the single remaining TFPT-specific object — by the closed ledger *equivalent* to Weil positivity $\iff$ RH (Weil 1952): an equivalence *typing*, not a progress claim. Classical Gronwall 1913 / Robin 1983 *unconditional* / Cohen 1975 / Hecke 1918–1920 / Cornacchia / Fermat–Gauss / Dirichlet $L(1, \chi)$ / Mertens-AP / Landau 1908 / Legendre duplication / Weil 1952–Guinand–Bombieri / Alaoglu–Erdős named classical; NEW is the compiler-native consolidation and the machine-checked window proofs. Honest fences: no RH-evidence language; *not* "almost RH"; ZETA.HP.CARRIER untouched; no marker moves. [E] (A–E exact) / [C] (frontier facts; two named limits open)

The margin-chain identities of phase 2 (PRIME.MARGIN.IDENT.01). The identity/theorem block of the discovery parts T128–T131 is now load-bearing as one *narrow* module ([verification/v542_margin_chain_identities.py]; 44 checks, ∼0.2 s; companion to RTF.GNS.LEDGER.01), recomputed on small frame-A windows: 24 seam instances ($n = 3 \ldots 97$, $\rho = 1.25 \ldots 4.00$, every inverted or diagonalised matrix ≤ 300), 43 graded/fine grid pairs and 400 randomised profiles, each identity a numerical residual against a preregistered tolerance *plus* a mutation control that must fail by $\ge 10^{-3}$ relative. *Border geometry.* $\sum_i(1 - g_i) = t_l + t_r$ exactly (5.5×10^{-13}), so the κ weights are a probability vector and the flat border vector gives $\kappa_{\text{flat}} = 1$; the (T) four-term identity $\tau_{\text{dn}} = \|y\|^2 - m_{\text{prot}} + m_{\text{fill}} - V_{\text{bord}}$ holds at 2.2×10^{-16}, with τ recomputed independently from the full odd overlap matrix (1.1×10^{-13}). *Profile and curvature.*

$p_{N-1} = 2 - p_0 + 2E$ at 2.3×10^{-16}, hence the equivalence $2E > p_0 \iff p_{N-1} > 2$ with 0 exceptions (both branches realised), and the Abel/Hölder chain $\kappa \le 2 - p_0 + \frac{N-2}{N} \sum_j |\Delta p_j - \overline{\Delta p}| + (p_{\max} - p_{N-1})$ holds per instance with 0 violations — while the same chain *without* the curvature term is violated on 88/400 profiles, which is what makes the curvature term load-bearing. *Assembly and compression.* The matrix-free two-scale assembly equals $J^{\mathsf{T}} Q J$ (5.2×10^{-16}), the graded overlap operator equals J, and the Céa/Strang defect identity $S_{\text{graded}} - S_{\text{uniform}} = R^{\mathsf{T}} X^{-1} R$ holds as a *matrix* identity on all 43 grids (2.1×10^{-14}) with a PSD defect — the compression error is one-sided *by* the identity. *Sandwich, sign, counting.* The secular sandwich $\varepsilon/(\|u\|^2 + \varepsilon/\mu_1) \le \lambda_{\min}(Q) \le \varepsilon/\|u\|^2$ holds on all 48 sections at which $A > 0$ and $\varepsilon > 0$ are *checked* (relative width ≤ 1.4448), with Albert's sign equivalence verified in both directions; $S^{-1} > 0$ entrywise (measured 48/48) implies a sign-constant and simple ground vector (Perron–Frobenius), with a control matrix where hypothesis and conclusion fail together; and $n_{\text{run}} \le n_{\text{cross}} + 1 \le s_2 + 2$ together with $\sum_j |\Delta p_j - \overline{\Delta p}| = \mathrm{TV}(P)$ hold without exception on a non-vacuous battery. **Named limits as content:** every statement is *per instance on small windows* — nothing is uniform in the zone index; the κ *law* is *not* claimed (T129 broke it), only the chain underneath it; positivity of the pole-free section at depth stays open and carries no floor value here; the Perron step has a *measured* hypothesis; and the defect identity says nothing about the *size* of the defect. Abel / Hölder / Schur–Haynsworth 1968 / Albert 1969 / Céa 1964–Strang / Yserentant 1986 / Perron–Frobenius–Ostrowski 1937 named classical; Weil 1952 cited, never used as a criterion. [E] (per-instance identities; nothing uniform, no marker moves)

The lumped M-matrix pair (`PRIME.MMATRIX.IDENT.01`). The identity block of the discovery part T136 is load-bearing as a second narrow module ([`verification/v543_lumped_pair_identities.py`]; 35 checks, $\sim$1 s; companion to `PRIME.MARGIN.IDENT.01`) on 20 frame-A seam windows ($n = 4 \ldots 59$, $\rho = 1.25 \ldots 4.00$, $h = 25 \ldots 299$, every factorised matrix ≤ 300) giving 39 matrices with positive off-diagonals, 312 shifted ladder rungs and 8 whitened two-grid rungs. *The pair and the congruence.* With Δ the positive off-diagonal part and $L_\Delta = \operatorname{diag}(\Delta \mathbf{1}) - \Delta$, the lumped pair $A_B = A + L_\Delta$ is Stieltjes and row-sum preserving (1.6×10^{-15}) with $A_B \succeq A$ — an *upper* bound on $\lambda_{\min}(A)$, never a floor; $A = A_B^{1/2}(I - W)A_B^{1/2}$ holds as a matrix identity (4.4×10^{-15}) with $\lambda_{\max}(W) = \rho(L_\Delta A_B^{-1})$, and the equivalence $\lambda_{\max}(W) < 1 \iff A \succ 0$ is checked in *both* directions, so the Loewner sandwich is circular by itself. *Anchor, ladder, Varga.* With $x = A_B^{-1}\mathbf{1} \ge 0$ and $A_B^{-1} \ge 0$ entrywise (Ostrowski 1937, checked), $\lambda_{\min}(A_B) \ge 1/\max_r x_r$ on all 39 rows (sharpest ratio 0.9070) from one solve and one sign check; lumping commutes with diagonal shifts and M-matrix inverses are entrywise monotone (Berman–Plemmons 1979), so the shifted ladder is *structurally* empty (0 exceptions of 312); and for the regular splitting $A_B = D_0 - N_0$ Varga's $\rho(J) = \tau/(1+\tau)$ is an identity (4.8×10^{-15}) whose Collatz–Wielandt bracket at the anchor vector gives an a-priori $\rho(J) \le 0.9861 < 1$ with no eigenvalue anywhere. *One negative result* [X]. The k-scanned M18 majorant is *valid* at every k (39 pairs, 0 violations) and still stays above its preregistered bar 1/2 (best 3.080) while the measured bad mass is ≤ 0.0120: the loss is the bound, the localisation term alone exceeds the bar, and the bar is not moved. **Named limits as content:** per instance on small windows, nothing uniform in the zone index, in D or in the gap; the T136 D-exponents are fits and stay in the sandbox; the anchor and Varga rows are implications with hypotheses checked at the instance. Stieltjes / Ostrowski 1937 / Fan 1958 / Berman–Plemmons 1979 / Varga 1962 / Collatz 1942–Wielandt 1950 named classical; Weil 1952 cited, never used as a criterion. [E] (per-instance identities) + [X] (one closed route; no marker moves)

The long-lag support structure (`PRIME.LONGLAG.SUPP.01`). The structure block of the discovery part T137 is load-bearing as the companion module ([`verification/v544_long_lag_support.py`]; 24 checks, $\sim$0.2 s) on the same 20 windows (3–25 prime-power atoms, 20555 positive off-diagonal edges, 455 loaded stripes, 39 bracketed lumped comparisons). *Support and amplitude.* The kernel splits exactly as $c = c_{\text{arch}} - c_{\text{comb}}$ with $c_{\text{comb}} \ge 0$, and the archimedean section alone has *all* off-diagonals ≤ 0 on 20/20 windows, so every positive off-diagonal is comb-generated; all 20555 positive edges have their Hankel index $M - 1 - r - t$ inside the atom-hat index set (per-window fraction exactly 1.000000), so the support is a union of anti-diagonal stripes at prime-power atom positions, each stripe a perfect matching with *exactly* orthogonal edge vectors; and $\Delta_{rt} \le c_{\text{comb}}[M - 1 - r - t]$ on all

20555 edges (sharpest ratio 0.99998532), with the Toeplitz lag in place of the Hankel index failing on 20415 of them. *The Green bracket.* For the lumped comparison, $q_J = 1 - \min_r 1/(d_r x_r) < 1$ is an identity (8.2×10^{-16}) needing no measurement, and the Neumann partial sum brackets the Green function entrywise from below *because the discarded terms are nonnegative*, the anchor-weighted tail from above, on 39 rows. *A death certificate* [**X**]. Gershgorin, every row-sum norm and every Collatz–Wielandt quotient at every *positive* weight are upper bounds for $\rho(|E|)$, $E = B^{-1}L_\Delta$; one rigorous lower bound therefore decides them all at once: $\rho(|E|) \geq 1.3141 \geq 1$ on 20/20 rows, so no member of the absolute-value envelope family can *ever* certify $\lambda_{\max}(W) < 1$ — while the *signed* radius $\rho(E) = \lambda_{\max}(W) \leq 0.99995781$ on the same rows. The absolute value alone destroys the cancellation. **Named limits as content:** per instance on small windows, no decay law and no amplitude law; the support statement is one inclusion; the amplitude bound does *not* bound $\lambda_{\max}(W)$, and the structure bound assembled from it does not beat the margin (T137: 0 of 35 windows) and is not promoted; the bracket is for the lumped comparison, not the target. Gershgorin 1931 / Neumann series / Haynsworth 1968 named classical; Demko–Moss–Smith 1984 cited as a model and *not* applied (the comparison is dense); Weil 1952 cited, never used as a criterion. [**E**] (per-instance structure) + [**X**] (one family closed; no marker moves)

The Hardy-core identities (`PRIME.HARDY.IDENT.01`). The identity blocks of the discovery parts T140 and T141 are load-bearing as the third companion module (36 checks, ∼1 s) on 11 frame-A windows ($n = 8 \dots 139$, $h = 58 \dots 285 \leq 300$, 193–5102 positive off-diagonal edges, D spanning a factor 3.15). *Form lifting and finite core.* With $C = \mathrm{diag}(\sqrt{\Delta})M$ and M the edge–interval incidence, Gram $= CHC^{\mathsf{T}}$ (3.0×10^{-15}), so rank(Gram) $\leq h - 1$; the covering kernel in closed geometric form $K_{rs} = W([r \wedge s,\, r \vee s])$ equals $M^{\mathsf{T}}\Delta M$ (1.3×10^{-15}); and $\rho(W) = \lambda_{\max}(K^{1/2}HK^{1/2})$ is an *exact* eigenvalue, agreeing with $1 - \lambda_{\min}(A, A_B)$ (5.4×10^{-14}) and with $\lambda_{\max}(\mathrm{Gram})$ (7.8×10^{-16}) — three independent routes, no truncation — while $H = \mathrm{diag}(s) + L_N$ (4.7×10^{-16}) reads the sign law as mass plus a long-range Dirichlet form. *The four Hardy identities.* $\rho(W) = u^{\mathsf{T}}Hu/u^{\mathsf{T}}K^{+}u$ at $u = K^{1/2}v$ (6.6×10^{-13}); $L_N = D^{\mathsf{T}}BD$ with the signed crossing kernel $B_{kl} = \sum_{r \leq k \wedge l,\, k \vee l < s} N_{rs}$ (4.9×10^{-16}); $Q = B_{+}\mathbf{1}$ (1.4×10^{-15}), so the Cauchy–Schwarz step is the Gershgorin diagonal step; and $DKD^{\mathsf{T}} = L_\Delta$ on the interior nodes (2.7×10^{-14}), whose diagonal is the endpoint edge mass. The crossing kernel of $|N|$ does *not* reproduce L_N: the signs are load-bearing. The closed Muckenhoupt quantity $A_{M0} = \max_k Q_k(\mathbf{\Delta 1})_{k+1}$ follows from two routes agreeing to 8.3×10^{-15}; its *size* is reported, not promoted. The checkerboard split $\mathrm{Band}_b = D + A_{\mathrm{even}} + A_{\mathrm{odd}}$ is entrywise exact on 9 (window, b) pairs, with three Weyl steps dominating. *Two shapes, promoted as valid and not by size.* Every Loewner step is Cholesky-certified; the additive Hardy chain dominates $\rho(W)$ on 11/11 (3.074–10.758×), its exact Weyl split dominates too (1.691–2.503×) and floors the additive family, and the joint shape $\Lambda(H, Y) \cdot \Omega(Y)$ dominates on 11/11 for a positive definite Jacobi Y — the same Y without its mass term is singular and the certificate refuses. *A negative typing* [**X**]. Over the D range the exactly bounded object spreads 1.36× while every absolute variant spreads $\geq 54.94\times$; the power laws ($D^{-0.138\pm0.094}$ signed against $D^{-0.776\pm0.352}$ and $D^{-2.748}$, $D^{-2.820}$, $D^{-2.918}$) are fits and labelled as fits. **Named limits as content:** per instance on small windows, nothing uniform in the zone index or in D; the bound rows promote *validity* and not size — neither shape beats the target — and the finite-core row is an identity, so the zone-uniform discrete Hardy inequality stays open. Abel summation / Cauchy–Schwarz / Weyl 1912 / Gershgorin 1931 / Muckenhoupt 1972 / Bradley 1978 / Opic–Kufner 1990 / Gantmacher–Krein 1950–1960 named classical; Weil 1952 cited, never used as a criterion. Machine-checked: [`verification/v545_hardy_core_identities.py`]. [**E**] (per-instance identities) + [**X**] (one route typed; no marker moves)

The capacity-chain identities (`PRIME.CAPCHAIN.IDENT.01`). The identity spine of the capacity cycle T142–T145 is load-bearing as the fourth companion module (26 checks, ∼0.5 s) on 12 frame-A windows ($n = 4 \dots 139$, $h = 26 \dots 285 \leq 300$, D spanning 7.6×10^{-3}–2.8×10^{-2}; the covering kernel positive definite on 11, where the first two items run). *The capacity decomposition.* $K^{-1} = D^{\mathsf{T}}J^{-1}D + xx^{\mathsf{T}}/\mathrm{cap}$ with $J = DKD^{\mathsf{T}}$ (3.8×10^{-15} entrywise), $x = K^{-1}\mathbf{1}$, $\mathrm{cap} = \mathbf{1}^{\mathsf{T}}K^{-1}\mathbf{1}$, as a matrix identity ($1.4 \times 10^{-13}$); under the congruence the Dirichlet half is an *orthogonal projection* — $\Omega = 1$ exactly (5.1×10^{-12}), where T141 had guessed 20.7–2724 — and its complement is the capacity rank-one. *The exact capacity–Rayleigh identity.* The assembled quotient for $1-\rho(W)$ equals $v^\mathsf{T}(K^{-1}-H)v/v^\mathsf{T}K^{-1}v$ for every v (9.5×10^{-14}), reproduces $1-\rho(W)$ at the exact top eigenvector under the declared 1/gap conditioning, and satisfies the inf property; on the constant vector the whole denominator *is* the capacity term. *Cholesky nestedness.* $\mathrm{cap}_E([a,a+j))$ is a prefix sum of squared forward-substitution entries of one triangular solve (3404 interval capacities at 2.0×10^{-14}, exhaustive on the median window); the reversed ordering fails at $\geq 4.6\times10^{-1}$. *The whitening certificate.* $\lambda_{\max}(W)\leq\kappa_{\rm up}\leq$ Gershgorin ceiling (1.2245–1.4236 against ≤ 2.1213), direction-correct, refusing at the shrunken shift. *The M4 split and the certified chain.* The mass half of Maz'ya's Markov step is a pointwise theorem, $\sigma_{\rm tot}=0.2197$–$0.4425<1$ on every window; licence 4 holds with $|R|$ and *not* R^+ (witness $[[1,-\frac{1}{2}],[-\frac{1}{2},1]]$ built in); and S1′ is *certified per window on both routes*, better route $c_0=2.5154$–4.2265 against the classical Dirichlet value 4, with Charikar's cited density constant bracketed by exhaustive enumeration. **Named limits as content:** per instance on small windows, nothing uniform in the zone index or in D; the c_0 is *measured* at the minimiser's own layer cake and the a-priori level lemma (T145's L1) stays open — Maz'ya's strong-type half is not used as a theorem anywhere. Maz'ya 1985 / Muckenhoupt 1972 / Fukushima–Ōshima–Takeda 1994 / Miclo 1999 / Charikar 2000 / Goldberg 1984 / Gershgorin 1931 named classical; Weil 1952 cited, never used as a criterion. Machine-checked: [`verification/v546_capacity_chain_identities.py`]. [E] (per-instance identities and certified inequalities; no marker moves)

The level-lemma identities (`PRIME.LEVEL.LEMMA.01`). The a-priori core of T146 is load-bearing as the fifth companion module (20 checks, ∼0.4 s) on 12 frame-A windows ($n=4\ldots139$, $m=26\ldots285\leq300$), 9 random positive definite forms for the form-independent items, and a stress ladder $m=64\ldots288$ — the companion to the capacity-chain module, whose certified chain carried a constant *measured* at the minimiser's layer cake: here the same constant is a functional of the form alone. *The union identity for nested cakes.* $\sum_{j,l}c_jc_l|S_j\cup S_l|=2T\|\psi_t\|_1-\|\psi_t\|^2$ — form-independent and $O(m)$, the reason the cake base is movable; exact (0.0) against the $K\times K$ double sum and against union sizes computed with no nestedness used (controls 3.2×10^{-2} / 4.8×10^{-3}). *The counting lemma at a general base.* $G(b)\leq 2b^2\|v\|_\infty\|v\|_1/\|v\|^2+\varepsilon$ for *any* vector at all 6 preregistered bases; the classical dyadic 8 recovered at $b=2$ exactly, the base free (gain $4/b_*^2=3.7612$); the one-level-down cake breaks the domination on every draw, so the $+1$ is load-bearing. *The resolvent delocalisation bound.* $\psi=\lambda R\psi$ (9.0×10^{-14}, an identity — no Davis–Kahan step anywhere) $\Rightarrow\|\psi\|_\infty\leq\lambda_{\rm up}\max_j\|Re_j\|$ with Rayleigh–Ritz $\lambda_{\rm up}$ (1.0592–1.2883 of the gap) and residual-paid certified column norms, within 1.061–1.337 of the truth; the lever: the columns sit 2.85–10.17 *below* the trivial ceiling $1/\lambda_{\min}$, giving $\Gamma=1.8356$–2.2797 against the useless $\sqrt{m}=5.1$–16.9. *The composite level lemma.* $c_0^{\rm ap}=2b^2\Gamma\min(1,\Gamma_1)+\varepsilon=3.9042$–$4.8488$ (slack 1.986–2.418 over the measured constant, under the ceiling $16=4\times$ Maz'ya's Dirichlet 4), the certified window inequality cert_lam_max$(R)\leq c_0^{\rm ap}\Psi_{\rm abs}$ on 12/12, chain loss 0.0423–0.1272 of the exact gap; the built-in no-go discriminator ($R=aa^\mathsf{T}+\varepsilon I$, $a_i=i^{-1/2}$: Γ grows $3.676\to6.798$, ratio $1.85\geq1.5$, $c_0^{\rm ap}=11.10$ exceeds every window value) fails exactly where it must, the theorem half surviving and the Dirichlet control flat (1.0063). **Named limits as content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in D, and explicitly *no* statement for all D (the asymptotic delocalisation of the Green columns stays open, typed open); the energy-route $\sigma_{\rm tot}$ stays measured. Maz'ya 1985 / Rayleigh 1877–Ritz 1909 / Cauchy–Schwarz / Miclo 1999 / Davis–Kahan 1970 (cited, deliberately not used) / Charikar 2000 / Goldberg 1984 / Gershgorin 1931 named classical; Weil 1952 cited, never used as a criterion. Machine-checked: [`verification/v547_level_lemma_identities.py`]. [E] (per-instance identities and certified inequalities; no marker moves)

The Green/Szegő identities (`PRIME.GREEN.SZEGO.IDENT.01`). The identity/certificate core of T147+T148 is load-bearing as the sixth companion module (21 checks, ∼0.4 s) on 12 frame-A windows ($n=4\ldots139$, $m=26\ldots285\leq300$), 7 random positive definite forms for the form-independent items, and a weight-class model ladder $m=64\ldots288$ — the companion to the level-lemma module: here the two scalars its constant is made of are certified. *The factorisation identity.* $\Gamma=\sqrt{m}\,\lambda_{\rm up}\max_j\|Re_j\|=\sqrt{Q^\star}\cdot S_w$ with $S_w=\lambda_{\rm up}\|R\|_F$ purely spectral and $Q^\star=$

$m\max_j(R^2)_{jj}/\operatorname{tr}(R^2)$ purely geometric — exact at 2.2×10^{-16}; the certified factorisation dominates the true Γ everywhere ($S_w = 1.1186$–1.5777, $Q^\star = 2.0879$–2.8634). *The bottom-block constant.* $\sum_j(\Pi_B)_{jj} = |B|$ exactly (3.3×10^{-16}; B the preregistered $\theta = 10$ block, $|B| = 3$–5), so $Q_B \ge |B|$ is a theorem; measured $Q_B/|B| = 1.5091$–1.7980 inside the preregistered band $[1, 2.2]$ against the Szegő/Widom sharp reference $2|B|$ — the coordinate-projector control overshoots by ≥ 3.9. *The second-difference ℓ^1 ceiling.* $|a_k| \le \min(1, \|s_k\|_\infty\|L_0^p\psi\|_1/\mu_k^p)$ from the exact Dirichlet eigenpair, with the m-free closed form $\|a\|_1 \le 2\sqrt{\nu}+1$ ($\nu = (m+1)^{3/2}\|L_0\psi\|_1/(2\sqrt{2})$) and the chain $m\|\psi\|_\infty^2 \le 2\|a\|_1^2$; Dirichlet calibration $\|a\|_1 = 1$ exact, $\nu \to \pi$. *The layer-cake S_w certificate.* LDL^T inertia counts equal the direct spectral counts on all 96 (window, rung) pairs, and the layer cake certifies $S_w \le 1.3288$–1.8220 (true 1.1186–1.5777); dropping the bottom layer breaks it everywhere. *The Liouville transform.* $m\|\psi\|_\infty^2 \le 2\kappa_\Lambda\,\mathrm{ceil}(\hat\varphi)^2$ with $\varphi = \Lambda^{-1/2}\psi$ and a-priori $\kappa_\Lambda = \max\Lambda/\min\Lambda = 1.3868$–$2.5805$, stepwise on every bottom-block mode; the weight-class discriminator at matched κ_Λ (mismatch 0.0013): the BV multiplier flat (1.003), the rough one (TV larger by 162.6) grows by 19.9 — the roughness, not the conditioning. **Named limits as content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in D, and explicitly *no* statement for all D (the one open lifting input, $\nu_L \le F(\text{form})$ m-free / bounded $\mathrm{TV}(\log\Lambda)$, stays open, typed open); the symbol-level order-2 hypothesis is refuted (T148: $f(0) < 0$ on every surface window) and enters nowhere as a theorem; σ_{tot} stays retired as a route. KMS 1953 / Widom 1958–Basor–Ehrhardt 2009–Böttcher–Silbermann (the mechanism's address, never an authority) / Sylvester 1852–Bunch–Kaufman 1977 / Euler 1735 / Rayleigh 1877–Ritz 1909 / Davis–Kahan 1970 (cited, computed vacuous, not used) / Gershgorin 1931 named classical; Weil 1952 cited, never used as a criterion. Machine-checked: [`verification/v548_green_szego_identities.py`]. **[E]** (per-instance identities and certified inequalities; no marker moves)

The gauge/parity identities (`PRIME.GAUGE.PARITY.IDENT.01`). The identity/certificate core of T149+T150 is load-bearing as the seventh companion module (21 checks, ~0.3 s) on 12 frame-A windows ($n = 4\ldots139$, $m = 26\ldots285 \le 300$), 48 gauge assemblies (Jacobi, const, moving-geometric-mean, and the arithmetic-free archimedean diagonal), the full-section parity items on the 6 windows with $M \le 300$, and a parity control model at $m = 2001$ — the companion to the Green/Szegő module: here the gauge *freedom* around that machinery and the parity *structure* underneath it are certified. *The gauge freedom.* For any positive diagonal $\tilde\Lambda$, $\lambda_{\min}(A) \ge \operatorname{cert}\lambda_{\min}(\tilde E)\cdot\min_j\tilde\Lambda_j$ is a valid certified lower bound (an identity plus one Rayleigh step; family maximum valid); $\Gamma = \sqrt{Q^\star S_w}$ holds gauge-invariantly at 2.4×10^{-16}; the const gauge has $\mathrm{TV}(\log\tilde\Lambda) = 0$ *exactly* with the certified entrywise sandwich $\sigma = 1.3868$–2.5805; on the shifted indefinite form the completed-Cholesky floor refuses on 12/12. *The two-regime discriminator.* The flutter smoother removes TV by $\ge 2.88\times$ while the Liouville-transported $\nu_{\tilde L}$ moves ≤ 0.0088 — even the rough re-gauge moves it ≤ 0.0046 — while the untransported functional jumps 0.41–9.76 under the same rough gauge: the roughness is in the form, not in the multiplier (T149's withdrawal of the TV hypothesis, wired per instance). *The explicit zone diagonal.* $\Lambda_r = c_0 - c_{M-1-2r} + \sum_{s\ne r}\max(c_{|r-s|} - c_{M-1-r-s}, 0)$ at residual 5.9×10^{-16}, the exact split $c = c^{\mathrm{arch}} + c^{\mathrm{atom}}$ with $c^{\mathrm{atom}} \le 0$, the closed atom budget $\|c^{\mathrm{atom}}\|_1 \le 4B\sqrt{N}$ ($B = 1.038821$ verified on the table); the archimedean section alone is indefinite on 12/12 (the atoms are co-responsible for the positivity — the additive perturbation reading is structurally empty). *The parity compression.* $U_-^\mathsf{T} T_M(c)\,U_- = T - H$ exactly (cross block 1.4×10^{-16}): the reflection-odd section is the compression of the full symmetric Toeplitz section onto its antisymmetric parity sector, and the negative inertia of the sign-changing symbol sits entirely in the *even* sector (odd sector zero, LDL^T) — $f(0) < 0$ explained rather than worked around. *The parity-sine calibration.* The parity sines are exact eigenpairs of the parity Laplacian (corner 3 forced by antisymmetry), $\nu_k^P = \pi k^2$ on the exact model to 3.0×10^{-5}, and the per-window ladder $\nu_k^P \le Ck^2$ holds with certified $C = 11.70$–17.28 and the derived per-mode ceiling $m\|\psi_k\|_\infty^2 \le 2(2\sqrt{C}k+1)^2$. **Named limits as content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in D, and explicitly *no* statement for all D (the one open input, an m-free C in the odd-sector ladder for a sign-changing symbol, stays open, typed open); $\mathrm{TV}(\log\Lambda)$ stays withdrawn as a hypothesis; σ_{tot} stays retired as a route. KMS 1953 (parity sector) / Widom 1958–Basor–Ehrhardt 2009–Böttcher–Silbermann (the parity sectors as the algebra's natural objects; the sign-changing symbol is exactly what Basor–Ehrhardt does not cover) / Sylvester 1852–Bunch–Kaufman 1977 / Chebyshev 1852 (verified on the table) / Abel / Weyl 1912–Bauer–Fike 1960–Davis–Kahan 1970 (cited, computed dead, not used) / Rayleigh 1877–Ritz 1909 / Gershgorin 1931 named classical; Weil 1952 cited, never used as a criterion. Machine-checked: [`verification/v549_gauge_parity_identities.py`]. **[E]** (per-instance identities and certified inequalities; no marker moves)

The odd-sector identities (`PRIME.ODD.SECTOR.IDENT.01`). The identity/certificate core of T151 is load-bearing as the eighth companion module (19 checks, $\sim$0.2 s) on 12 frame-A windows ($n = 4\ldots139$, $m = 26\ldots285 \leq 300$), parity control models at $m = 241$ and $m = 2001$, and a no-go size ladder $m = 64\ldots256$ — the companion to the gauge/parity module: here the *reroute* off the quadratic ladder is certified — through the pencil against the parity Laplacian and one elementary Sobolev step, to a per-mode bound *linear* in k. *The grid step-over.* The parity sines live on the shifted grid $\theta_k = 2\pi k/N$ with $\mu_k^P = 2 - 2\cos\theta_k$ exactly; the symbol has $f(0) = -47.6\ldots-6.3 < 0$ with a nonempty negative window $[0,\theta_c)$, and $\theta_c/\theta_1 = 0.371$–$0.485$ on every window: the odd grid steps over the negative window — odd-sector positivity is a grid fact. The honest negative beside it, promoted as content: the symbol-only pencil floor is vacuous ($\rho \geq 1$ on 12/12) and the symbol moves by 0.45–3.27 across one mode spacing — the local model is a *matrix* statement, not a symbol statement. *The certified bottom ladder.* $\lambda_k(A) \leq S\mu_k^P$ for $k \leq 8$ by completed LDL^T counts (Sylvester 1852), $S = 1.1019$–6.4725 — order 2 in k, with $f(0) < 0$ absorbed by the Rayleigh floor $L_P \geq \mu_1^P I$. *The Sobolev step.* $\|\psi\|_\infty^2 \leq 2\|\psi\|_2\|\nabla\psi\|_2$, licensed by the odd sector's virtual node $\psi_{-1} = 0$; on the exact parity model the pencil is $(1,1)$ (certified 1.1×10^{-12}) and the per-mode price is exactly π (5.1×10^{-8}). *The closed archimedean minimum.* $\min_r \Lambda_r^{\mathrm{arch}} = c_0^{\mathrm{arch}} - c_{M-1}^{\mathrm{arch}}$ exactly (residual 0.0), attained at $r = 0$, from two per-window-checked shape properties of the smooth kernel. *The linear per-mode bound.* With the certified pencil floor $\kappa = 0.1939$–0.3599 and $K_{\mathrm{bot}} = S$, $b_k = 2m\sqrt{K_{\mathrm{bot}}\mu_k^P/\kappa} \leq C_S k$ with $C_S = 12.081$–$26.953 \leq 2\pi\sqrt{K_{\mathrm{bot}}/\kappa}$ — linear where the quadratic ladder grew, nothing additive anywhere; on the T145 no-go the bottom pencil ratio explodes ($\times15.8$ over $m = 64\ldots256$) while the parity model holds 1. **Named limits as content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in D, and explicitly *no* statement for all D (the one open input, the m-freeness of the bottom pencil ratio $R = K_{\mathrm{bot}}/\kappa = 3.7362$–$18.6607$, stays open, typed open); the symbol route stays certified vacuous. KMS 1953 / Widom 1958–Basor–Ehrhardt 2009–Böttcher–Silbermann (the symbol question's address — the computed answer: the pointwise symbol language does not extend to the bottom mode) / Weyl 1912 / Parlett / Hardy–Littlewood–Pólya 1934–Agmon 1965 / Sylvester 1852–Bunch–Kaufman 1977 / Rayleigh 1877–Ritz 1909 / Wilkinson 1968–Higham 2002 / Chebyshev 1852 named classical; Weil 1952 cited, never used as a criterion. Machine-checked: [`verification/v550_odd_sector_identities.py`]. **[E]** (per-instance identities and certified inequalities; no marker moves)

The fixed-size Ritz ceiling certificate (`PRIME.RITZ.CEIL.01`). The certificate core of T154 is load-bearing as the ninth companion module (16 checks, $\sim$0.4 s) on the 12 *deepest* prime-power zones admitting a frame-A window inside the cap ($n = 29\ldots139$, $m = 96\ldots291 \leq 300$ — declared, not tuned: on the shallowest zones $n \leq 11$ more than one bottom direction leaves the sine block and the fixed-depth certificate is not claimed there), a no-go size ladder $m = 48\ldots288$, parity controls at $m = 64, 128, 256$ and 3 Dirichlet control windows — the companion to the odd-sector module: here the *fixed-size* replacement for the ceiling step is certified. *The direction lemma.* $\lambda_k(A) \leq \theta_k$ for every $k \leq \dim$ — Ritz values are upper bounds for the eigenvalues of the *same index* (Courant–Fischer 1920 / Cauchy 1829), verified against exact spectra with zero one-sided excess — so the ceiling needs *no* residual argument; the reversed reading is refuted on 12/12 (overshoot ≥ 18.6). *The sixteen-column certificate.* $\mathrm{span}\{t_1..t_8\} + A^{-1}L_P\,\mathrm{span}\{t_1..t_8\}$ (dimension 16 fixed, independent of m; 8 completed Choleskys of matrices $\leq 8\times8$) carries $K^F = 1.1019$–1.5845 attaining the inertia-certified K_{bot} to $\leq 9.9\times10^{-7}$ against the declared cap 10^{-4} on 12/12 — the 8 size-m

$\mathrm{LDL}^{\mathsf{T}}$ counts retired from the ceiling step (controls: the sines alone overshoot 41.6–739; corrupted Green columns break the attainment on 12/12). *The one-direction anatomy.* Median of the seven aligned angles 0.03–0.14°, the largest 83.79–89.70° (measured): the single misaligned direction *is* the collapse price $\lambda_1(A)/(t\,\mu_1^P) = 4.408$–$7.042$, recovered in full per window by one Cholesky of $A-\gamma I$ (size m, certified, not fixed size). *The no-go discriminator.* On the T145 reconstruction $c(l) = 1/(1+l)$ the floor survives ($t = 0.05$) but K_{bot} explodes $\times 35.4$ and the ceiling explodes with it $\times 57.5$ — no manufactured closure; the Dirichlet control (the Hankel half removed) is certified non-positive on 3/3: the reflection half is load-bearing for positivity. **Named limits as content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in D, and explicitly *no* statement for all D; the attainment is not promoted to a theorem (the no-go overshoots — it is a property of the real section); the two open inputs after T154 (R1′, an m-free floor for $\lambda_{\min}(B_{HH})$ on the bulk parity block; R2′, an m-free lower bound for the L_P floor on the Ritz complement, arithmetic-free and worth 91–100% of the collapse price) stay open, typed open. Courant–Fischer 1920–Cauchy 1829 / KMS 1953 / Schur 1917–Haynsworth 1968 / Sylvester 1852–Bunch–Kaufman 1977 / Weyl 1912 / Temple 1928–Kato 1949 (a floor device, not used here) / Kaniel 1966–Paige 1971 (quoted, never a bound) / Björck–Golub 1973 / Wilkinson 1968–Higham 2002 / Chebyshev 1852 named classical; Weil 1952 cited, never used as a criterion. Machine-checked: [`verification/v551_ritz_ceiling_certificate.py`]. **[E]** (per-instance theorems and certified inequalities; no marker moves)

The four fixed-size angle instruments (`PRIME.ANGLE.INSTR.01`). The instrument cores of T155 and T157 are load-bearing as the tenth companion module (18 checks, ∼0.4 s) on the same declared surface as the Ritz module (the 12 deepest in-cap frame-A windows, $n = 29\ldots139$, $m = 96\ldots291 \le 300$; a no-go size ladder $m = 48\ldots288$; parity and Dirichlet controls at $m = 64, 128, 256$) — none of the four closes an angle, and none is promoted as anything but a per-instance instrument. *The complement-floor certificate (T155).* $\min_{v\perp W} v^{\mathsf{T}} L_P v \ge \mu_{13}^P - \lambda_{\max}(M^{1/2}(I - GG^{\mathsf{T}})M^{1/2})$ with $G = T_{12}Q_W$ — an identity plus one certified Rayleigh bound, a theorem for every m and every subspace; never above the exact size-m complement bottom, agreement $\le 1.01\times10^{-3}$ against the declared cap 2×10^{-3} on this small surface, both closed-form controls to 2.0×10^{-11} (including the *worthless* configuration $W = \mathrm{span}\{t_2..t_{13}\} \to \mu_1^P$: the instrument reports its own worthlessness); a corrupted overlap row moves it by ≥ 0.549. *The sine-block confinement (T157).* From the certified pencil floor and ladder ceiling alone, $\|\gamma_H\|^2 \le \lambda_1/(t\,\mu_{17}^P) \le (S/t)/\rho_{17} = 0.0153$–$0.0245 < 1$ on 12/12 — the bottom eigenvector lives 97.5–98.5% inside the first 16 parity sines by a per-instance theorem (measured tail 6.7×10^{-6}–3.0×10^{-5}); with the pencil ceiling κ instead of the ladder S the bound is vacuous (17.6–477), and the inflated floor is refuted at $e_1(A)$ by 3.3×10^3–1.0×10^5. *The Schur-entry identity (T157).* $s = \mu_1^P\, t_1^{\mathsf{T}} A^{-1} t_1 = (B^{-1})_{11}$ exactly (1.1×10^{-11}), $(B^{-1})_{LL} S_L = I$ to 5.8×10^{-11} with S_L the fixed-size 16×16 Schur complement the chain already forms: $1/s = 4.0460$–5.8962 is an $O(1)$ number carried by *one* entry, while the inversion-free Cauchy–Schwarz ceiling misses by 250–7.27×10^3 (the cancellation is nearly complete); a one-index shift breaks the identity by 1.55–2.15. *The adaptive Lipschitz ceiling (T157).* Interval bisection with the local Lipschitz constant certifies $\inf f^{\mathrm{arch}}/(4\sin^2(\theta/2)) \ge 0.25$ on 12/12 windows in 519–1201 evaluations (a uniform grid at the global constant would need up to 61× more); fed a target above the true grid minimum it *refuses* on 12/12. *The no-go discriminator.* On $c(l) = 1/(1+l)$ the floor survives ($t = 0.05$) while the confinement goes vacuous ($\times 52.0$) and $(S_L)_{11}$ explodes ($\times 60.0$) against the flat 4.05–5.90 real band; parity control exact to 4.9×10^{-14}, the Dirichlet residual hits $2/\sqrt{N}$ to 7.4×10^{-5}. **Named limits as content:** per instance on small windows — nothing uniform in the zone index or in D, and explicitly *no* statement for all D; nothing about the angles is closed — the two open terms after T157 (an m-free lower bound on $\hat{g}_1^2$, equivalently an m-free upper bound on the one Schur diagonal entry; an m-free R1″ domination with a non-shrinking margin) stay open, typed open. KMS 1953 / Schur 1917–Haynsworth 1968 / Courant–Fischer 1920–Cauchy 1829 / Cauchy–Schwarz (refuted as a ceiling, wired as a control) / Kantorovich 1948 / Temple 1928–Kato 1949 (floor devices, not used here) / Sylvester 1852–Bunch–Kaufman 1977 / Wilkinson 1968–Higham 2002 / Chebyshev 1852 / Szegő 1915–Widom 1958 (a named address, never a licence) named classical; Weil 1952 cited, never used as a criterion. Machine-checked: [`verification/v552_angle_instruments.py`]. **[E]** (per-instance identities and certified inequalities; no marker moves)

The exact-form identities (`PRIME.EXACT.FORM.IDENT.01`). The theorem cores of T158 and T159 are load-bearing as the eleventh companion module (25 checks, $\sim 0.2\,\mathrm{s}$), recomputed on the same declared surface as the two modules above (12 deepest in-cap frame-A windows, $m = 96 \ldots 291 \leq 300$; no-go ladders $m = 48 \ldots 288$ on *both* T145 forms; parity and Dirichlet controls at $m = 64, 128, 256$) — the algebra of the sixteen-form itself, and it closes neither open term. *The Thomson dual form and the positive Cholesky ladder (T158).* $s = (B^{-1})_{11} = \max_x (2x_1 - x^\intercal Bx)$ with the direction *executed* (8 random trials per window never exceed s); the ladder g_K from one Cholesky of the leading 16×16 block: terms strictly positive, partial sums monotone, $1/s \leq 1/g_{16} \leq 1/g_1 = B_{11}$ literally, tight to 1.0642–1.1522; a 5% corruption of one off-diagonal entry makes the Cholesky *refuse* on 12/12. *The two-dimensionality (T158).* $\mathrm{span}\{t_1, A^{-1}L_P t_1\}$ attains s exactly (9.0×10^{-13}) because $L_P t_1 = \mu_1^P t_1$; without the Green column the two-sine span misses by 1.42–4.8, the corrupted column by 255–7.7×10^3. *The gauge identity and the signature (T159).* The reflection-odd section annihilates constant lag vectors entrywise with *zero* tolerance, hence $\sum_d w_d = 0$ to 1.1×10^{-14} of $\sup |w|$: the form is blind to the lag mass of c — exactly where the archimedean and atom halves are h^2-sized and cancel; the Toeplitz-minus-Hankel signature holds to 1.4×10^{-16}; a one-index Hankel shift breaks the identity loudly, and the T145 form is not even a parity section (signature defect 7.4×10^{-2}). *The exact lag sum and the two closed scalars (T159).* $x^\intercal B_{LL} x = \sum_d c_d w_d$ to 2.7×10^{-16} of the absolute scale, with the honest price measured (relative to the cancelled total only 1.4×10^{-11} on this small surface); $w_0 = \sum_k x_k^2/\mu_k^P$ and $2w_0 - w_1 = \|x\|^2$ exact. *The Z-matrix law and the Collatz–Wielandt floor (T159).* B_{HH}^{arch} is a symmetric Z-matrix *raw* (no tolerance, 12/12); the closed sign-based floor $\min_k (B_{HH}^{\mathrm{arch}} \mathbf{1})_k = 0.9093$–$1.1284 \geq 0.25$ with the theorem direction verified against the exact $\lambda_{\min} = 0.9138$–1.1432; the atom block obeys *no* sign law (a coin toss) and the full-block row-sum floor is negative — admissible and vacuous; the checkerboard similarity (an isometry) destroys the raw law: an entrywise fact, not a spectral one. *The no-go discriminator.* The T145 family breaks on three axes of this module's structure, while on the $c(l) = 1/(1+l)$ reconstruction the ladder theorem survives and $1/g_{16}$ explodes ($1187 \to 7.11 \times 10^4$) — the instrument reports the collapse. **Named limits as content:** per instance on small windows — nothing uniform in the zone index or in D, and explicitly *no* statement for all D; the two open terms after T159 (the one explicit pairing inequality for R2″, needing correlation instead of sizes; a third device for the atom off-band block for R1″) stay open, typed open. Maz'ya 1985 / Miclo 1999 / Schur 1917–Haynsworth 1968 / KMS 1953 / Dirichlet 1829 / Perron 1907–Frobenius 1912 / Collatz 1942–Wielandt 1950 / Gantmacher–Krein 1950 named classical; Weil 1952 cited, never used as a criterion. Machine-checked: [`verification/v553_exact_form_identities.py`]. **[E]** (per-instance identities and certified inequalities; no marker moves)

The sampling/harmonics identities (`PRIME.SAMPLING.HARM.01`). The closed theorem cores of T160 and T161 are load-bearing as the twelfth companion module (21 checks, $\sim 0.5\,\mathrm{s}$), recomputed on the same declared surface as the module above (12 deepest in-cap frame-A windows, $m = 96 \ldots 291 \leq 300$) plus a declared Λ-battery of six log-spaced cut-offs $X = 10^3 \ldots 3.2 \times 10^5$ (finite von-Mangoldt sums only, no matrix anywhere) — what the sixteen-form *pairs against*, and it closes no open term. *The sampling identity (T160).* The atom half of the pairing equals $-\frac{1}{2} \sum_{n \leq X} (2\Lambda(n)/\sqrt{n})\, \widehat{w}(\log n)$ ($\widehat{w}$ the linear interpolant of the lag weights; exact to 5.6×10^{-15} of the absolute atom scale on 12/12): identically a finite combination of sums $\sum_n \Lambda(n) n^{-1/2} \cos(t \log n)$ at 32 explicit frequencies; a one-index grid shift breaks the identity loudly. *The moment laws and the TV theorem (T160).* $m_0 = 0$, $m_1 = -[S_0^2 + 2\sum_j P_j^2] \leq 0$ and $m_2 = -[2S_1 - (M-1)S_0]^2 \leq 0$ (a perfect square), both strictly negative on every window; and U3 as a theorem with *checked* hypotheses: $\mathrm{TV}(c^{\mathrm{arch}}) = 3.98$–$4.90 \leq |c_0| + 2\sup < 6$, closed and window-independent — the full atom-including kernel violates the monotone hypothesis on 12/12 (the wired mutation). *The harmonic-frequency theorem and the PNT main term (T161).* $t_j(2\alpha) = 2\pi j$ *exactly*: the 32 frequencies are the Fourier harmonics of the log-window, and the closed parameter-free prediction $(\sqrt{X} - 1)/(\frac{1}{4} + t^2)$ (de la Vallée Poussin 1896) matches the finite sums to 0.0007–0.097 of the trivial mass against the declared cap 0.12; a wrong Mellin pole and a mass-matched uniform fake measure both fail loudly — the agreement is a property of the von-Mangoldt measure. *The head split and the scale-free Bernstein rate (T161).* $c_d^{\text{arch}} = \Psi_d + D\widehat{G}(dD, D)$ with $\Psi_d = -\frac{1}{2}[(d+1)\log(d+1) - 2d\log d + (d-1)\log(d-1)]$ closed, D-free and m-free, exact to 3.2×10^{-16} at every lag; $\rho^* = (3+\sqrt{5})/2 = 2.618034$ is the root of x^2-3x+1, scale-free (interval ratio 5 only), with every window's $\rho(D) = 2.588$–2.608 below it. *The fraction bound and the sign refutation (T161).* $|\text{off-diagonal}| \le \frac{1}{4}|Q^{\text{arch}}|$ certified per window (0.171–0.212 on 12/12), with the *refuted* sign law wired as the negative control (arch half negative, off-diagonal block positive, 176/240 pairs violate the per-pair form) — nothing single-signed is promoted. **Named limits as content:** per instance on small windows — nothing uniform in the zone index or in D; finite Λ-sums allowed and used, *no* RH statement, and the δ triage of T161 (required depth against RH strength) is a documentation item, not a module claim; the open terms after T161 (R-A′ the prime-free log-moment; R-B′ the m-free 16×16 Gram fraction bound; R-C′ one split of the pairing with $\delta_{\text{eff}} < \frac{1}{2}$; R-D a fifth R1″ device) stay open, typed open. Dirichlet 1829 / Abel 1826 / Fejér 1915–Schur 1917 / Bernstein 1912 / Chebyshev 1852–Mertens 1874 / de la Vallée Poussin 1896 / Clausen–Lerch / KMS 1953 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion. Machine-checked: [`verification/v554_sampling_harmonics_identities.py`]. **[E]** (per-instance identities and certified inequalities; no marker moves)

The Pareto/total-variation identities (`PRIME.PARETO.TV.01`). The closed theorem cores of T162 and T163 are load-bearing as the thirteenth companion module (23 checks, $\sim$0.4 s), recomputed on the same declared surface as the module above (12 deepest in-cap frame-A windows, $m = 96\ldots291 \le 300$) with the one arithmetic input $\kappa = 0.038821$ *verified* at every jump point of $\psi(x)/x$ on the von-Mangoldt table — the *price structure* of the pairing, and it closes no open term beyond the negative closure that the floor inequality itself is. *The exchange law (T163).* $\delta_{\text{bnd}}(x) = \frac{1}{2} + \log(2\kappa g_{16}\mathrm{TV}(x)/P(x))/\log X$ is an identity at all 612 grid points of the declared Fejér knob (5.1×10^{-16}), with the crossing criterion $\delta_{\text{bnd}} < \frac{1}{2} \iff P > 2\kappa g_{16}\mathrm{TV}$ an exact equivalence at every point: price and demand are two coordinates of one object; the wrong constant breaks the law loudly. *The total-variation floor (T163).* $w_0 = \|a\|^2$ (4.9×10^{-16}), $\mathrm{TV} \ge |w_0|$ by telescoping, and $x_1 = 1$ forces $\mu_1^P\mathrm{TV} = 5.00$–$28.84 \ge 1$ on all 708 trial vectors built — the closed floor $1/\mu_1^P = 1/(4\sin^2(\pi/N)) \sim h^2$ — so every sub-$\frac{1}{2}$ point pays $P > 2\kappa g_{16}/\mu_1^P$: a flat-price sub-$\frac{1}{2}$ demand is impossible in the parity sector on this surface (closes R-C‴ negatively); the inadmissible vector $0.01x^*$ undercuts the floor and is recognised (the wired negative control). *The chain-derived price axis (T163).* $\sigma = 1$ is exactly the $K = 1$ rung of the T158 Cholesky ladder (4.1×10^{-16}) and $\sigma = \infty$ the $K = 16$ top rung (1.2×10^{-12}); the knob is monotone in both coordinates (its curve *is* the Pareto front); a corrupted sixteen-block makes the Cholesky refuse. *The Mellin cell moments and the Abel step (T162).* The k-th Mellin rung $\sum_d w_d C_d(-(2k+\frac{1}{2}))$, closed, agrees with an independent quadrature to 2.8×10^{-13} ($k = 1\ldots4$); the boundary-free Abel identities hold at levels 1–3 (8.3×10^{-17}), the optimal level is one for the closed reason $32\pi/\alpha = 40.6$–$59.4 > 1$, and a gauge violation breaks by its closed prediction (1.2×10^{-15}). *The Lerch/Frullani log-moment (T162).* $L = \sum_d \Psi_d w_d$ on three routes (Wronskian form 2.8×10^{-14}, integral with the $d = 1$ peel 1.8×10^{-15}, $2\log2$ to 2.2×10^{-16}, $\Psi_1 = -\log2$ exact); plus the monotone no-go staircase ($1.00 \to 3.49 \to 10.81 \to 24.98$, exact null control). **Named limits as content:** per instance on small windows — nothing uniform in the zone index or in D; the T163 h-exponents are sandbox fits, *not* consumed; finite Λ-sums allowed and used, *no* RH statement; the open terms after T163 (R-E the successor, both arms prime-free; R-B‴ the h-uniform positivity margin; R-D a fifth R1″ device) stay open, typed open. Fejér 1915 / Mellin 1896 / Abel 1826 / Chebyshev 1852–Rosser–Schoenfeld 1962 / Legendre / Dirichlet 1829 / KMS 1953 / Frullani 1828–Lerch 1887–Clausen 1832 / Schur 1917–Haynsworth 1968 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion. Machine-checked: [`verification/v555_pareto_tv_identities.py`]. **[E]** (per-instance identities and certified inequalities; no marker moves)

The gauge/P_{pr} identities (`PRIME.GAUGE.PPR.01`). The closed theorem cores of T164 and T165 are load-bearing as the fourteenth companion module (23 checks, $\sim$0.3 s), recomputed on the same declared surface as the module above (12 deepest in-cap frame-A windows, $m = 96\ldots291 \le 300$)

with the one arithmetic input $\kappa = 0.038821$ *verified* at every jump point of $\psi(x)/x$ on the von-Mangoldt table — the *gauge structure* of the pairing, and it does not close the one remaining quantifier. *The gauge theorem (T164).* $x_1 = 1$ fixes only the *scale* of a; Q and TV are homogeneous of degree two; hence the demand ratio Q/TV of the same a-direction is the *same number in every sector* — unchanged to 2.9×10^{-11} on 2520 sector-by-trial-vector combinations (5 weight families × 7 shifts × 6 taper widths), with the pure shift scaling the certified value by exactly $\theta = \mu_1^P/(\mu_1^P + \kappa_s)$; the full space is settled strictly worse ($\mu_0 = 0$ exactly); the wrong transport map breaks loudly (2.0×10^2–5.4×10^3). *The h^2 conservation law (T164).* floor × transfer $= 1/(\mu_1^P + \kappa_s) \cdot (1 + \kappa_s/\mu_1^P) = 1/\mu_1^P$ *identically* at every shift (2.2×10^{-16}): the h^2 is conserved, never removed — it lives in the ratio, not in the floor; the wrong transfer misses by 0.9995 at the largest shift. *The power-one spend (T164).* The r-ceiling $r \le U/L$ with $L = \lambda_1(A)/\mu_1^P$ is linear in the entry ceiling, so $d\log r/d\log U = +1$ exactly (1.1×10^{-14}); the quadratic consumption reads slope 2 and is recognised. *The P_{pr} identity and the predicate equivalence (T165).* $P_{\mathrm{pr}} = g_{16} \cdot R \cdot (\mathrm{TV}/(t_1 v)^2)/\mu_1^P$ exact to 4.3×10^{-16} on all 192 battery vectors, with the predicates $R > 2\kappa$ and $P_{\mathrm{pr}} > 2\kappa g_{16}\,\mathrm{tvf}/\mu_1^P$ agreeing as booleans everywhere and every crossing vector paying $P_{\mathrm{pr}} > 2\kappa g_{16}/\mu_1^P = 15.4$–$142.8$ above the preregistered tolerance 10 on 12/12 — the successor question R-F closes negatively by an identity plus a floor; the price with $(t_1 v)$ unsquared breaks the gauge invariance loudly. *The Schur-cascade direction and the exponent closure (T164/T165).* $s = \mu_1^P t_1^\top A^{-1} t_1 \ge g_{16} \ge g_1$ window by window with $P_K = \hat{a}\, s \ge 1$ (Kantorovich 1948), and the exponent closure $\log(1/g_1) = \log(g_{16}/g_1) + \log(1/g_{16})$ is exact per instance and on the fitted lists (1.3×10^{-15}); plus the monotone no-go staircase and exact Dirichlet/parity controls. **Named limits as content:** per instance on small windows — nothing uniform in the zone index or in D; the T164/T165 h-exponents and the measured anatomy are sandbox, *not* consumed; finite Λ-sums allowed and used, *no* RH statement; **the one open object after T165 stays open, typed open**: $\inf_m g_{16}(m) > 0$, equivalently a lower bound on the sixteen-step Schur-cascade gain g_{16}/g_1 uniform in m — a quantifier over a certified flat list. Schur 1917–Haynsworth 1968 / KMS 1953 / Abel 1826 / Chebyshev 1852–Rosser–Schoenfeld 1962 / Kantorovich 1948 / Fejér 1915 / Dirichlet 1829 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion. Machine-checked: [`verification/v556_gauge_ppr_identities.py`]. **[E]** (per-instance identities and certified inequalities; no marker moves)

The cascade/vector identities (`PRIME.CASCADE.VECT.01`). The closed theorem cores of T166 and T167 are load-bearing as the fifteenth companion module (23 checks, ∼0.1 s), recomputed on the same declared surface as the module above (12 deepest in-cap frame-A windows, $m = 96 \ldots 291 \le 300$) with the one arithmetic input $\kappa = 0.038821$ *verified* at every jump point of $\psi(x)/x$ on the von-Mangoldt table — the *anatomy of the cascade* and the *exactness of the closed vector at* $K = 2$, and it does not close the one remaining scalar. *The cascade identities (T166).* The prefix property (one Cholesky of the sixteen-block yields all sixteen rungs; 192 comparisons, 1.1×10^{-12}), the minor form $1/g_K = \det G_{[1..K]}/(\mu_1^P \det G_{[2..K]})$ on the raw arithmetic Gram block (2.7×10^{-12}; Schur 1917–Haynsworth 1968), and the regression form $g_K/g_1 = 1/(1 - R_K^2)$ (8.8×10^{-13}) — the open object *is* a near-collinearity statement; the wrong minor misses by 0.99–1.45. *The diagonal invariance (T166).* The gain $g_K/g_1 = B_{11}(Q_K^{-1})_{11}$ is unchanged under $B \to DBD$ on 144 random positive diagonals (2.7×10^{-12}): the cascade does not see the KMS normalisation — the gain belongs to the arithmetic Gram block alone; a non-diagonal congruence moves it by 2.7–3.8. *The exact $K = 2$ identity (T167).* The closed vector $u = (1, -Q_{21}/Q_{22})$ *attains* $1/g_2 = Q_{11}(1 - r_{12}^2)$ exactly (1.1×10^{-12}) and $B_{11} g_2 = 1/(1 - r_{12}^2)$ identically (1.2×10^{-12}): at $K = 2$ the fast-null-vector is free — the third dress collapses onto the second, one scalar built from three closed lag sums; the wrong-sign vector pays $4r_{12}^2/(1 - r_{12}^2) = 6.7 \times 10^2$–$1.5 \times 10^4$. *The pivot identity and the unification (T167).* $u^\top Q_K u = 1/g_K + \delta^\top Q_K \delta$ for $\delta_1 = 0$ (2.0×10^{-12} on 252 evaluations) — every candidate excess is an exact number; the sign-aligned entry attainment $|x^{*\top} E x^*| = \varepsilon S_K$ (3.8×10^{-16}) makes the entry threshold exact, and the unification $(1/g_2)/S_2 = (1 - r_{12}^2)/(1 + 3r_{12}^2)$ (1.2×10^{-12}) makes the three dresses *one* inequality; the $\delta_1 = 0.1$ mutation breaks by exactly the closed cross term $2\delta_1$. *The trial theorem and the two must-breaks (T166/T167).* All 288 random trials with $u_1 = 1$ bound g_K from below (min ratio 15.14; Schur 1917); the L_P no-go gives cascade gain exactly 1 (deviation 0.0); the scramble type-change makes B_{11} itself lose positivity on a majority of draws on 12/12 windows — at $K = 2$ the collapse under scrambling is a change of type, not a factor; plus exact Dirichlet/parity controls. **Named limits as content:** per instance on small windows — nothing uniform in the zone index or in D; the T166/T167 h-exponents and the measured anatomy are sandbox, *not* consumed; finite Λ-sums allowed and used, *no* RH statement; **the one open object after T167 stays open, typed open**: R1, the single scalar — an m-free upper bound on $1 - r_{12}^2 = 1 - Q_{12}^2/(Q_{11}Q_{22}) \le C\,h^{-3+\varepsilon}$, three closed lag sums and nothing else. Schur 1917–Haynsworth 1968 / KMS 1953 / Rayleigh 1877–Courant–Fischer 1920 / Cauchy 1829 / Kato 1949 (cited as the closed-off route, never used) / Chebyshev 1852–Rosser–Schoenfeld 1962 / Dirichlet 1829 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion. Machine-checked: [`verification/v557_cascade_vector_identities.py`]. [**E**] (per-instance identities and certified inequalities; no marker moves)

The bilinear/rank identities (`PRIME.BILINEAR.RANK.01`). The closed theorem cores of T168, T169 and T170 are load-bearing as the sixteenth companion module (27 checks, $\sim$0.2 s), recomputed on the same declared surface as the module above (12 deepest in-cap frame-A windows, $m = 96 \ldots 291 \le 300$) with the one arithmetic input $\kappa = 0.038821$ *verified* at every jump point of $\psi(x)/x$ on the von-Mangoldt table — the *exact shape of the one remaining scalar*, and it does not close that scalar. *The Lagrange/Wronskian anatomy (T168).* The lag-sum identity $\hat{a}_{ij} = t_i^{\mathsf{T}} A_h t_j = \sum_d c_d W_d^{ij}$ with the closed correlation weights (1.6×10^{-15}; the $1/\sqrt{\mu}$ factors cancel, T169-TH3); Euclidean orthogonality $t_1 \cdot t_2 = 0$ (4.4×10^{-16}) — the near-parallelism is created *entirely* by the arithmetic metric; the Wronskian telescope $(\mu_1 - \mu_2) u_r v_r = W_{r-1} - W_r$ with $W_{-1} = W_{h-1} = 0$ (1.9×10^{-17}) and $\frac{1}{2}\sum M^2 = 1/(\mu_1\mu_2)$ exactly (4.6×10^{-16}; Lagrange 1773); the wrong eigenvalue pair misses by 1.66–1.67 relative. *The exponent account and the Gershgorin certificate (T168/T169).* $1 - r_{12}^2 = \nu_1\nu_2/(\hat{a}_{11}\hat{a}_{22})$ identically (2.5×10^{-13}); $\nu_1 \le \max(\hat{a}_{11}, \hat{a}_{22}) + |\hat{a}_{12}|$ (Gershgorin 1931, unconditional) on 12/12 with loss 1.1885–1.1947; dropping $|\hat{a}_{12}|$ fails on every window. *The det-collapse theorem (T169-TH7).* $R(K7) = 2\sqrt{\hat{a}_{11}\hat{a}_{22}}\,\det \hat{A}/[(\hat{a}_{11} + \hat{a}_{22})(\sqrt{\hat{a}_{11}\hat{a}_{22}} + \hat{a}_{12})]$ in closed form (8.7×10^{-13}) — the loop T167 scalar $\to$ T168 factor $\to$ T169 candidate closes as an identity; usable form $\nu_2 \le 2\det \hat{A}/(\hat{a}_{11} + \hat{a}_{22})$, tight to < 2 exactly; the closed constant $1/\sqrt{2}$ overshoots by 17–2.3×10^2. *The exact bilinear von Mangoldt form (T170).* $\hat{A} = B - S$ against the direct block (1.7×10^{-14}); $\det \hat{A} = \det B - D(B, S) + \det S$ (1.0×10^{-10}); $\det S = \sum_{n,m} \Lambda(n)\Lambda(m)K(n,m)/\sqrt{nm}$ with $K = \frac{1}{2}D(X_n, X_m)$ through the *explicit* kernel matrix (7.5×10^{-16}; Cauchy–Binet); the wedge reading to the measured rank-one defect 4.6×10^{-3}–3.1×10^{-2}. *The rank-3 theorem and the Type II collapse (T170).* The polarisation matrix has eigenvalues $\{-2, -1, +1\}$ exactly — rank 3, signature $(1, 2)$ — so $\sigma_4/\sigma_1 \le 1.6 \times 10^{-16}$ on every window: the bilinear form *is* the rank-3 polynomial $S_{11}S_{22} - S_{12}^2$ in three linear Λ-sums; Vaughan's identity is coefficient-exact for every $U = V \in \{3, 4, 5, 6\}$ and the Type II kernel has 99.9%-energy rank $[2, 2, 2, 2]$ against generic $[20, 15, 11, 9]$. *The R4-free chain and the discriminators (T170/T169).* $1 - r_{12}^2 = \det \hat{A}/(\hat{a}_{11}\hat{a}_{22})$ is an identity — no A-positivity, the Weil fence never approached; the L_P no-go gives $\hat{a}_{12} = 1.1 \times 10^{-17}$ identically (no decay); the scramble leaves the rank-3 collapse *untouched* on all 60 draws while the value moves $\times 834.8$ median — the arithmetic sits in the joint values of (S_{11}, S_{22}, S_{12}) against B; plus exact Dirichlet/parity controls. **Named limits as content:** per instance on small windows — nothing uniform in the zone index or in D; the T168–T170 h-exponents and route-table deltas are sandbox, *not* consumed; finite Λ-sums allowed and used, *no* RH statement — the one open object after T170 (R1, now finally *classified* but not closed: an m-free unconditional certificate that the two rows of $\hat{A}$ become collinear at rate $h^{-3+\varepsilon}$ — a joint near-degeneracy of three explicit linear Λ-sums, not a size) stays open, typed open. Lagrange 1773 / Cauchy–Binet 1812–1815 / Gershgorin 1931 / Rayleigh 1877 / Montgomery–Vaughan 1973–Vaughan 1977 (the priced route) / KMS 1953 / Chebyshev 1852–Rosser–Schoenfeld 1962 / Dirichlet 1829 named classical. Machine-checked: [`verification/v558_bilinear_rank_identities.py`]. [**E**] (per-instance identities and certified inequalities; no marker moves)

The phase-2 capstone (`PRIME.PHASE2.CAPSTONE.01`). The capstone *verifies the map and claims nothing new*: T126–T170 reduced the I5 uniformity question through sixteen exact reformulations to one object (R1), T171 (contract `FINAL.MAP`, verdict MAP-COMPLETE) reproduced every link of that reduction in one connected sandbox run, and the seventeenth companion module (35 checks, ~ 0.3 s) is the load-bearing version of that run on the same declared surface as the module above (12 deepest in-cap frame-A windows, $m = 96 \ldots 291 \leq 300$; $\kappa = 0.038821$ *verified* at every jump point of $\psi(x)/x$; the 16×16 low block positive definite on 12/12 — a numerical fact, never routed through the Weil criterion). *The chain, sixteen links in one pass* (13 THEOREM / 3 CERT): the T152 Schur two-block criterion sharp on both sides (holds at $0.5\lambda_{\min}$, fails at $1.02\lambda_{\min}$; KMS 1.3×10^{-15}, Haynsworth 3.8×10^{-15}); the T151 Sobolev bound $b_k \leq C_S k$ (worst ratio 0.4816); the T153 Ψ-collapse $\Psi = \sum_d c_d w_d = x^{\mathsf{T}} G x$ (6.3×10^{-16}; $\sum_d w_d = 0$ to 9.0×10^{-17}); the T154 Ritz ceiling from *below* ($g_{32}/s = 0.9520$–0.9849); the sharp T155 12×12 floor ($t = 0.2295$–0.3719); the T156 $F(P, r)$ chain ($L(16) = 6.6$–15.6); the T158 Thomson/ladder duality (3.1×10^{-12}; 36/36 trials below); the T160 sampling identity (1.1×10^{-15} / 9.9×10^{-15}); the T163 Abel swap with the TV floor (overshoot $\times 5.2 \times 10^{4}$–1.1×10^{6}); the T164 gauge/quantifier collapse ($\Psi(x^*) = 1/g_{16}$ to 7.0×10^{-12}; 48/48 perturbed vectors larger); the T165 P_{pr} identity ($= 1$ at x^*); the T166 cascade $1/(1 - R_{16}^2)$ (1.8×10^{-12}, diagonal-invariant); the T167 $K = 2$ exactness (milder $\times 1.62$–5.20); the T169 det collapse, R4-*free* ($\leq 4.2 \times 10^{-12}$, no positivity of A_h consumed — the Weil fence never approached); the T170 rank-3 theorem (J-eigenvalues $\{-2, -1, +1\}$ exact; $\sigma_4/\sigma_1 \leq 1.6 \times 10^{-16}$); and the T170 R1 shape via the 2×2 Lagrange identity ($\sin\angle = 1.169 \times 10^{-4}$–$2.467 \times 10^{-3} \leq 10^{-2}$ per window — *rate and quantifier typed OPEN*). *The no-go battery*, all eight failing as classified: size budget $\times 2.7 \times 10^{3}$–1.6×10^{4}; symbol infimum < 0 on 12/12; sector change an invariance ($\leq 4.2 \times 10^{-12}$); perturbation $\times 9.8 \times 10^{2}$–2.0×10^{4} off scale (Kato 1949); atom band dominates $\times 1.71$–2.83; sieve operator-norm $\times 8.8 \times 10^{3}$–7.5×10^{4} over (Montgomery–Vaughan 1973 / Vaughan 1977 the priced route); scramble $\times 679$ median with rank-3 *untouched* on 60 draws; L_P gain $= 1$ exactly. *The precision ledger*: needed 2.752×10^{-4} at the deepest window — finer than the RH yardstick $h^{-0.5}$ by $215\times$ and the best unconditional $h^{-0.996}$ by $13\times$ on this surface; both self-similarity identities $\leq 6.8 \times 10^{-13}$; mutations fail loudly. **Named limits as content:** per instance on small windows — nothing uniform in the zone index or in D; the T171 trend fits (pooled $h^{-2.33}$, per-frame $h^{-1.86}/h^{-2.87}$, $\delta = +1.044$ against 3.0, the full-surface numbers 2.284×10^{-7} / $1.2 \times 10^{5}\times$ / $3.1 \times 10^{3}\times$) are sandbox, *not* consumed; *zero* new uniform-in-m statements — which is exactly why the capstone is promotable; the one open phase-2 object (R1, an m-free unconditional certificate that the two rows of $\hat{A}$ become collinear at rate $h^{-3+\varepsilon}$) stays open, typed open; *no* RH statement. Schur 1917–Haynsworth 1968 / KMS 1953 / Cauchy 1829–Courant–Fischer 1920 / Rayleigh 1877 / Abel 1826 / Dirichlet 1829 / Lagrange 1773 / Cauchy–Binet 1812–1815 / Kato 1949 (the closed-off route) / Montgomery–Vaughan 1973–Vaughan 1977 / Chebyshev 1852–Rosser–Schoenfeld 1962 / Fejér 1915 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion. Machine-checked: [`verification/v559_phase2_capstone.py`]. [**E**] (per-instance identities and certified inequalities; no marker moves)

3.4 Glue-norm arithmetic, and three readings of 41, 60, 120

The two glue norms generate the carrier integers through sum, difference and product. With $\dim S^+ = 16$:

$$\boxed{\begin{aligned}&16(q_D + q_A) = 32 = 2^{g_{\mathrm{car}}} = h(D_5)h(A_3), \qquad 16(q_D - q_A) = 8 = r(E_8) = h(D_5),\\ &16\, q_D q_A = 15 = \dim S^+ - 1\end{aligned}} \quad [\mathbf{E}], \tag{12}$$

so the glue already contains the full carrier exhaustion $2^{g_{\mathrm{car}}}$, the E_8 rank, and the non-trivial transition count $\mathcal{A}_H + \mathcal{A}_\Lambda = 15$; and $|\mu_4| = (q_D - 1)^{-1} = (1 - q_A)^{-1} = 4$ is the inverse displacement of the two norms from the unit norm.

Spine Quotient Lemma: the recurring factors are adjacent quotients of $(2, 3, 4, 5)$ [E] ([verification/v74_compiler_micro_lemmas.py])

The integers of the spine are $(|\mathbb{Z}_2|, N_{\rm fam}, |\mu_4|, g_{\rm car}) = (2, 3, 4, 5)$. Every "mysterious" $O(1)$ factor in TFPT is an *adjacent quotient* of this chain, not a separate special number:

$$\frac{|\mathbb{Z}_2|}{N_{\rm fam}} = \frac{2}{3}, \quad \frac{N_{\rm fam}}{|\mu_4|} = \frac{3}{4} = q(A_3), \quad \frac{|\mu_4|}{g_{\rm car}} = \frac{4}{5}, \quad \frac{g_{\rm car}}{|\mu_4|} = \frac{5}{4} = q(D_5), \quad \frac{|\mu_4|}{N_{\rm fam}} = \frac{4}{3}.$$

These are exactly the factors that recur across the theory: $\lambda_2 = (\frac{2}{3})^6$ (the transport gap = Page recovery rate, and the Koide target $Q_\star = \frac{2}{3}$); $\varepsilon = \frac{3}{4}\varphi_0 = q(A_3)\varphi_0$ (the solar misalignment); $\varphi_0 = \frac{4}{3}c_3 + 48c_3^4$ with leading $\frac{4}{3}c_3 = \frac{1}{6\pi}$; and $q(D_5)+q(A_3) = \frac{5}{4}+\frac{3}{4} = 2$ (the E_8 root norm). So $(\frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{4}, \frac{4}{3})$ are five projections of one discrete chain — not scattered constants. The spine is moreover *one* finite object: as a tetrahedron on $T = \{e_3(a), p_0(a), e_1(a), e_2(a)\}$ its edge and face products and its volume $120 = |R^+(E_8)| = 5!$ reproduce the central integer grammar, with the honest negative control that the graph reading $240 = |\mu_4| \cdot |E(K_4)| \cdot |E(K_5)|$ breaks at K_6 ([verification/v91_spine_tetrahedron.py]).

The carrier adjoint $\dim D_5 = |R(D_5)| + g_{\rm car} = 45$ then gives a second reading of the $U(1)$ integer,

$$\boxed{10b_1 = \dim D_5 - |\mu_4| = 45 - 4 = 41 = \underbrace{40}_{\sum L} + \underbrace{(g_{\rm car} - |\mu_4|)}_{=N_\Phi=1}} \qquad \text{[E]}, \tag{13}$$

i.e. the single Higgs doublet is the carrier rank minus the glue corners, $5 - 4 = 1$.

A fourth, metric reading of 41 *([verification/v222_cm_norm_duality.py]).* On the square seam (j=1728, complex multiplication by the Gaussian integers $\mathbb{Z}[i]$) the $U(1)$ index is the Gaussian *norm* of the carrier-glue vector,

$$10b_1 = N_{\mathbb{Z}[i]}(g_{\rm car}{+}i|\mu_4|) = g_{\rm car}^2 + |\mu_4|^2 = 5^2 + 4^2 = 41,$$

and the hexagonal partner (j=0, Eisenstein $\mathbb{Z}[\omega]$) reads the same $(3, 2)$ carrier split as the scalaron exponent $N_{\mathbb{Z}[\omega]}(3{+}2\omega) = 3^2{-}3{\cdot}2{+}2^2 = 7$. So 41 (EM) and 7 (scalaron) are the two exceptional CM norms of the one carrier, square vs. hexagonal — not two unrelated $O(1)$ numbers [E]. The start defect 60 and the hypercharge moment 120 are "bilingual":

$$\begin{aligned} D_{\rm start} = \dim D_5 + \dim A_3 = 45 + 15 = 60 = \mathcal{A}_\Lambda \cdot |R^+(A_3)| = 10 \cdot 6, \\ \mathrm{Tr}_{S^+} X^2 = 120 = \underbrace{60}_{\text{adjoint}} + \underbrace{60}_{\text{mixed (10,6)}}, \end{aligned} \tag{14}$$

and $\dim E_8 = 248 = \mathrm{Tr}_{S^+} X^2 + 2 \dim S^+ |\mu_4| = 120 + 2 \cdot 16 \cdot 4$. Finally the hypercharge norm and the EW exponent are A_3-rational,

$$\boxed{\gamma_{\rm car} = \frac{\mathcal{A}_\Lambda}{|R(A_3)|} = \frac{10}{12} = \frac{5}{6}, \qquad I_1^{\rm EW} = \frac{|E(K_5)|}{|R(A_3)|} + \frac{1}{|\mu_4||R(A_3)|} = \frac{10}{12} + \frac{1}{48} = \frac{41}{48}} \qquad \text{[E]}. \tag{15}$$

Transport reduction. The flavor matrix L has its mod-6 content fixed by a single combinatorial datum: $\{0, 1, 3\}$ is the *unique* (up to D_6) 3-subset of $C_6 = |R^+(A_3)|$ with all distinct cyclic distances $\{1, 2, 3\}$, and the three sectors are its D_6 orbit ($u{\equiv}\{0, 1, 3\}$, $d{\equiv}2{-}\{0, 1, 3\}$, $e{\equiv}2{+}\{0, 1, 3\}$). The total is fixed by the carrier, $\sum L = 40 = |R(D_5)|$, with winding unit $6 = |R^+(A_3)|$. Given the residue *set* of a sector, the full ordered packet is reproduced by two universal rules — the lightest generation carries the single $+6$ winding, and the entries are sorted in decreasing length (monotone hierarchy). The *only* remaining datum is then, per sector, which residue is the first-generation anchor that receives the winding:

$$\boxed{\text{anchors } (a_u, a_d, a_e) = (1, 1, 2) \;\Rightarrow\; L_u = (7, 3, 0),\; L_d = (7, 5, 2),\; L_e = (8, 5, 3)} \qquad \text{[E] (given anchors)}. \tag{16}$$

So the open step of a full "Clebsch walk" theorem is sharply isolated: it is exactly these three per-sector anchors. They are *not* free: the three transport cusps are exactly the D_5 dual-root combinations, and each sector is tied to its cusp by the right-handed hypercharge,

$$\boxed{\{1, \tfrac{2}{3}, \tfrac{1}{3}\} = \{2y_+,\ -2y_-,\ 2(y_+ + y_-)\}, \qquad \text{lepton} \leftrightarrow 2y_+,\ \text{up} \leftrightarrow -2y_-,\ \text{down} \leftrightarrow 2(y_+ + y_-)} \quad \textbf{[E]}. \tag{17}$$

The canonical-transport-kernel theorem (document 3, Part II) derives the residue packets — and hence the anchors — as the *word-lengths of the family-connection holonomy* $\mathrm{Hol}_{\mathcal{G}_{D_4}}(\nabla_F^\star)$ along the D_4-invariant geodesic spine of $\mathbb{P}^1 \setminus \mu_4$ with these cusps as boundary data. The anchors are therefore a genuine $D_5 \times A_3$ readout (cusps from the D_5 carrier, spine holonomy from the A_3 four-punctured sphere), not an extra parameter. A compact candidate closed form is the nearest-integer cusp map

$$\boxed{a_f = \mathrm{round}(2\,\mathrm{cusp}_f): \quad (a_u, a_d, a_e) = \mathrm{round}(\tfrac{4}{3}, \tfrac{2}{3}, 2) = (1, 1, 2)} \qquad \textbf{[C]}, \tag{18}$$

which reproduces the matrix exactly. The remaining open step is now precise and purely geometric: an *independent* closed-form evaluation of the D_4-spine holonomy word-lengths from the lattice data alone (the same $\mathbb{P}^1 \setminus \mu_4$ whose residue pairing is Cartan(A_3) and whose corner rotation is the A_3 Coxeter element), rather than from the family-connection theorem (document 3).

Riemann–Hilbert check, and what it does (and does not) fix. Setting up the rigid $\mathbb{Z}_4$-symmetric rank-3 $SU(3)$ local system explicitly — four puncture monodromies $M_k = R^k C R^{-k}$ with the order-3 cusp class $C \sim \mathrm{diag}(1, \omega, \omega^2)$ ($\omega = e^{2\pi i/3}$, the three cusps as eigenvalue-exponents) and the rotation R ($R^4 = \mathbb{1}$) — the closure $M_0 M_1 M_2 M_3 = \mathbb{1}$ has a solution to machine precision, and the unique R has eigenvalues $\{i, -1, -i\}$: the A_3 Coxeter element. This *independently* confirms the rigidity of the family connection and the corner-rotation = Coxeter identity. However, the flat monodromy *alone* does not fix the integer word-lengths $\{0, 1, 3\}$: those are *parabolic degrees* (the cusp exponents as parabolic weights, together with the $\mathbb{Z}_2$ sheet parity), i.e. Hodge/metric data on the spine, not topological monodromy. *Status update (the computation has since been carried out):* the exact residue normal form $A_0^\star$ with spec $= \{0, \tfrac{1}{3}, \tfrac{2}{3}\}$ is pinned algebraically ([`verification/v115_anchor_residue.py`], [`verification/v116_resonance_uniqueness.py`]), the unitarised holonomy is the exact 24-element $W(A_3) = S_4$ ([`verification/v117_monodromy_weyl_a3.py`]), and the parabolic weights are the μ_4-character degrees of $H^1(\mathbb{P}^1 \setminus \mu_4)$ — one spectrum in four readouts ([`verification/v137_qplus_cohomology.py`]); the deck itself is *derived* — the integer model selects the sheet-twisted class, no discrete freedom left ([`verification/v141_deck_selection.py`]), and the full Möbius D_4 of the curve matches the integer model parity by parity ([`verification/v146_moebius_d4.py`]). What remains of the old residual is the much smaller selector establishment R4′ (research contracts): derive the *one* lift map of the torsion normal — the pairing values are atom identities ([`verification/v142_frame_integrality.py`]/[`verification/v145_pairing_atoms.py`]). **[E]**/**[C]**

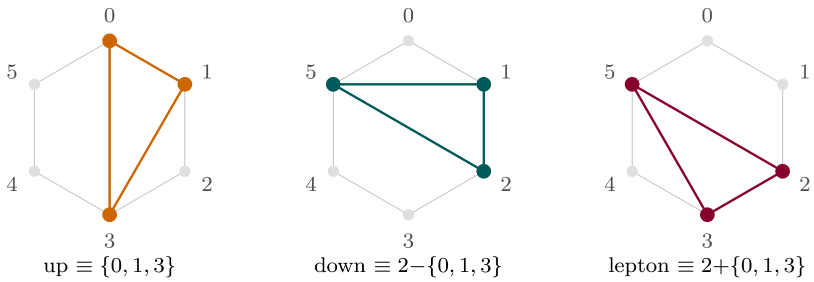


Figure 5: Flavor transport on the hexagon $C_6 = Z_6 = |R^+(A_3)|$. The unique distinct-distance triple $\{0, 1, 3\}$ (cyclic distances $\{1, 2, 3\}$) and its D_6 orbit give the three sectors' residues mod 6; the total $\sum L = 40 = |R(D_5)|$ and the first-generation $+6$ winding come from the carrier.

3.5 What the compiler outputs for the Standard Model and our reality

Every entry below is an *output* of $(c_3, g_{\rm car})$ through the $D_5 \times A_3$ compiler; none is an independent input. "Reads" names the lattice object the observable comes from.

SM / cosmology observable	Compiler output	Reads	St.
Number of generations	$N_{\rm fam} = 3 = \operatorname{rank} A_3 = \dim H^1(\mathbb{P}^1 \setminus \mu_4)$	A_3	[E]
Gauge algebra dimension	$\dim \mathfrak{g}_{\rm SM} = 12 = \lvert R(A_3)\rvert = \gcd(6r, 2h)$	A_3	[E]
One generation (16, incl. ν^c)	$S^+ = D_5$ half-spinor $= \Lambda^{0,2,4} E$	D_5	[E]
Hypercharge spectrum	all Y from $y_\pm = \{\frac{1}{2}, -\frac{1}{3}\}$; $\operatorname{Tr} X^2 = 120$	D_5 root data	[E]
Abelian index b_1	$10 b_1 = 1 + \lvert\mu_4\rvert\lvert E(K_5)\rvert = 1 + 4 \cdot 10 = 41$	$\mu_4 \times K_5$	[E]
Fermion family occupancy	$\Omega_{\rm adm} = N_{\rm fam} \dim S^+ = (N_{\rm fam} + N_\Phi)\lvert R(A_3)\rvert = 48$	A_3	[E]
Fine structure $\alpha_\star^{-1} = 137.0359992168$	root of $F_{U(1)}(\alpha) = 0$ (explicit, unique)	EM closure	[E]
Cabibbo angle	$\lambda_C = \sqrt{u(1-u)}$, $u = \frac{4}{3} c_3 + 48 c_3^4$	seed	[E]
CKM CP phase	$\delta_{\rm CKM} = \frac{2\pi}{\lvert R^+(A_3)\rvert} + N_{\rm fam} u(1-u) = \frac{\pi}{3} + 3\lambda_C^2$	A_3^+, seed	[E]/[C]
PMNS angles/phase	$\sin^2\theta_{12} = \frac{1}{3} - \frac{u}{2}$, $\sin^2\theta_{13} = e^{-5/6} u$, $\delta^\nu = \frac{4\pi}{3}$	A_3^+, seed	[E]/[C]
Charged-lepton hierarchy	$\det M_\ell \propto u^{h/N_{\rm fam}} = u^{10}$; $(5, 3, 2)$	decuple	[E]/[C]
Mass-hierarchy transport	$\sum L = 40 = \lvert R(D_5)\rvert$; entries on $C_6 = \lvert R^+(A_3)\rvert$	$D_5 \times A_3$	[E]
Strong CP	$\theta_{\rm eff} = 0$ (determinant erasure, given RP; document 3, Part III)	—	[E]/[C]
Weak CP	holonomy: $\delta_{\rm CKM} - \frac{\pi}{3} = N_{\rm fam}\, u(1-u)$	A_3^+	[C]
Gravity tree normaliser	$\xi_{\rm tree} = \frac{\dim \mathfrak{g}_{\rm SM}}{\dim S^+} = \frac{3}{4} = q(A_3)$	A_3 glue	[E]
Cosmological constant	$\rho_\Lambda / \bar{M}_{\rm Pl}^4 = \frac{3}{4\pi^2} e^{2\delta_{\rm ph}} [v_{\rm geo}/(5\beta_{\rm rad}^2 \bar{M}_{\rm Pl})]^{10}$	decuple	[E]
Hubble anchor	$H_0 \sqrt{\Omega_\Lambda} = \bar{M}_{\rm Pl} e^{-\alpha_\star^{-1}}/(2\pi)$	action	[E]
Baryon density	$\omega_b = \lambda_C / \mathcal{A}_\Lambda = \lambda_C / 10$	decuple	[E]/[C]
Inflation (scalaron)	$M_{\rm scal}^2 / \bar{M}_{\rm Pl}^2 = c_3^{\Omega_{\rm adm} - 10 b_1} = c_3^7$	deficit $r-1$	[E]
Inflation amplitude	$A_s = N_\star^2 c_3^7 / 24\pi^2 \approx 2.2 \times 10^{-9}$	deficit	[C]/[O]

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

Reading. The discrete sector — generations, gauge dimension, family content, hypercharge, b_1, occupancy — is now *group-theoretic output* of $D_5 \times A_3$. The continuous sector — α^{-1}, λ_C, the mixings, ρ_Λ, H_0, inflation — is the seam seed c_3 and the action exponentials read on the *same* carrier graphs (K_5 pairs, the C_6 hexagon $= \lvert R^+(A_3)\rvert$, the decuple $\mathcal{A}_\Lambda = \binom{g_{\rm car}}{2} = 10$). What used to be a list of separate hits is, structurally, one machine: D_5 carries the family and the flavor budget, A_3 carries the generations and the gauge dimension, μ_4 glues them (and reappears as the 41 chain, the discriminant norms, and the CP phase step $2\pi/6$), and c_3 dresses everything.

4 Grammar I — the carrier grammar $(3+2)$

Machine checks for this section: [`verification/v2_carrier_pascal.py`]

From $g_{\rm car} = 5$ and the split $(b, s) = (3, 2)$ the carrier data follow:

$$\gamma_{\rm car} = \frac{g_{\rm car}}{g_{\rm car}+1} = \frac{5}{6}, \qquad N_{\rm fam} = \frac{g_{\rm car}+1}{2} = 3, \qquad \dim S^+ = 2^{g_{\rm car}-1} = 16,$$
$$\Omega_{\rm adm} = N_{\rm fam}\, 2^{g_{\rm car}-1} = 48, \qquad b_1 = \frac{g_{\rm car}\, 2^{g_{\rm car}-2} + 1}{10} = \frac{41}{10}. \tag{19}$$

4.1 Dual-root dictionary (everything from two charges)

Two carrier roots generate the family and the transport cusps [E]. With $y_+ = \frac{1}{2}$ and $y_- = -\frac{1}{3}$ (the determinant-normalised carrier eigenvalues), the full one-family hypercharge spectrum and the flavor-transport cusps are sums of the two roots:

$$Y(Q_L) = y_+ + y_- = \tfrac{1}{6}, \quad Y(u^c) = 2y_- = -\tfrac{2}{3}, \quad Y(d^c) = -y_- = \tfrac{1}{3},$$
$$Y(L_L) = -y_+ = -\tfrac{1}{2}, \quad Y(e^c) = 2y_+ = 1, \quad Y(\nu^c) = 0,$$

$$\{1, \tfrac{2}{3}, \tfrac{1}{3}\} = \{\, 2y_+,\ -2y_-,\ 2(y_+ + y_-)\,\}, \qquad \gamma_{\rm car} = \mathrm{Tr}_E\, Y^2 = 3y_-^2 + 2y_+^2 = \tfrac{5}{6},$$
$$\varphi_{\rm base} = \frac{1 - \gamma_{\rm car}}{\pi} = \frac{1}{6\pi}.$$

All residuals are exactly zero. The Standard-Model charge table is not assumed; it is built by addition from $\{1/2, -1/3\}$.

The $X = 6Y$ quadratic and the hypercharge Pascal-sign duality [E]

Clearing denominators with $X = 6Y$ turns the two base charges into the integer roots $x_+ = 3$, $x_- = -2$ of

$$\boxed{x^2 - x - 6 = 0}: \qquad x_+ + x_- = 1 = N_\Phi, \quad x_+ - x_- = 5 = g_{\rm car},$$
$$-x_+x_- = 6 = |R^+(A_3)|, \qquad \sqrt{\Delta} = 5,$$

so the Higgs singleton, the carrier rank and the A_3 hexagon all fall out of one quadratic. Counting the signs of X over a generation ($X = 1^{\times 6}, 2^{\times 3}, 6^{\times 1}, (-4)^{\times 3}, (-3)^{\times 2}, 0^{\times 1}$) gives the *Pascal triple* again:

$$\boxed{(\#X{=}0,\ \#X{<}0,\ \#X{>}0) = (1, 5, 10) = \tbinom{5}{0}, \tbinom{5}{1}, \tbinom{5}{2}},$$

$$10{-}5{=}g_{\rm car}, \quad 5{-}1{=}|\mu_4|, \quad 10{-}1{=}\mathrm{tr}\, R,$$

with $\nu^c \leftrightarrow \binom{5}{0}$ the neutral singleton — the right-handed neutrino is the Pascal zero with a job. The moment ladder adds $\mathrm{Tr}\, X^2 = 120 = 5!$ and $\mathrm{Tr}\, X^4/\,\mathrm{Tr}\, X^2 = 2280/120 = 19$ (an E_8 exponent; audit). [E]

4.2 Carrier arithmetic (proved identities)

The discrete coefficients are not isolated numbers; they are exact carrier traces:

$$10\, b_1 = 41 = g_{\rm car}^2 + (g_{\rm car} - 1)^2 = \Omega_{\rm adm}\, \gamma_{\rm car} + 1 = 1 + h_{E_8} \tfrac{\dim S^+}{\dim \mathfrak{g}_{\rm SM}} = 1 + 30 \cdot \tfrac{16}{12} = 40 + 1, \quad \text{[E]} \tag{20}$$

$$I_1^{\rm EW} = \frac{41}{48} = \mathrm{Var}_{S^+}(Y) \cdot b_1 = \frac{5}{24} \cdot \frac{41}{10} = \gamma_{\rm car} + \frac{1}{\Omega_{\rm adm}}, \quad \text{[E]} \tag{21}$$

$$\mathrm{Tr}_{S^+}(X^2) = 120 = 5! \quad (X = 6Y), \qquad \delta_2 = \tfrac{5}{4}\delta_{\rm top}^2 = 2880\, c_3^8 = 4! \cdot 5!\, c_3^8 = 4!\ \mathrm{Tr}_{S^+}(X^2)\, c_3^8, \quad \text{[E]} \tag{22}$$

$$\frac{5}{4} = \underbrace{\tfrac{3}{2} \cdot \tfrac{5}{6}}_{B\gamma} = \underbrace{\tfrac{D_{\rm start}}{\Omega_{\rm adm}}}_{60/48} = \underbrace{\tfrac{g_{\rm car}}{g_{\rm car}-1}}_{5/4} = \frac{\delta_2}{\delta_{\rm top}^2}. \quad \text{[E]} \tag{23}$$

So: b_1 is a Pythagorean carrier identity $(5^2 + 4^2)$; equivalently the anchor-product form $10b_1 = p_1p_3 + (g_{\rm car} - p_1) = 4 \cdot 10 + (5 - 4) = 41$ reads "all four glue corners times all ten carrier pairs, plus the one leftover Higgs slot"; the electroweak exponent $41/48$ is the hypercharge variance times the $U(1)$ budget; the second seam defect $2880\, c_3^8$ is a product of factorial traces $4! \cdot 5!$; and the universal compression $5/4 = g_{\rm car}/(g_{\rm car} - 1)$ recurs whenever the five-slot carrier projects onto four effective directions. As a minor curiosity, the SU(5)-tree weak mixing angle rewrites as

$\sin^2\theta_W^{\text{tree}} = \frac{3}{8} = N_{\text{fam}}/(g_{\text{car}} + N_{\text{fam}}) = N_{\text{fam}}/\operatorname{rank}(E_8)$ (the value is the standard one; the rewrite is a coincidence to note, not a claim). [C]

5 Grammar II — the seed grammar (one decoder u)

Machine checks for this section: [`verification/v106_review_validation.py`]

The seed is, exactly, a pure function of c_3:

$$\boxed{u = \varphi_0^{\text{ret}} = \frac{4}{3}\,c_3 + \Omega_{\text{adm}}\,c_3^4 = \underbrace{\frac{1}{6\pi}}_{\varphi_{\text{base}}} + \underbrace{\frac{3}{256\pi^4}}_{\delta_{\text{top}}} = 0.0531719521768} \qquad [\mathbf{E}]. \tag{24}$$

Its two coefficients are themselves carrier data: $\frac{4}{3} = \dim S^+/\dim \mathfrak{g}_{\text{SM}} = 16/12 = 8/(g_{\text{car}} + 1)$ (so $\varphi_{\text{base}} = 1/((g_{\text{car}} + 1)\pi)$) and $48 = \Omega_{\text{adm}}$. The seed is thus a pure function of c_3 and the carrier counts.

One seed, four readouts (and their inverses) [E].

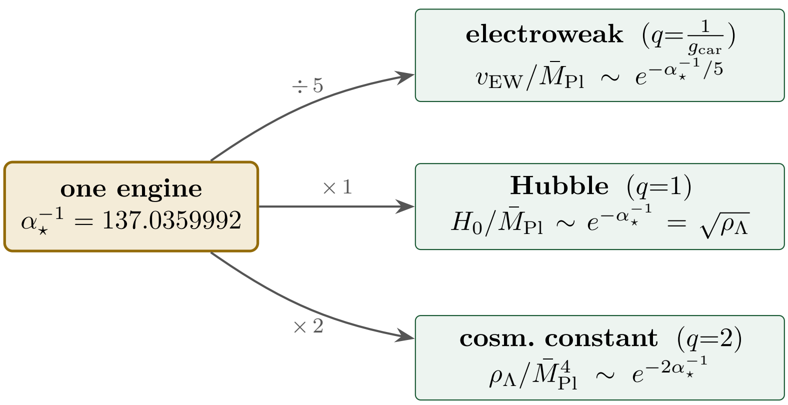
$$\lambda_C^2 = u(1-u), \quad \beta_{\text{rad}} = \frac{u}{4\pi}, \quad \Omega_b = \Big(1 - \frac{1}{4\pi}\Big)u, \quad \sin^2\theta_{13} = e^{-5/6}\,u,$$

$$\begin{aligned} u = \frac{1 - \sqrt{1 - 4\lambda_C^2}}{2} &= 4\pi\,\beta_{\text{rad}} = \frac{\Omega_b}{1 - 1/(4\pi)} = e^{5/6}\sin^2\theta_{13} \\ &= \sum_{n\ge 0} C_n\,\lambda_C^{2n+2} = \lambda_C^2 + \lambda_C^4 + 2\lambda_C^6 + 5\lambda_C^8 + \cdots \end{aligned}$$

One knob u turns four dials simultaneously: Cabibbo mixing (the Bernoulli width $\sqrt{u(1-u)}$), cosmic birefringence, the baryon fraction, and the reactor angle. Turning one forces the others; there is no free per-observable tuning. The exponent $5/6$ is the carrier trace $\gamma_{\text{car}} = \operatorname{Tr}_E Y^2$, *not* the Euler–Mascheroni constant $0.5772\ldots$ (audit [O]).

6 Grammar III — the action grammar (exponentials of $\alpha_\star^{-1}$)

The relation in plain words. The fine-structure constant is not just “a number near 137”: it is the *one exponential engine* behind three wildly separated scales. Take the same $\alpha_\star^{-1} = 137.036$ and divide the carrier into it: $\frac{1}{5}$ of it sets the *electroweak* scale, $1\times$ it sets the *Hubble* scale, $2\times$ it sets the *cosmological constant*. The three action charges are the Pascal row $1 : 5 : 10$ of the five-slot carrier, and the Hubble rung is just the square root of the vacuum energy. So one constant fixes the electroweak hierarchy, dark energy and the expansion rate at once.



action charges $1 : g_{\text{car}} : 2g_{\text{car}} = 1 : 5 : 10 = \binom{5}{0} : \binom{5}{1} : \binom{5}{2}$ (Pascal row of K_5)

Read every dimensionless ratio through its *action* and ask for the action charge:

$$\boxed{x = R_x\, e^{-A_x}, \qquad A_x = q_x\, \alpha_\star^{-1} + \Delta_x, \qquad q_x \in \left\{\tfrac{1}{g_{\rm car}}, 1, 2\right\}.} \tag{25}$$

The robust ladder and its residues are

sector	action A_x	charge q_x	residue R_x	value of $-\ln x$
electroweak $v_{\rm geo}/\bar{M}_{\rm Pl}$	$(\alpha_\star^{-1} + \delta_{\rm ph})/g_{\rm car}$	$1/g_{\rm car}$	$g_{\rm car}\beta_{\rm rad}^2$	36.85
Hubble $H_0/\bar{M}_{\rm Pl}$	$\alpha_\star^{-1}$	1	$1/(2\pi\sqrt{\Omega_\Lambda})$	138.68
cosm. const. $\rho_\Lambda/\bar{M}_{\rm Pl}^4$	$2\alpha_\star^{-1}$	2	$3/(4\pi^2)$	276.65

$$A_{\rm EW} = \frac{\alpha_\star^{-1} + \delta_{\rm ph}}{g_{\rm car}} = 27.5259, \qquad \mathcal{A}_\Lambda = 2\alpha_\star^{-1} = 274.072, \qquad \boxed{A_{\rm EW}{:}\mathcal{A}_H{:}\mathcal{A}_\Lambda = 1{:}5{:}10}. \tag{26}$$

The leading ratio is exact: $\mathcal{A}_\Lambda/(\alpha_\star^{-1}/g_{\rm car}) = 2g_{\rm car} = 10$ (with the transport correction $\mathcal{A}_\Lambda/A_{\rm EW} = 9.957$). The claim is that the *action charges*, not the raw logarithms, form the ladder.

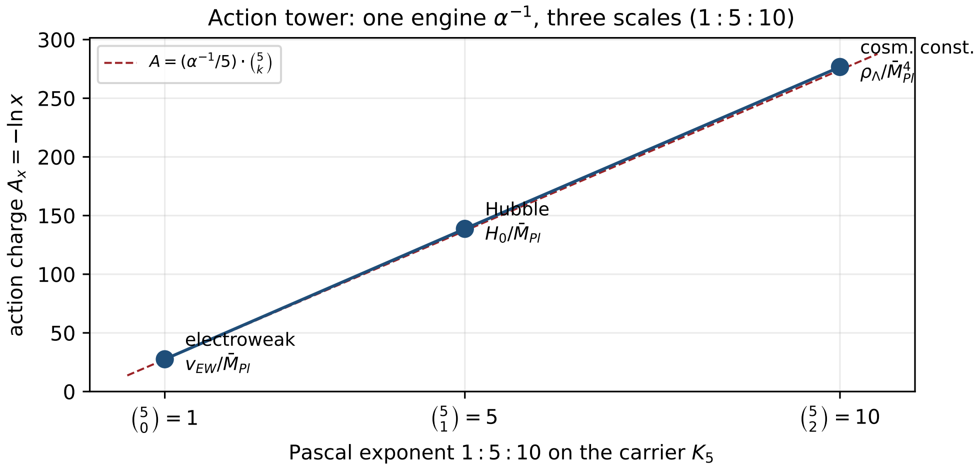


Figure 6: Action tower (data plot, `verification/make_figures.py`). The three action charges $A_x = -\ln x$ for the electroweak, Hubble and cosmological-constant ratios fall on the line $A = (\alpha_\star^{-1}/5)\binom{5}{k}$: one engine $\alpha_\star^{-1}$ generates all three scales with the Pascal exponents $1 : 5 : 10$.

6.1 The EM closure: $\alpha_\star^{-1}$ as an explicit fixed point (made explicit here)

The engine $\alpha_\star^{-1}$ is not an input — it is the unique root of an explicit closure equation (the exact electromagnetic closure theorem). With $c_3 = 1/(8\pi)$, $\varphi_{\rm base} = 1/(6\pi)$, $q(\alpha) = \delta_{\rm top} e^{-2\alpha}$ and the seam response $\varphi_{\rm seam}(\alpha) = \varphi_{\rm base} + q(\alpha)(1 - q(\alpha))^{-5/4}$,

$$\boxed{F_{U(1)}(\alpha) = \alpha^3 - 2c_3^3\alpha^2 - \tfrac{4}{5}c_3^6\Big(\textstyle\sum_{f,j} L_{f,j} + N_\Phi\Big)\log\frac{1}{\varphi_{\rm seam}(\alpha)} = 0} \qquad \textbf{[E]}/\textbf{[C]}, \tag{27}$$

where the coefficient is $\frac{4}{5}(\sum L + N_\Phi)c_3^6 = \frac{4}{5}\cdot 41\, c_3^6 = 8b_1c_3^6$, so the fine-structure constant reads the *flavor transport budget* $\sum L = 40$ and the $U(1)$ index $b_1 = 41/10$.

Lemma 1 (Existence and uniqueness of the EM fixed point)**.** $F_{U(1)}$ has *exactly one* positive root, and it is simple.

Proof. Write $F_{U(1)}(\alpha) = h(\alpha) - C\,g(\alpha)$ with $h(\alpha) = \alpha^3 - 2c_3^3\alpha^2$, $C = \frac{4}{5}c_3^6 M > 0$ ($M = 41$), and $g(\alpha) = -\log\varphi_{\rm seam}(\alpha)$. Since $q(\alpha) = \delta_{\rm top}e^{-2\alpha} \in (0,1)$ is strictly decreasing, $\varphi_{\rm seam} = \varphi_{\rm base} + q(1 -$

$q)^{-5/4}$ is C^1 and decreasing, so g is C^1, bounded, and slowly increasing with $g'(\alpha) = O(q)$ tiny; numerically $g \in [2.9342, 2.9343]$ on the admissible window. The cubic h has its interior critical point at $\alpha_c = \frac{4}{3}c_3^3 = 8.399 \times 10^{-5}$, with $h(0) = 0$ and $h < 0$ on $(0, \alpha_c)$. *Monotonicity (corrected endpoint).* Since $F'_{U(1)}(\alpha) = 3\alpha^2 - 4c_3^3\alpha - C\,g'(\alpha)$ and at α_c the first two terms *cancel* ($3\alpha_c = 4c_3^3$), one has $F'_{U(1)}(\alpha_c) = -C\,g'(\alpha_c) \approx -5.89 \times 10^{-10} < 0$: the derivative is *not* yet positive at α_c. Its first zero lies slightly above, at $\alpha_0 = 8.6264 \times 10^{-5}$ (where $3\alpha_0^2 - 4c_3^3\alpha_0 = C\,g'(\alpha_0)$), with $F'_{U(1)} < 0$ on (α_c, α_0) and $F'_{U(1)} > 0$ on (α_0, ∞). Now $F_{U(1)}(0) = -C\,g(0) < 0$, and $F_{U(1)}$ *decreases* on $(0, \alpha_0]$ (both h and $-Cg$ decrease on $(0, \alpha_c]$, then $F'_{U(1)} < 0$ on $[\alpha_c, \alpha_0]$), so $F_{U(1)}(\alpha_0) \approx -3.82 \times 10^{-7} < 0$; thereafter $F_{U(1)}$ is strictly increasing to $+\infty$. Hence $F_{U(1)} < 0$ on $(0, \alpha_0]$ and strictly increasing on $[\alpha_0, \infty)$, so it crosses zero *exactly once*. The crossing has $F'_{U(1)}(\alpha_\star) = 1.58 \times 10^{-4} > 0$, so the root is simple. A sign scan on $(0, 0.05)$ confirms a single change. *Rigorous version:* an **interval-arithmetic** enclosure (`mpmath.iv`) brackets the root in $[0.00729735256220970, 0.00729735256221000]$ with $F_{U(1)} < 0$ on the lower and > 0 on the upper endpoint (definite signs), and a 200-cell interval partition of $(\alpha_c, 0.05)$ contains exactly *one* indeterminate (root-bearing) band — a verified uniqueness, not merely a sampled one. [`verification/v3_em_alpha.py`] □

Solving:

$$\boxed{\alpha_\star = 0.00729735256220985\ldots, \qquad \alpha_\star^{-1} = 137.0359992168407\ldots} \quad \textbf{[E]} \tag{28}$$

[`verification/v3_em_alpha.py`] versus CODATA-2022 $\alpha_\star^{-1} = 137.035999177(21)$ — a relative deviation 2.9×10^{-10} ($\approx 1.9\sigma$ on the tiny experimental error; the older 2018 value 137.035999084 gave 9.7×10^{-10}, so the agreement *improved* in relative terms). The exponent $-5/4$ in φ_{seam} is the carrier discriminant norm $q(D_5)$, and $\delta_{\text{top}} = 48c_3^4$, so the closure is built only from c_3, the carrier, and the flavor budget. This discharges the former open audit point: the EM closure is a stated, existence-/uniqueness-proven, numerically reproduced equation, not a fit.

Why *this* function: the boundary $U(1)$ Ward reading (Theorem C) [E]/[C]**.** $F_{U(1)}$ is not a chosen ansatz but the reduced Ward identity of the boundary $U(1)$ partition function $Z_\partial = Z_{\text{Maxwell}}\, Z_{\text{Calderón}}\, Z_{\text{transport}}$, with three terms whose coefficients are compiler atoms, not fits:

$$\underbrace{\alpha^3}_{\text{Maxwell moment}} \quad \underbrace{-\,2c_3^3\alpha^2}_{\text{Calderón sheet }(2=|\mathbb{Z}_2|)} \quad \underbrace{-\,8b_1c_3^6 \log \tfrac{1}{\varphi_{\text{seam}}}}_{C_6 \text{ transport det.}} = 0, \qquad 8b_1 = \tfrac{4}{5}M = \tfrac{4}{5}(\textstyle\sum L + N_\Phi) = \tfrac{164}{5}.$$

The Calderón factor $2 = |\mathbb{Z}_2|$ (sheet), the transport power c_3^6 is the C_6 hexagon, and the seam exponent $-\frac{5}{4} = -q(D_5)$ is the carrier discriminant norm. **Typing:** the three-term decomposition and the coefficient identities ($8b_1 = \frac{4}{5}M$, $-\frac{5}{4} = q(D_5)$) are exact [E]; the *physical origin* as a boundary Ward closure (the lemmas $\partial_\tau \log Z_{\text{Maxwell}} = \alpha^3$, $\partial_\tau \log Z_{\text{Calderón}} = -2c_3^3\alpha^2$, $\partial_\tau \log \det_\zeta(1 - \mathcal{T}_{C_6,U(1)}) = -8b_1c_3^6 \log \frac{1}{\varphi_{\text{seam}}}$) is a conjectured [C] interpretation, not yet machine-proven. [`verification/v48_em_ward.py`]

The determinant-line (Quillen) form. ([`verification/v341_alpha_quillen.py`]) The three terms assemble into a single *stationarity* statement for the boundary $U(1)$ determinant line,

$$\underbrace{\left[\alpha^3 - 2c_3^3\alpha^2\right]}_{\delta \log \det \Delta_{U(1)}} = \underbrace{\left[8b_1c_3^6\right]}_{\text{ctr.}} \underbrace{\left[\log \tfrac{1}{\varphi_{\text{seam}}}\right]}_{\text{seam resp.}},$$

so $F_{U(1)} = 0$ is $\delta \log \det \Delta_{U(1)} + \text{counterterm} + \text{seam response} = 0$. Every factor is a *named* object, not a knob: the counterterm coefficient is $8b_1c_3^6 = (\text{rank } E_8)\, b_1\, c_3^6$ (index × heat-kernel), $c_3 = 1/(8\pi)$ is the Gauss–Bonnet boundary coefficient, the exponent $-\frac{5}{4} = -q(D_5)$ is one carrier discriminant norm and the seam base $\varphi_{\text{base}} = 1/(6\pi) = \frac{4}{3}c_3$ carries the *other*, $c_3/\varphi_{\text{base}} = \frac{3}{4} = q(A_3)$ — both glue forms appear, and the split is verified at the root. **Typing (and: no new gate).** The determinant-line assembly and all coefficient identities are exact [E]; $\alpha_\star^{-1}$ itself stays closed [E] (unique root + ablation). This adds *no* open item — it *sharpens* the pre-existing [O]/physical-origin obligation `EM.WARD.01` (the "why *this* function" residual, already conjectural) by naming every coefficient. The one still-open part *is* that residual, now *named* as its own tracked target `ALPHA.QUILLEN.EXACT.01` ([`verification/v382_alpha_quillen_exact.py`]): the from-first-principles AQFT/ζ proof that $\delta_\tau(\log\det_\zeta \Delta_{U(1)} + 8b_1c_3^6 \log\varphi_{\text{seam}}) = 0$ — i.e. that $F_{U(1)}$ is the *exact* Quillen functional of the seam $U(1)$, the variational principle *derived*, not assembled. It is a *face* of `SEAM.EQUIV.01` (v336) — the seam $U(1)$ determinant line lives on the same holomorphic net, so once the net is $(E_8)_1$ its $U(1)$ sub-determinant is fixed — hence no second independent gate, just the sharpest remaining EM-Ward obligation, now citable on its own. *Honest progress towards it ([`verification/v391_alpha_quillen_progress.py`]), without closing it:* a solvable model $\det_\zeta(-\partial_\tau^2+m^2)$ on a circle $= (2\sinh(\beta m/2))^2$ has the closed m-variation $\beta\coth(\beta m/2) = 2/m + (\beta^2/6)m - (\beta^4/360)m^3 + \ldots$, a heat-kernel/Laurent series — so a ζ-determinant's variation is *structured* heat-kernel data, and the counterterm $8b_1c_3^6$ (a Gauss–Bonnet heat-kernel power) is the right *kind* of object. This *reduces* `ALPHA.QUILLEN.EXACT.01` to the named sub-target "the seam $U(1)$ Seeley–DeWitt coefficient $= 8b_1c_3^6$", but the target itself stays [O] (an external math proof, v384); the value $\alpha_\star^{-1}$ stays [E]. *A second honest step ([`verification/v433_alpha_quillen_heatkernel.py`]) bridges this reduction to the heat-kernel origin below, without closing it:* v391's circle reached only the low orders a_0, a_1, whereas a solvable *4D* model — the flat $\Delta = -\partial^2+m^2$ on T^4, $\operatorname{Tr} e^{-t\Delta} = V/(4\pi)^2\, t^{-2} e^{-tm^2}$ — has $a_4 = Vm^4/(2(4\pi)^2) \neq 0$, *reaching* the t^0 (conformal-anomaly) order the circle could not, the very order in which the quadratic term is grounded below (v342, Gilkey a_4). So a ζ-determinant's variation is heat-kernel data *at the a_4 order*, and the measure itself is c_3-built, $(4\pi)^{-2} = (2c_3)^2 = 4c_3^2$; the *actual* seam $a_4 = 8b_1c_3^6$ and the cubic α^3 Chern moment stay [O] (v384), the seam F-normalisation [C] (retyped by the fifth step below, v470, as the embedding index k_Y=5/3).

The heat-kernel origin of the terms ([`verification/v342_em_ward_heatkernel.py`]). The determinant-line form is more than bookkeeping: its terms are textbook heat-kernel coefficients. [E] $c_3 = 1/(8\pi)$ *is* the boundary Gauss–Bonnet coefficient ($\oint_{S^2} K = 2\pi\chi = 4\pi$, $c_3 = 1/(|\mathbb{Z}_2|\oint K)$, the same χ=2 as the marks, v216). [E] the α^2 (Calderón) term *is* the Gilkey gauge-curvature coefficient: the $U(1)$ Laplacian's Seeley–DeWitt a_4 contains $30\,\Omega_{ij}\Omega^{ij}/360 = \frac{1}{12}\Omega^2$ (Gilkey, *Invariance Theory*, Thm 4.8.16), and with $\Omega = i\alpha F$ this forces an $-(\alpha^2/12)F^2$ contribution to $\log\det\Delta_{U(1)}$ — so the Calderón $-2c_3^3\alpha^2$ is the gauge-curvature (Quillen-curvature) piece, not an ansatz. [E] the c_3-orders $\{0,3,6\}$ are an arithmetic ladder (step 3 = the three channels; $6 = |R^+(A_3)|$, the C_6 hexagon), and the π-power test confirms it: $2c_3^3 = 1/(256\pi^3)$ carries π^3, i.e. *three* boundary c_3-insertions, not a single bulk coefficient. **Typing:** the boundary structure and the gauge-curvature origin of the quadratic term are heat-kernel-forced [E]; the exact coefficient value is [C] (the seam F-normalisation — retyped as the embedding index k_Y=5/3 by the fifth step below, v470); the cubic α^3 Maxwell moment (a topological/Chern level) and the full "this *is* the seam $U(1)$ ζ-determinant" claim stay [O] — exactly the `EM.WARD.01` residual, unchanged. So the quadratic/curvature half of the EM-Ward origin is now derived; the cubic-moment half remains the honest open part. *A third honest step ([`verification/v434_alpha_quillen_betafunction.py`]) settles those residuals and shows they are not three independent problems.* The counterterm's *matter* factor is heat-kernel-forced too: by the $\beta = a_4$ theorem the one-loop $U(1)_Y$ β *is* the a_4 coefficient, and the carrier hypercharge content gives $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$ (SM) $\Rightarrow b_1 = 41/10$ (GUT, v159/v246), so $8b_1c_3^6 = b_1\,(\operatorname{rank} E_8)\,c_3^6$ is an a_4 content coefficient times a boundary geometry — the same *kind* of object as the Calderón a_4. The exact value then turns only on the seam F-normalisation, so the "actual seam a_4" residual *is* the F-normalisation [C] (one obligation, not two); and the cubic factors $a^3 - 2c_3^3a^2 = a^2(a - 2c_3^3)$, exhibiting a^3 as the first moment of the Chern density a^2 — a level whose from-first-principles origin stays [O]. The EM-Ward residual is thus *one* [C] + *one* [O], not three. *A fourth honest step ([`verification/v435_alpha_quillen_chernlevel.py`]) sharpens that single remaining [O] — residual (2), the cubic α^3 Chern level — without closing it.* A π-power / metric-independence test partitions the closure: the three coefficients carry π-powers exactly $\{0,3,6\}$ ($k_0 = 1$; $-2c_3^3 = -1/(256\pi^3)$; $-8b_1c_3^6 = -41/(327680\pi^6)$), so the Maxwell a^3 is the *unique* π^0/c_3^0 (metric-*independent*) term. A heat-kernel boundary coefficient always carries the $(4\pi)^{-d/2}$ measure; a metric-independent term carries none — the signature of a *topological* term. So the test confirms, now from a test rather than a claim, that a^3 is a level and not a Seeley–DeWitt curvature integral (the other two terms are heat-kernel boundary data). [E] *Conditional* on the Chern–Simons/η reading of the Maxwell moment (the `EM1` interpretation, v48), large-gauge invariance quantises the seam $U(1)$ level to $\mathbb{Z}$, so $k_0 = 1$ is the *unit* (minimal) level — an integer, not a tunable real. [C] What stays [O] is the `EM1` lemma itself — the from-first-principles Chern–Simons-boundary proof that $\partial_\tau \log Z_{\text{Maxwell}} = a^3$ — external math (type A, v384). *A fifth honest step ([`verification/v470_alpha_inflow_level.py`]) upgrades both leftovers, still without closing the target.* [C] *The level is computed, not minimal:* under the same `EM1` reading, the a^3 coefficient $k_0 = 1$ *equals* the FHS bulk Chern invariant $|C| = 1$ of the very p+ip collar that realises S3 of `SEAM.EQUIV.01` (v367/v461; control $C(M{=}3) = 0$) — "why exactly 1" is answered by a computation plus the cited inflow identification (TKNN / Avron–Seiler–Simon quantisation; Callan–Harvey inflow; the APS/Witten η=CS reading of $\delta \log \det$; Quillen/Bismut–Freed determinant-line integrality), *replacing* the minimality assumption of the fourth step. [E] *The F-normalisation is the embedding index:* $k_Y = \mathrm{tr}_{\bar{5}}(Y^2)/\mathrm{tr}(T_3^2) = (5/6)/(1/2) = 5/3$ (Ginsparg 1987), so $(3/5) \times \frac{41}{6} = \frac{41}{10} = b_1$ exactly — the "GUT 3/5" *is* the inverse embedding index, and once the seam net is $(E_8)_1$ at level 1 the $U(1)_Y$ normalisation $\langle J(z)J(w)\rangle = k/(z{-}w)^2$ is *fixed* by it (level-1 current-algebra rigidity, no continuous dial): the seam F-normalisation [C] has *zero independent content* — a face of `SEAM.EQUIV.01`, per the v382 typing. [C] *One phase, two responses:* the collar phase carries $c_- = 8$ (gravitational, feeding `SEAM.EQUIV.01/S3`, v456) *and* $C = 1$ ($U(1)$, feeding `EM1`) — so discharging the one realisation input feeds both named targets at once. *A sixth honest step ([`verification/v472_quillen_detline_moduli.py`]) exhibits that bridge lemma at the finite level.* `ALPHA.QUILLEN.EXACT.01` names the level as the curvature of the *determinant line over the $U(1)$ moduli* (Quillen/Bismut–Freed curvature, Dai–Freed section) — and the fifth step computed the Chern invariant only over the Bloch Brillouin zone, the translation-invariant shortcut. The sixth step computes the Quillen-shaped object itself: the same collar Hamiltonian in real space with twisted boundary conditions — the twist torus (θ_x, θ_y) *is* the moduli space of flat $U(1)$ connections — and the Fukui–Hatsugai–Suzuki curvature of the determinant line of the occupied frame over that torus integrates to $2\pi \cdot 1$ exactly (10^{-9}), size-independently ($L = 4, 6$), with clean controls (trivial collar $\to 0$, orientation flip $\to -1$), the twist-moduli integer equal to the Bloch-BZ integer for all three phases (Niu–Thouless–Wu), and the Fermi gap open over the whole torus (the Dai–Freed section has no zero). So "$\delta \log \det(\text{seam model}) =$ the inflow response" *holds verbatim at the finite/model level*, where $\log \det$ of the occupied frame replaces $\log \det_\zeta$. [E] (computation) / [C] (the EM1 reading). What stays [O] is the *continuum* identification "$\delta \log \det_\zeta(\text{seam}) =$ the inflow response" on the abstract seam (the `SEAM.EQUIV.01` face); `ALPHA.QUILLEN.EXACT.01` stays [O], and $\alpha_\star^{-1}$ stays [E] regardless. *A seventh step unifies this target with the φ_0^{ret}-puncture target ([`verification/v484_seam_contact_unit.py`]).* The $\{0, 3, 6\}$ c_3-ladder above and the puncture's per-mark weight c_3 (v483) are *one* rule — the seam contact unit — and it is not a new unknown: "c_3 per insertion" *is* the KMS seam unit $2\pi = 1/(4c_3)$ with $\frac{1}{4} = 1/|\mu_4|$ (v239), i.e. one bare boundary propagator ($\frac{1}{2\pi} = 4c_3$) orbit-averaged over the four marks. [E] derived on the seam circle for the finite cycle sector (the bare Green function takes integer multiples of $c_3 \ln 2$ at the μ_4 separations; the log-det contact expansion is exactly graded in c_3 per insertion), with the grammar already suite-wide (the Λ prefactor $\frac{3}{4\pi^2} = 48c_3^2$, v60, carries the same $\Omega_{\text{adm}} = 48$ at two insertions). [C]/[O] Consequence: `ALPHA.QUILLEN.EXACT.01` and `HYP.PHI0.PUNCTURE.01` *merge* into one remaining analytic step — the diagonal zeta-renormalisation at the marks plus the state-multiplicity matching ($41 = 10b_1$ here, $48 = \Omega_{\text{adm}}$ there). *An eighth step settles that remaining step at the computable level ([`verification/v485_contact_diagonal_closed.py`]).* [E] The renormalised coincident-point Green function on a circle of circumference ℓ is $G_{\text{reg}}(0; \ell) = \frac{1}{\pi} \ln(\ell/2\pi)$ exactly — it *vanishes* precisely at $\ell = 2\pi = 1/(4c_3)$, the KMS seam circumference (v239): the seam scale is the unique zero-self-energy scheme point, and the log is the pure *scale* channel (matching the fact that $F_{U(1)}$ carries exactly *one* log term while the φ_0^{ret} split is log-free). [E] With the diagonal zero the mark contact calculus *resums in closed form* (the BFK/gluing route, v151): $\det(I - uC) = (1 - 4u)(1 + 2u)^2$ with $u = \varepsilon c_3 \ln 2$ — all orders, the linear term absent because $\mathrm{Tr}\, C = 0$. [E] And the two multiplicities are *one* state set under two response weights: the same $48 = 3 \times 16$ admissible states counted flat give the $\varphi_0^{\rm ret}$ puncture count $48 = \Omega_{\rm adm}$, and Y^2/loop/GUT-weighted give the alpha budget $41 = 10b_1 = 40 + 1$ ($\mathrm{Tr}_{16}\, Y^2 = \frac{10}{3}$ exact, $\nu^c = 0$; Ginsparg, v470). [C]/[O] Net: every finite/computable piece of the merged target is proven; the single remaining [O] is the identification of the abstract seam ζ-det with this glued mark determinant — *exactly* the v382 typing (a face of `SEAM.EQUIV.01`), now substantiated computationally rather than asserted.

The full ablation — which inputs are load-bearing and which only set the last digit — is in the Ablation appendix below.

Inversion test (this round). Running the closure backwards — inserting the *measured* α and solving for the integer budget $M = \sum L + N_\Phi$ — returns $M = 41.0000001\ldots$, i.e. the external measurement reconstructs the internal discrete budget $41 = 40 + 1$ to 10^{-7}. This is stronger than "we hit α": the equation reads the discrete 41 out of nature. [E]

6.2 The decuple bridge (new, exact headline)

Eliminating $\alpha_\star^{-1} = g_{\rm car}\, A_{\rm EW} - \delta_{\rm ph}$ between the electroweak and the cosmological-constant lines gives a direct relation between the two scales:

Decuple bridge: Λ is the tenth power of the stripped EW scale [E]

$$\boxed{\frac{\rho_\Lambda}{\bar M_{\rm Pl}^4} = \frac{3}{4\pi^2}\, e^{2\delta_{\rm ph}} \left[\frac{v_{\rm geo}}{5\,\beta_{\rm rad}^2\, \bar M_{\rm Pl}}\right]^{10}} \tag{29}$$

where the exponent $10 = 2g_{\rm car}$ and $5 = g_{\rm car}$. Verified to a relative residual $\sim 10^{-159}$.

Proof. From $v_{\rm geo}/\bar M_{\rm Pl} = g_{\rm car}\beta_{\rm rad}^2 e^{-A_{\rm EW}}$, $e^{-A_{\rm EW}} = (v_{\rm geo}/\bar M_{\rm Pl})/(g_{\rm car}\beta_{\rm rad}^2)$. From $\rho_\Lambda/\bar M_{\rm Pl}^4 = \frac{3}{4\pi^2} e^{-2\alpha_\star^{-1}}$ and $\alpha_\star^{-1} = g_{\rm car} A_{\rm EW} - \delta_{\rm ph}$, $e^{-2\alpha_\star^{-1}} = e^{2\delta_{\rm ph}} e^{-2g_{\rm car} A_{\rm EW}} = e^{2\delta_{\rm ph}} (e^{-A_{\rm EW}})^{2g_{\rm car}}$. Substituting and using $2g_{\rm car} = 10$, $g_{\rm car} = 5$ gives the boxed form. $\square$

In words: the electroweak scale is not directly the Λ scale, but once the electroweak ratio is stripped of its small $\beta_{\rm rad}$ prefactor, Λ is its tenth power, and the "ten" is $2g_{\rm car}$, not a fitted exponent. This is the direct algebraic compression of the action ladder.

6.3 The full power tower: x^1, x^5, x^{10} on the carrier graph K_5

The decuple bridge is one rung of a tower. Writing $x := e^{-A_{\rm EW}} = v_{\rm geo}/(5\beta_{\rm rad}^2 \bar M_{\rm Pl})$ for the stripped electroweak unit, all three cosmological scales are powers of the *same* x with binomial exponents (verified exactly):

$$\boxed{\frac{v_{\rm geo}}{5\beta_{\rm rad}^2 \bar M_{\rm Pl}} = x^{\binom{5}{0}} = x, \qquad \frac{H_0}{\bar M_{\rm Pl}} = x^{\binom{5}{1}} R_{\rm H} = x^5 R_{\rm H}, \qquad \frac{\rho_\Lambda}{\bar M_{\rm Pl}^4} = x^{\binom{5}{2}} R_\Lambda = x^{10} R_\Lambda} \tag{30}$$

with residues $R_{\rm H} = e^{\delta_{\rm ph}}/(2\pi\sqrt{\Omega_\Lambda})$ and $R_\Lambda = \frac{3}{4\pi^2} e^{2\delta_{\rm ph}}$. [E]. The exponents are the complete-graph counts of the five-slot carrier K_5:

$$H \leftrightarrow \text{vertices of } K_5\ \left(\# = \tbinom{5}{1} = 5\right), \qquad \Lambda \leftrightarrow \text{edges of } K_5\ \left(\# = \tbinom{5}{2} = 10\right).$$

So the cosmological constant is the product of one stripped-electroweak unit over every carrier *pair* (K_5 edge product), and the Hubble scale the product over every carrier *slot* (K_5 vertex product). This is the cleanest single-variable statement of the action ladder.

6.4 Hubble is the square root of Λ (corollary, exact)

On the comparison layer $\rho_\Lambda/\bar M_{\rm Pl}^4 = 3\Omega_\Lambda (H_0/\bar M_{\rm Pl})^2$; equating to the TFPT line gives

$$\boxed{\frac{H_0}{\bar M_{\rm Pl}} = \frac{e^{-\alpha_\star^{-1}}}{2\pi\sqrt{\Omega_\Lambda}}}, \qquad \mathcal{A}_H = -\ln \frac{H_0}{\bar M_{\rm Pl}} = \alpha_\star^{-1} + \ln(2\pi\sqrt{\Omega_\Lambda}) = 138.68. \tag{31}$$

So the Hubble rung is *not* an independent third hit: it is the square root of the vacuum energy plus a geometric cosmology factor. The genuine TFPT content is the single $e^{-\alpha_\star^{-1}}$. (The numerical coincidence $\mathcal{A}_H - \alpha_\star^{-1} \approx \pi^2/6$ is just $\ln(2\pi\sqrt{\Omega_\Lambda})$ and is Ω_Λ-dependent; not a claim. [O])

6.5 The ~ 123 orders of the cosmological constant

$$\log_{10}\frac{M_{\rm Pl}^4}{\Lambda_{\rm IR}} = \underbrace{\frac{2\alpha_\star^{-1}}{\ln 10}}_{119.028} + \underbrace{\log_{10}\frac{1}{\delta_{\rm top}}}_{3.920} = 122.948 \approx 123. \tag{32}$$

The "119" is the action $2\alpha_\star^{-1}$; the "≈ 4" is the seam defect $\delta_{\rm top} = \Omega_{\rm adm} c_3^4$.

High-precision cosmological constant: a $\sim 10^{-13}$ prediction [E]

Because $\alpha_\star^{-1}$ is a *derived* number (the EM fixed point, known to 13 digits) and the residue $3/(4\pi^2)$ is exact, the dimensionless vacuum energy is predicted essentially exactly:

$$\boxed{\frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} = \frac{3}{4\pi^2}\, e^{-2\alpha_\star^{-1}} = 7.12533 \times 10^{-121}} \qquad \textbf{[E]}\ [\texttt{verification/v7\_gravity\_cosmo.py}], \tag{33}$$

i.e. a dark-energy scale $\rho_\Lambda^{1/4} = \left(\frac{3}{4\pi^2}\right)^{1/4} \bar{M}_{\rm Pl}\, e^{-\alpha_\star^{-1}/2} = 2.23747$ meV (reduced Planck mass $\bar{M}_{\rm Pl} = 2.435323 \times 10^{18}$ GeV). The *dimensionless* ratio carries no observational input — its uncertainty is that of $\alpha_\star^{-1}$, $\sim 10^{-13}$ relative; the absolute meV value inherits only the Planck-mass uncertainty ($\sim 2 \times 10^{-5}$). Both are *far* sharper than the measured vacuum energy (Planck $\rho_\Lambda^{1/4} \simeq 2.24$ meV, known to $\sim 1\%$ and entangled with the H_0 tension). So TFPT pins the cosmological constant several orders of magnitude more precisely than it is currently measured — a genuine, falsifiable forward prediction: any future sub-percent ρ_Λ must land on 2.2375 meV.

6.6 The ladder is binomial; $2g_{\rm car} = \binom{g_{\rm car}}{2}$ singles out $g_{\rm car} = 5$

In units of the electroweak action $\alpha_\star^{-1}/g_{\rm car}$, the three rungs are the first three binomial coefficients of the five-slot carrier:

$$\frac{A_{\rm EW}}{\alpha_\star^{-1}/g_{\rm car}} : \frac{\mathcal{A}_H}{\alpha_\star^{-1}/g_{\rm car}} : \frac{\mathcal{A}_\Lambda}{\alpha_\star^{-1}/g_{\rm car}} = 1 : g_{\rm car} : 2g_{\rm car} = \binom{g_{\rm car}}{0} : \binom{g_{\rm car}}{1} : \binom{g_{\rm car}}{2} = 1 : 5 : 10. \tag{34}$$

The first two are automatic ($\binom{g_{\rm car}}{0} = 1$, $\binom{g_{\rm car}}{1} = g_{\rm car}$). The non-trivial content is the cosmological-constant charge:

$$\boxed{2g_{\rm car} = \binom{g_{\rm car}}{2} \quad\Longleftrightarrow\quad g_{\rm car} = 5} \qquad \textbf{[E]}\ [\texttt{verification/v2\_carrier\_pascal.py}], \tag{35}$$

since $\binom{g_{\rm car}}{2} = \frac{g_{\rm car}(g_{\rm car}-1)}{2} = 2g_{\rm car} \iff g_{\rm car} = 5$. So the "double overlap" charge $2g_{\rm car}$ equals the number of carrier-slot *pairs* $\binom{g_{\rm car}}{2}$ *only* at $g_{\rm car} = 5$: the cosmological constant is suppressed by one action unit per carrier pair, and this combinatorial reading is exclusive to the physical rank. It is an independent overdetermination of $g_{\rm car} = 5$ from the action grammar (cf. the landscape section).

6.7 Discriminant action theorem: the ladder reads the hypercharge discriminant

The $g_{\rm car}$ in the action ladder is not merely a rank: it is the square root of the discriminant of the carrier (hypercharge) polynomial. For a general split $b + s = g$ the carrier polynomial is $bs\,Y^2 + (s-b)Y - 1 = 0$ with discriminant

$$\Delta_Y = (s-b)^2 + 4bs = (b+s)^2 = g_{\rm car}^2, \qquad \text{so for } 6Y^2 - Y - 1 = 0:\ \Delta_Y = 25,\ \sqrt{\Delta_Y} = 5 = g_{\rm car}. \tag{36}$$

Hence the action ladder is, exactly,

$$\boxed{A_{\mathrm{EW}} = \frac{\alpha_\star^{-1}}{\sqrt{\Delta_Y}}, \qquad \mathcal{A}_H = \sqrt{\Delta_Y}\, A_{\mathrm{EW}}, \qquad \mathcal{A}_\Lambda = \left(\frac{\sqrt{\Delta_Y}}{2}\right) A_{\mathrm{EW}}} \quad \text{[E]}. \tag{37}$$

The electroweak hierarchy therefore reads the *discriminant of the hypercharge polynomial*, tying the Standard-Model hypercharge derivation directly to the EW-hierarchy and Λ problems.

6.8 A two-dimensional action lattice

All scales sit on the integer lattice spanned by two generators — the gauge unit $\alpha_\star^{-1}/g_{\mathrm{car}}$ and the flavor unit $L_0 = \ln(1/u)$ — as $S_x = a_x(\alpha_\star^{-1}/g_{\mathrm{car}}) + b_x L_0 + (\text{residue})$:

scale	a_x	b_x	axis
$v_{\mathrm{geo}}/\bar{M}_{\mathrm{Pl}}$	$1 = \binom{5}{0}$	0	gauge
$H_0/\bar{M}_{\mathrm{Pl}}$	$5 = \binom{5}{1}$	0	gauge
$\rho_\Lambda/\bar{M}_{\mathrm{Pl}}^4$	$10 = \binom{5}{2}$	0	gauge
$m_e/\bar{M}_{\mathrm{Pl}}$	1	5	flavor
$m_\mu/\bar{M}_{\mathrm{Pl}}$	1	3	flavor
$m_\tau/\bar{M}_{\mathrm{Pl}}$	1	2	flavor

The cosmological/gauge sector runs along $b = 0$ with binomial charges; the charged-fermion sector runs along $a = 1$ with the integer flavor ladder $n_f = (5, 3, 2)$. This is the compact "periodic table" of TFPT scales: two integer labels per scale. [E] (charges) / [C] (residues).
A third (seam/gravity) generator completes the lattice: $L_c = \ln(1/c_3) = \ln(8\pi)$. The scalaron lives purely on it, $M_{\mathrm{scal}}/\bar{M}_{\mathrm{Pl}} = c_3^{7/2} = e^{-\frac{7}{2}L_c}$, and the seam defect is $\delta_{\mathrm{top}} = e^{-(4L_c - \ln 48)}$. So the three log-generators are $\{\alpha_\star^{-1}$ (hierarchy), $L_0 = \ln(1/u)$ (flavor), $L_c = \ln(1/c_3)$ (seam/gravity)$\}$. [E]

7 The Einstein normaliser as a corrected tree value

Machine checks for this section: [`verification/v1_e8_glue.py`]

The gravitational normaliser $\xi = c_3/u$ has a transparent exact form. Writing $u = \frac{1}{6\pi}[1 + \frac{9}{128\pi^3}]$,

$$\boxed{\xi = \frac{c_3}{u} = \frac{3}{4}\cdot\frac{1}{1+\dfrac{9}{128\pi^3}} = 0.748303} \quad \text{[E]}. \tag{38}$$

So $\xi_{\mathrm{tree}} = \frac{3}{4} = \dim\mathfrak{g}_{\mathrm{SM}}/\dim S^+ = 12/16$ is the exact leading value (the inverse of the $\frac{4}{3} = \dim S^+/\dim\mathfrak{g}_{\mathrm{SM}}$ family/gauge compression that sits in the seed $u_{\mathrm{base}} = \frac{4}{3}c_3$, so $\xi_{\mathrm{tree}}\, u_{\mathrm{base}} = c_3$), and the deviation is precisely the small topological seed correction $9/(128\pi^3)$, i.e. 0.226%. Gravity does not pick up a mysterious factor: first comes 3/4, then the seam defect shifts it minimally.

8 Λ-anchored metrology closure, and two anchors

Machine checks for this section: [`verification/v60_lambda_metrology_branch.py`]

The absolute SI scale is a calibration, not a compiler output. The dimensionless vacuum quotient lets one anchor it to the observed geometric curvature without circularity.

Theorem 1 (Λ-anchored metrology closure)**.** With $\lambda_\Lambda := \rho_\Lambda/\bar{M}_{\mathrm{Pl}}^4 = \frac{3}{4\pi^2}e^{-2\alpha_\star^{-1}}$ and the observed geometric curvature $\Lambda_{\mathrm{geom}}^{\mathrm{obs}} > 0$,

$$\bar{M}_{\mathrm{Pl}}^2 = \frac{\Lambda_{\mathrm{geom}}^{\mathrm{obs}}}{\lambda_\Lambda}, \qquad G_N^{\mathrm{TFPT}} = \frac{c^3}{8\pi\hbar}\frac{\lambda_\Lambda}{\Lambda_{\mathrm{geom}}^{\mathrm{obs}}} = \frac{3c^3}{32\pi^3\hbar\,\Lambda_{\mathrm{geom}}^{\mathrm{obs}}}\, e^{-2\alpha_\star^{-1}}.$$

Proof. $\bar{M}_{\rm Pl}^2 = 1/(8\pi G_N)$ and $\Lambda_{\rm geom} = \rho_\Lambda/\bar{M}_{\rm Pl}^2$; the branch sets $\rho_\Lambda = \lambda_\Lambda \bar{M}_{\rm Pl}^4$, so $\Lambda_{\rm geom} = \lambda_\Lambda \bar{M}_{\rm Pl}^2$, giving $\bar{M}_{\rm Pl}^2$ and then $G_N = \lambda_\Lambda/(8\pi\Lambda_{\rm geom}^{\rm obs})$; restoring $c^3/\hbar$ closes it. □

Non-circularity ([O], essential). The anchor must be the curvature $\Lambda_{\rm geom}^{\rm obs}$ (units m^{-2}), *not* the SI vacuum energy density $\rho_\Lambda^{\rm obs} = \Lambda_{\rm geom}c^4/(8\pi G_N)$, which already contains G_N. The curvature follows from $\Lambda_{\rm geom}^{\rm obs} = 3\Omega_\Lambda H_0^2/c^2$.

Two independent anchors (crosscheck with teeth). The metrology can be pinned in two ways that should agree:

1. *Electroweak anchor*: the dimensionless $G_N v_{\rm phys}^2 = 4.06717 \times 10^{-34}$ fixes G_N once $v_{\rm phys}$ is calibrated non-gravitationally.
2. Λ-*curvature anchor* (this note): $G_N = \frac{3c^3}{32\pi^3\hbar\Lambda_{\rm geom}^{\rm obs}} e^{-2\alpha_\star^{-1}}$.

With the example anchor $H_0 = 67.36\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_\Lambda = 0.6847$ ($\Lambda_{\rm geom}^{\rm obs} = 1.08914 \times 10^{-52}\,\mathrm{m^{-2}}$) the Λ anchor gives

$$G_N^{\rm TFPT} = 6.65071 \times 10^{-11}\,\mathrm{m^3 kg^{-1} s^{-2}}, \qquad G_N^{\rm TFPT}/G_N^{\rm CODATA} - 1 = -3.5 \times 10^{-3}. \tag{39}$$

Closure algebra [E]; SI anchor and numerical match [C] (the cosmological anchor is model dependent; the Hubble tension applies). EW crosscheck: the Λ-pinned $\bar{M}_{\rm Pl} = 2.43964 \times 10^{18}\,\mathrm{GeV}$ gives $v_{\rm geo} = 242.6\,\mathrm{GeV}$, needing $Z_{\rm EW} = (246.22/242.61)^2 = 1.030$.

Prefactor branch ([O]). This note uses $\Lambda_{\rm IR}/M_{\rm Pl}^4 = \delta_{\rm top} e^{-2\alpha_\star^{-1}}$. A competing prefactor $2c_3 e^{-2\alpha_\star^{-1}}$ would differ by $2c_3/\delta_{\rm top} = 661.5$ and misscale G_N by the same factor; the branch must be versioned.

Mandatory Λ-layer typing ([O]). Three distinct Λ objects must never be conflated:

$$\underbrace{\frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} = \frac{3}{4\pi^2} e^{-2\alpha_\star^{-1}}}_{\text{vacuum density quotient (dimensionless)}} \neq \underbrace{\frac{\Lambda_{\rm IR}}{M_{\rm Pl}^4} = \delta_{\rm top} e^{-2\alpha_\star^{-1}}}_{\text{seam-determinant / IR scale (dimensionless)}} \neq \underbrace{\Lambda_{\rm geom}^{\rm obs} = \frac{3\Omega_\Lambda H_0^2}{c^2}}_{\text{observed curvature } [\mathrm{m}^{-2}]}.$$

Only $\Lambda_{\rm geom}^{\rm obs}$ carries a dimension and is the non-circular anchor. Keeping the three typed protects the decuple bridge and the metrology closure from a spurious prefactor objection.

Measurement-near (physical) decuple bridge. Since $v_{\rm phys} = v_{\rm geo}\sqrt{Z_{\rm EW}}$ (charged-current residue, not a fit), the decuple bridge in the observable vev reads

$$\frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} = \frac{3}{4\pi^2} e^{2\delta_{\rm ph}} \left[\frac{v_{\rm phys}}{5\beta_{\rm rad}^2\sqrt{Z_{\rm EW}}\,\bar{M}_{\rm Pl}}\right]^{10}. \tag{40}$$

This is the same identity, written in the low-energy charged-current scale rather than the geometric source scale; both forms should appear side by side. [E]/[C]

9 The landscape: $g_{\rm car} = 5$ is overdetermined

Machine checks for this section: [verification/v6_bootstrap.py] [verification/v14_carrier_uniqueness.py]

Varying the carrier rank and rebuilding every constant from the same machinery (c_3 fixed), the EM-closure value $\alpha^{-1}(g_{\rm car})$ is monotone and meets the observed value only at $g_{\rm car} = 5$:

$g_{\rm car}$	$N_{\rm fam}$	$\dim S^+$	$\Omega_{\rm adm}$	$10b_1$	$\alpha^{-1}(g_{\rm car})$	verdict
3	2	4	8	7	258.15	wrong α, wrong family count
4	2.5	8	20	17	187.31	non-integer families
5	**3**	**16**	**48**	**41**	**137.036**	**SM, 3 families, observed α**
6	3.5	32	112	97	101.30	non-integer families
7	4	64	256	225	75.61	wrong α

Several distinct requirements converge on $g_{\rm car}=5$: integer families ($g_{\rm car}$ odd); three families $(g_{\rm car}{+}1)/2=3$; the observed α^{-1} on the monotone curve; the SM structure $(b,s)=(3,2)$, $\dim S^+=16$; and $\Omega_{\rm adm}=48$, $10b_1=41$. A clean algebraic shadow: $\frac{g(g+1)}{2}=2^{g-1}-1$ has $g=5$ as its non-trivial solution ([C]).

10 The E_8 carrier atlas — stated unambiguously

Machine checks for this section: [`verification/v5_e8_cascade.py`]

In the series E_8 is a *downstream scale grammar*, not a primitive cause: the boundary kernel fixes α, the carrier rank, families and admissibility without E_8 upstream. Three levels.

10.1 Level 1 — original E_8 data of the series ([E])

$$|R(E_8)|=248-8=240=5\cdot48=5\,\Omega_{\rm adm}={\rm lcm}(\Omega_{\rm adm},D_{\rm start})={\rm lcm}(48,60), \tag{41}$$
$$D_{\rm start}=g_{\rm car}\cdot\dim\mathfrak{g}_{\rm SM}=5\cdot12=60,\qquad \gcd(\Omega_{\rm adm},D_{\rm start})=12=\dim\mathfrak{g}_{\rm SM}, \tag{42}$$
$$\frac{|R(E_8)|}{D_{\rm start}}=\frac{240}{60}=4=\frac{\Omega_{\rm adm}}{\dim\mathfrak{g}_{\rm SM}},\qquad 2880=D_{\rm start}\,\Omega_{\rm adm}=|R(E_8)|\cdot\dim\mathfrak{g}_{\rm SM}, \tag{43}$$
$$\kappa_{E_8}=\frac{\gamma_{\rm car}}{\ln(248/60)}=0.58723,\qquad X(n,r,I_1)=\frac{\bar M_{\rm Pl}}{8}\sin^2\theta_{13}\Big(\frac{60-2n}{60}\Big)^{\kappa_{E_8}}e^{-(8-r)/64}e^{-12\pi I_1}. \tag{44}$$

Coxeter/gauge typing of the lcm–gcd pair. Writing $D_{\rm start}=60=2h_{E_8}$ and $\Omega_{\rm adm}=48=6\,{\rm rank}(E_8)$, the relations become $\gcd(2h_{E_8},6r_{E_8})=\dim\mathfrak{g}_{\rm SM}=12$ and ${\rm lcm}(2h_{E_8},6r_{E_8})=|R(E_8)|=240$: the local SM algebra is the gcd of the Coxeter double-period and the $\mathbb{Z}_6$ rank-occupancy, and the E_8 root count is their lcm. Likewise the EW exponent reads $I_1^{\rm EW}=\frac{41}{48}=\frac{h_{E_8}}{36}+\frac{1}{6r_{E_8}}=\frac{5}{6}+\frac{1}{48}$ (Coxeter principal part plus a rank singlet). [E] (arithmetic) / [C] (typing).

10.2 Level 2 — new exact counting identities ([E])

$$|R(E_8)|=240=16\cdot15=\dim S^+\,(\dim S^+-1), \tag{45}$$
$$\dim E_8=248=8\cdot31=(g_{\rm car}+N_{\rm fam})(2^{g_{\rm car}}-1), \tag{46}$$
$$|R(E_8)|=8\,(31-1)={\rm rank}(E_8)\,(2^{g_{\rm car}}-2),\qquad {\rm rank}(E_8)=8=g_{\rm car}+N_{\rm fam}=5+3, \tag{47}$$
$$10\,b_1=41=\frac{|R(E_8)|}{6}+1,\qquad I_1^{\rm EW}=\frac{41}{48}=\gamma_{\rm car}+\frac{1}{\Omega_{\rm adm}}, \tag{48}$$

with $31=2^{g_{\rm car}}-1$ the number of non-empty occupation words of the $[5,4,2]$ carrier code.

10.3 Level 2.5 — the even-flip transition atlas (a falsifiable program)

Identify S^+ with the even-weight binary code on five bits. A transition $x\to y\neq x$ has flip $F=x\oplus y$, even and nonempty, so $|F|\in\{2,4\}$; per state there are $\binom{5}{2}+\binom{5}{4}=10+5=15$ flips, giving

$$240=16\cdot(10+5)=\underbrace{160}_{|F|=2}+\underbrace{80}_{|F|=4}, \tag{49}$$

and the 120 undirected flips match the 120 E_8 root *pairs* ([E], counting). This makes the atlas conjecture *testable*: a genuine root map must send each flip to a norm-2 vector in $\mathbb{R}^8$ with inner products in $\{-2,-1,0,1,2\}$ reproducing the E_8 profile (per root: 1 self, 56 at +1, 126 orthogonal, 56 at −1, 1 anti).

Honest obstruction ([O]). The flip atlas splits $160+80$ by flip size; the standard E_8 (D_8) split is $112+128$. Since $160+80\neq112+128$, no flip-size-preserving map is an E_8 root system; a metrization, if it exists, must mix flip sizes. Until a norm-2, E_8-profile embedding is exhibited, E_8 stays a *counting* atlas, not a derived root origin.

Sharper two-stage audit (Coxeter before Weyl). A full root-system isomorphism is hard; a much earlier, cheaper test is the Coxeter spectrum. *Stage A:* find a natural order-30 operator C_{TFPT} on the even-flip atlas (or its 8-dimensional primitive-character space) with characteristic polynomial $\chi_{C_{\mathrm{TFPT}}}(t) = \Phi_{30}(t)$. *Stage B (only if A holds):* the norm-2, E_8-profile, Weyl-closure metrization. Stage A can fail or carry far sooner than B, so it is the right first falsifier.

Stage A is closed on the primitive-character space ([`verification/v531_coxeter_tensor_stage_a.py`]**).** The explicit integer operator

$$T_{30} = \mathrm{Comp}(\Phi_5) \otimes \mathrm{Comp}(\Phi_6)$$

(the 4×4 carrier clock of v418 tensor the 2×2 family/CP hexagon) has *exact* order 30 ($T_{30}^{30} = I$, $T_{30}^k \neq I$ for $1 \leq k < 30$) and $\chi_{T_{30}} = \Phi_{30} = t^8 + t^7 - t^5 - t^4 - t^3 + t + 1$ exactly: by CRT ($\gcd(5,6) = 1$) its eigenvalues are the products (primitive 5th)×(primitive 6th) = exactly the eight primitive 30th roots — the carrier clock times the family clock *is* the Coxeter clock, as one integer operator. Its traces are the Ramanujan sums, $\mathrm{tr}(T_{30}^k) = c_{30}(k) = c_5(k)\,c_6(k)$ for all k, with the signed divisor table $(-1,1,2,4,-2,-4,-8,8)$ at $k = (1,2,3,5,6,10,15,30)$ and the atom readout $(|\,\mathrm{tr}\,T^3|, |\,\mathrm{tr}\,T^5|, |\,\mathrm{tr}\,T^{15}|) = (2,4,8) = (|\mathbb{Z}_2|, |\mu_4|, \mathrm{rank}\,E_8)$ — eight integer traces type the clock *without* diagonalisation, a cheap kill test for any flip-operator candidate (honest scope: Φ_{15} passes the unsigned triple since $\Phi_{30}(t) = \Phi_{15}(-t)$; the *signed* table is the discriminator, and $\Phi_{16}, \Phi_{20}, \Phi_{24}$ die on the triple already). [E] *What stays open ([O]):* a canonical intertwiner from the concrete even-flip atlas onto this primitive space (`COX.FLIP.INTERTWINER`), and Stage B.

Audit only — *non-load-bearing* speculation: the dim 496 **heterotic count** [O]**.** *This box feeds no derivation; it is an arithmetic coincidence kept for completeness only, under the no-free-pattern rule.* The even half S^+ and odd half S^- of the exterior algebra each have dimension $2^{g_{\mathrm{car}}-1} = 16$ and each carry $16 \cdot 15 = 240$ transitions. Together the full carrier ($2^{g_{\mathrm{car}}} = 32$ states) gives

$$\dim S^+(2^{g_{\mathrm{car}}} - 1) = 16 \cdot 31 = 496 = \dim(E_8 \times E_8) = \dim SO(32) = \binom{32}{2} \qquad \text{(count)},$$

i.e. the two halves read as two E_8 root atlases, all $\binom{32}{2} = 496$ pairs as $SO(32)$ — the two anomaly-free heterotic gauge dimensions. The *dimension* equality is exact; that the carrier algebra *is* either gauge group is unproven and not used anywhere. [O] (audit-only). The same fingerprint recurs at checksum level: the dual products of the E_8 invariant degrees sum to $744 = 3 \cdot 248$ (the j-function constant), and the identical construction for D_{16} gives $1488 = 3 \cdot 496 = 2 \cdot 744$ ([`verification/v532_e8_degree_modular_checksum.py`]) — the checksum measures the common heterotic rank-16/Coxeter-30 structure, *not* E_8 uniqueness. [E] (audit-only).

10.4 Level 2.7 — the carrier Coxeter theorem (the strongest new bridge)

Use the universal root-system identity $|R| = \mathrm{rank} \cdot h$ (Coxeter number h). With the carrier values $\mathrm{rank}(E_8) = g_{\mathrm{car}} + N_{\mathrm{fam}} = 8$ and $|R(E_8)| = \dim S^+(\dim S^+ - 1) = 240$,

$$\boxed{h_{E_8} = \frac{|R(E_8)|}{\mathrm{rank}(E_8)} = \frac{240}{8} = 30 = N_{\mathrm{fam}}\binom{g_{\mathrm{car}}}{2} = N_{\mathrm{fam}}\,A_\Lambda = 2g_{\mathrm{car}}N_{\mathrm{fam}} = 2^{g_{\mathrm{car}}} - 2} \quad \text{[E]}. \tag{50}$$

The Coxeter number of E_8 is exactly three families times the cosmological pair action: each family contributes one Λ-decuple ($A_\Lambda = \binom{5}{2} = 10$) to the Coxeter period. Equivalently the action ladder is a Coxeter ladder,

$$A_H = \frac{h_{E_8}}{2N_{\mathrm{fam}}} = 5, \qquad A_\Lambda = \frac{h_{E_8}}{N_{\mathrm{fam}}} = 10, \tag{51}$$

so Hubble reads half a Coxeter period per family and Λ a full Coxeter period per family. [C]

Rank–Coxeter coupling. Equivalently $\dim S^+ = 2\,\mathrm{rank}(E_8) = 16$ and $h_{E_8} = 2(\dim S^+ - 1) = 2 \cdot 15 = 30$, so $\dim E_8 = (g_{\mathrm{car}} + N_{\mathrm{fam}})(h+1) = 8 \cdot 31$. The two counting forms $16 \cdot 15 = 8 \cdot 30 = 240$ coincide because $\dim S^+ = 2r$. Dropping ν^c sends $\dim S^+ \to 15$, which breaks *both* $\dim S^+ = 2r$ and $\dim S^+(\dim S^+ - 1) = 240$: ν^c is the state that closes the rank–Coxeter coupling, not an optional singlet. [E]/[C]

10.5 Level 2.8 — positive-root moment: $|R^+(E_8)| = \mathrm{Tr}_{S^+} X^2 = 5!$

The number of positive E_8 roots equals the master hypercharge moment:

$$\boxed{|R^+(E_8)| = 120 = \mathrm{Tr}_{S^+} X^2 = 5!} \qquad [\mathbf{E}], \tag{52}$$

so the second hypercharge moment of one family *is* the positive-root half-system. The whole integer chain then reads off $|R^+|$:

$$\sum_{f,j} L_{f,j} = \frac{|R^+(E_8)|}{3} = 40, \qquad 10b_1 = 1 + \frac{|R^+(E_8)|}{3} = 41, \qquad \delta_2 = 4!\,|R^+(E_8)|\,c_3^8. \tag{53}$$

The $\mathbb{Z}_6$ slice is then $|R(E_8)|/6 = \frac{|R^+|}{3} = 40 = \Omega_{\mathrm{adm}}\gamma$: one $\mathbb{Z}_6$ phase slice of the E_8 root atlas is exactly the full flavor transport budget. [E] (arithmetic) / [C] (interpretation).

10.6 Level 2.9 — the 2·3·5 Coxeter cyclotomic compiler (meta-theorem)

The three hard carrier atoms — weak rank 2, color rank / family count 3, carrier rank $g_{\mathrm{car}} = 5$ — generate the E_8 data *cyclotomically*, without taking E_8 as input. Set

$$\boxed{h := 2 \cdot 3 \cdot g_{\mathrm{car}} = 30, \qquad r := \varphi(h) = \varphi(30) = 8 = g_{\mathrm{car}} + N_{\mathrm{fam}},} \tag{54}$$

where h is the Coxeter number and $r = \varphi(2\cdot3\cdot5) = 8$ the rank. Then the standard $|R| = rh$, $\dim = r(h+1)$ give

$$|R(E_8)| = rh = 8\cdot30 = 240, \qquad \dim E_8 = r(h+1) = 8\cdot31 = 248, \qquad h+1 = 2^{g_{\mathrm{car}}} - 1 = 31. \tag{55}$$

Moreover the E_8 *exponents* are exactly the primitive residues modulo $h = 30$:

$$\mathrm{Exp}(E_8) = (\mathbb{Z}/30\mathbb{Z})^\times = \{1, 7, 11, 13, 17, 19, 23, 29\}, \qquad \sum_{m \in (\mathbb{Z}/30)^\times} m = 120 = \mathrm{Tr}_{S^+} X^2 = |R^+(E_8)|, \tag{56}$$

so the second hypercharge moment equals the sum of the E_8 exponents and the positive-root count. The $\mathbb{Z}_6 \times \mathbb{Z}_5 \simeq \mathbb{Z}_{30}$ tact has $\varphi(30) = 8$ primitive characters, whose phases $e^{2\pi i m/30}$ are the Coxeter eigenphases of E_8; its characteristic polynomial is the 30th cyclotomic polynomial

$$\chi_C(t) = \Phi_{30}(t) = t^8 + t^7 - t^5 - t^4 - t^3 + t + 1. \tag{57}$$

Reading: E_8 here is not primarily a root count but the *Coxeter cyclotomic readout of the* $2, 3, 5$ *carrier*. [E] (arithmetic) / [C] (interpretation).

10.7 Level 3 — conjectural geometric readings ([C])

Conjecture 1 (Carrier transition atlas)**.** $R(E_8) \cong \{(a,b) \in S^+ \times S^+ : a \neq b\}$ as a structure-bearing transition set, since $16 \cdot 15 = 240$. *Open:* a structure-preserving map (roots, lengths, Weyl data), not only the count.

Conjecture 2 (Least common refinement / rank-8 over the exhausted carrier)**.** $240 = \mathrm{lcm}(\Omega_{\mathrm{adm}}, D_{\mathrm{start}})$ reads E_8 as the least common refinement of the occupancy lattice and the $\mathbb{Z}_6$ phase lattice; $248 = (g_{\mathrm{car}} + N_{\mathrm{fam}})(2^{g_{\mathrm{car}}} - 1)$ reads it as a rank-8 structure over the 31 non-empty carrier words.

Two routes to E_8 — keep their status separate. Levels 1–2 are exact arithmetic. There are now *two* distinct routes to E_8, and they must not be conflated:

- **Glue route (closed, [E]):** $E_8 = (D_5 \oplus A_3) + \mu_4$-glue is built explicitly in Section 3 ($\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4$, glue index 4, 240 roots of norm 2, simple-root Gram det 1). This closes the former "blind 240-metrisation".
- **Direct flip-atlas route (open, [O], optional bonus):** the even-flip transition set $240 = 160 + 80$ is *not* a flip-size-preserving E_8 map, since $160 + 80 \neq 112 + 128$. A direct structure-preserving metrisation of the flip graph remains open — a nice-to-have, not load-bearing, because the glue route already delivers E_8.

So E_8 is closed via the glue; only the direct flip atlas is open.

11 $\mathbb{Z}_6$ monodromy, flavor and CP readouts

Machine checks for this section: [`verification/v88_cp_phase_audit.py`]

A single $\mathbb{Z}_6$ motif recurs in the quotient $G_{\text{phys}} = (SU(3)\times SU(2)\times U(1)_Y)/\mathbb{Z}_6$, the transport cusps $\{1, (2/3)^6, (1/3)^6\}$, $\delta_{\text{ph}} = z_\star^{1/6}$, and the CP phases

$$\delta^{(0)}_{\text{CKM}} = \frac{\pi}{3}, \quad \delta_{\text{CKM}} = \frac{\pi}{3} + 3\lambda_C^2 = 1.19823\,\text{rad}, \quad \delta^{\text{PMNS}}_{\text{CP}} = \frac{4\pi}{3} = 240^\circ. \tag{58}$$

From the documented inputs $s_{12} = \lambda_C$, $s_{23} = u/(1+\lambda_C)$, $s_{13} = \lambda_C^3/3$ one gets the new derived readouts $|V_{td}| = 0.00908$, $|V_{ts}| = 0.04264$, $J_{\text{CKM}} = 3.33\times 10^{-5}$. On the neutrino side, with $\sin^2\theta_{13} = e^{-5/6}u = 0.02311$, $\sin^2\theta_{23} = 0.4557$, $\delta_{\text{CP}} = 4\pi/3$ and the solar candidate $\sin^2\theta_{12} = \frac{1}{3} - \frac{u}{2} = 0.30675$, a near-maximal $|J^\nu_{\text{CP}}| \approx 0.030$ (sharp readout for DUNE/Hyper-Kamiokande). [C]

11.1 CP as triple seed variance; a closed observable quadrilateral

The Cabibbo angle is the Bernoulli width of the seed. The *exact* CKM phase is the family transport holonomy (document 3), $\delta_{\text{CKM}} = \arg \operatorname{Tr} \Pi_\triangle(\nabla^\star_F)$; on the small-area branch this has the compact readout form (with an $O(\lambda_C^4)$ remainder)

$$\lambda_C = \sqrt{u(1-u)}, \qquad \boxed{\delta_{\text{CKM}} - \frac{\pi}{3} = N_{\text{fam}}\, u(1-u) = 3\lambda_C^2} \qquad \text{[E]/[C] (readout; exact = holonomy).} \tag{59}$$

The coefficient 3 is the family number N_{fam}: weak CP is the $\mathbb{Z}_6$ base angle $\pi/3$ plus N_{fam} times the seed (Bernoulli) variance. The $\mathbb{Z}_6$ base angle now has an A_3 root: since $\mathbb{Z}_6 = |R^+(A_3)| = 6$ (the positive-root hexagon of the family algebra), the CP phases are integer steps of the A_3 holonomy quantum $2\pi/|R^+(A_3)|$,

$$\boxed{\delta_{\text{CKM}} = \frac{2\pi}{|R^+(A_3)|} + N_{\text{fam}}\, u(1-u), \qquad \delta^{\text{PMNS}}_{\text{CP}} = 4\cdot\frac{2\pi}{|R^+(A_3)|} = \frac{4\pi}{3}} \qquad \text{[E]/[C].} \tag{60}$$

So weak CP = (A_3 positive-root phase step) + (family count × seed variance), while strong CP is the *determinant* sector, erased. The solar angle reads as a carrier root plus a half-seed, $\sin^2\theta_{12} = \frac{1}{3} - \frac{u}{2} = -y_- - \frac{u}{2}$ (since $-y_- = \frac{1}{3}$), so PMNS = carrier root + seed + A_3 holonomy phase. [C]Strong CP is killed by the determinant selector ($\arg\det M_u = \arg\det M_d = 0$, $\bar\theta = 0$), while weak CP survives as the seed-deformed $\mathbb{Z}_6$ holonomy: CP is not forbidden, only *determinant* CP is. [C] — the strong-CP closure lives on the admissible QFT branch and is *conditional* on the analytic hypotheses of the QFT closure (document 3, Part III: reflection positivity, temperedness, clustering, mass gap, OS reconstruction); its status is therefore weaker than the discrete lattice identities, even though the semantics "determinant CP forbidden, holonomy CP allowed" is robust.

Since $u = \Omega_b + \beta_{\rm rad}$ exactly, eliminating the seed closes the four UV-shadow observables into one cycle:

$$\lambda_C^2 = (\Omega_b + \beta_{\rm rad})(1 - \Omega_b - \beta_{\rm rad}), \quad \sin^2\theta_{13} = e^{-5/6}(\Omega_b + \beta_{\rm rad}), \quad \Omega_b = \Big(1 - \tfrac{1}{4\pi}\Big) e^{5/6}\sin^2\theta_{13}. \tag{61}$$

One measured member of $\{\lambda_C, \theta_{13}, \Omega_b, \beta_{\rm rad}\}$ fixes the other three. [E] (the four rows remain operationally distinct readout classes).

12 Mass hierarchy as a hexagon with one extra turn

Machine checks for this section: [verification/v18_quark_yukawa.py] [verification/v20_lepton_c_derivation.py]

The transport length packets are

$$L_u = (7,3,0), \qquad L_d = (7,5,2), \qquad L_e = (8,5,3), \qquad \sum_{f,j} L_{f,j} = 40 = \Omega_{\rm adm}\,\gamma_{\rm car} = 10b_1 - 1. \tag{62}$$

Modulo 6 they lie on related triples of the C_6 hexagon; the first generation carries one extra full turn ($u, d: 1+6$; $e: 2+6$), costing $\lambda_Y^6 = (u(1-u))^3 \approx 1.28 \times 10^{-4}$ and explaining why the first generation is so light — "one turn more", not a free fit. [C]*Flagged proximity ([O], not exact): the* full-turn cost is close to the topological seam defect, $\lambda_C^6/\delta_{\rm top} = 1.061$ ($\sim 6\%$), hinting that one flavor winding $\approx$ one seam-defect unit $\delta_{\rm top} = \Omega_{\rm adm} c_3^4$ — a structural smell, not a theorem. Writing the length matrix $L = R + W$ with $R = L \bmod 6$ the residue shadow and W the full-winding part,

$$L = \begin{pmatrix} 7 & 3 & 0 \\ 7 & 5 & 2 \\ 8 & 5 & 3 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 3 & 0 \\ 1 & 5 & 2 \\ 2 & 5 & 3 \end{pmatrix}}_{R} + 6\underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}}_{W}, \qquad \sum R = 22,\ \sum W = 18 = 3\cdot 6,\ \sum L = 40,$$

and the column sums are $(22, 13, 5)$ with $S_1 = 22 = \sum R$ and $S_2 + S_3 = 18 = \sum W$. So the first generation is light because it stores the *entire* residue budget plus the three full C_6 windings; generations 2 and 3 are the winding complement. [E] (arithmetic).

12.1 Lepton flavor decuple: the charged-lepton determinant carries u^{10}

The source masses on the $v_{\rm geo}$ branch are $(\hat m_e, \hat m_\mu, \hat m_\tau) = \frac{v_{\rm geo}\pi}{\sqrt 2}\big(\frac{16}{7}u^5, \frac{4}{3}u^3, \frac{7}{6}u^2\big)$. Their product is a pure decuple in the seed:

$$\boxed{\frac{\hat m_e \hat m_\mu \hat m_\tau}{(v_{\rm geo}\pi/\sqrt 2)^3} = \Big(\tfrac{16}{7}\cdot\tfrac{4}{3}\cdot\tfrac{7}{6}\Big)\,u^{5+3+2} = \frac{32}{9}\,u^{10}} \qquad \text{[E]}, \tag{63}$$

with $5+3+2 = g_{\rm car} + b + s = 10 = \binom{5}{2}$. So the charged-lepton Yukawa powers $(n_e, n_\mu, n_\tau) = (5,3,2)$ *are* the three carrier numbers $(g_{\rm car}, b, s)$, and the lepton determinant carries the same decuple (u^{10}) that the cosmological constant carries in the gauge sector ($A_\Lambda = 10$): Λ and the charged-lepton determinant are two realisations of the same degree-$\binom{5}{2}$ sector.

Sum–product reading. The lepton powers $(5,3,2)$ encode *both* decuple numbers at once: their sum is the Λ action, their product is the E_8 Coxeter number,

$$5+3+2 = 10 = A_\Lambda, \qquad 5\cdot 3\cdot 2 = 30 = h_{E_8} = 2\cdot 3\cdot 5, \tag{64}$$

and $(5,3,2)$ is the prime factorisation of $h_{E_8} = 30$. So $\det M_\ell$, ρ_Λ and the E_8 Coxeter period read the same carrier core additively ($= 10$) and multiplicatively ($= 30$). [E]/[C]

12.2 Why the top Yukawa is ≈ 1: the $L=0$ anchor (Froggatt–Nielsen ladder)

The transport length L_f is a Froggatt–Nielsen-type charge with expansion parameter λ_C:

$$\boxed{y_f \simeq \lambda_C^{L_f} \times \Lambda_f \ (\Lambda_f = \mathcal{O}(1)), \qquad y_t \simeq 1 \text{ because } L_t = 0} \qquad \text{[E]/[C].} \tag{65}$$

The top sits at $L_t = 0$ (the *unwound* carrier direction), so $y_t = \Lambda_t \simeq 1$ ($\sqrt{2}\,m_t/v = 0.991$); every other charged fermion is suppressed by integer powers of $\lambda_C = 0.2244$. Numerically (with $\mathcal{O}(1)$ sector prefactors $\Lambda_f \in [0.43, 1.07]$):

f	t	b	τ	c	μ	s	u, d	e
L_f	0	2	3	3	5	5	7	8
$y_f/\lambda_C^{L_f}$	0.99	0.48	0.90	0.65	1.07	0.94	~ 0.4–0.9	0.46

So the entire charged-fermion hierarchy is one λ_C-ladder anchored by the top at $L = 0$; the lightness of the first generation ($L = 7, 8$) and the heaviness of the top ($L = 0$) are the two ends of the same integer winding spectrum. This is the standard Froggatt–Nielsen mechanism with λ_C as the flavon ratio and L_f as the FN charge — a clean bridge between TFPT transport and an established flavor mechanism.

13 Inflation interface

Machine checks for this section: [`verification/v7_gravity_cosmo.py`] [`verification/v86_nstar_reheating.py`] [`verification/v494_cosmo_running_mu_record.py`]

The documented scalaron is Starobinsky R^2; with the pivot $N_\star = 57.1$:

$$\begin{aligned} n_s &= 1 - \frac{2}{N_\star} = 0.96497, \qquad r = \frac{12}{N_\star^2} = 0.00368, \\ \frac{M_{\rm scal}}{\bar{M}_{\rm Pl}} &= \sqrt{8\pi}\, c_3^4 = c_3^{7/2} = (8\pi)^{-7/2} \ \Rightarrow\ M_{\rm scal} \approx 3.06 \times 10^{13}\,\text{GeV}. \end{aligned} \tag{66}$$

n_s matches Planck; $r = 0.0037$ is the falsifiable target for LiteBIRD/CMB-S4.

Amplitude (status-clean). The scalaron mass is a pure seam power,

$$\boxed{\frac{M_{\rm scal}^2}{\bar{M}_{\rm Pl}^2} = c_3^{r-1} = c_3^{\Omega_{\rm adm} - 10 b_1} = c_3^7} \qquad \text{[E],} \tag{67}$$

an *exact* identity (the exponent $7 = 48 - 41$ is the post-flavor, post-Higgs occupancy deficit). The cosmology interface reduces the geometric gravity action to an FRW R^2/scalaron branch in which $M_{\rm scal}$ *is* the Starobinsky mass M; the identification is therefore derived in that sector, not posited. With the standard large-N amplitude $A_s = \frac{N_\star^2}{24\pi^2}(M/\bar{M}_{\rm Pl})^2$ this gives

$$A_s = \frac{N_\star^2\, c_3^7}{24\pi^2} = \frac{N_\star^2\, \delta_2}{8640\pi} = 2.17 \times 10^{-9} \qquad \text{[C]/[O],} \tag{68}$$

matching Planck ($\simeq 2.1 \times 10^{-9}$); $\delta_2 = 2880\, c_3^8$ is the second seam defect. The seam-power form is exact [E]; the amplitude is then determined *within* the R^2 sector, the only residual being the geometric-gravity (Hodge) closure on which that sector rests [C]. *Convention note:* $A_s \propto N_\star^2$, so the quoted 2.17×10^{-9} uses the pivot $N_\star = 57.1$; the alternative seed pivot $N_\star = 3/u - 1 = 55.4$ gives 2.05×10^{-9}. A number should only be quoted with $N_\star$ fixed.

Running (consistency relation). The same attractor also fixes the *running* of the tilt parameter-free: $\alpha_s = \mathrm{d}n_s/\mathrm{d}\ln k = -2/N_\star^2 = -r/6$ and $\beta_s = -4/N_\star^3$ at leading order; the exact slow roll on the Starobinsky potential gives $\alpha_s = -7.09 \times 10^{-4}$, $\beta_s = -2.70 \times 10^{-5}$ at the reheating point $N_\star$=51.4 (leading-order band $[-8.0, -5.6] \times 10^{-4}$ over $N_\star \in [50, 60]$). Current data are consistent (Planck 2018 -0.0045 ± 0.0067: $+0.57\sigma$; P-ACT-LB $+0.0062 \pm 0.0052$: -1.33σ), and since plateau potentials give $\alpha_s < 0$, any robust *positive* running would kill the branch. [C] [`verification/v494_cosmo_running_mu_record.py`]

14 Further simple bridges

Machine checks for this section: [`verification/v74_compiler_micro_lemmas.py`]

- **Dark-energy equation of state $w = -1$ (exact).** The TFPT Λ is a genuine vacuum constant $\rho_\Lambda = \frac{3}{4\pi^2} e^{-2\alpha_\star^{-1}} \bar{M}_{\rm Pl}^4$, not a rolling field, so $w = -1$ identically (no quintessence). A robust measurement $w \neq -1$ would falsify the Λ-branch. [E] (structural).
- **Effective neutrino number $N_{\rm eff} \simeq 3$.** Three families ($N_{\rm fam} = 3$) give three light neutrinos and the standard $N_{\rm eff} \simeq 3.04$. [E].
- **Top Yukawa** $\simeq 1$ as the $L = 0$ anchor (above): a bridge to the Froggatt–Nielsen flavor mechanism.
- *Flagged (not clean):* $m_H/v = 0.509$ is near $\frac{1}{2}$ (1.7%); $m_W/m_Z = 0.881$ matches the running $\sqrt{1 - \sin^2\theta_W(M_Z)}$ (not the tree $\sqrt{5/8} = 0.79$). No clean closed bridge for the Higgs/W/Z masses yet. [O]

15 Closures that the architecture rests on

Machine checks for this section: [`verification/v36_spectral_action_g2.py`] [`verification/v4_flavor_matrix.py`]

Five results close or sharpen the discrete core. They are *proved in this document set* — the $\varphi_0^{\rm ret}$-ladder, the H2 realisation and the strong-CP/dynamics gap in document 2 (`tfpt_2_standard_model`), the $E_6 \times A_2$ shadow in document 3 (`tfpt_3_e8_audit_bootstrap`) — and are collected here because the architecture above depends on them.

1. **The $\varphi_0^{\rm ret}$-ladder (whole spectrum, one formula).** Every fermion mass is $\hat{m}_{f,j} = \frac{v_{\rm geo}}{\sqrt{2}} \lambda_Y^{L_{f,j}} \Lambda_{f,j}$ with $\lambda_Y = \sqrt{\varphi_0^{\rm ret}(1 - \varphi_0^{\rm ret})}$, and the closed evaluation gives the universal form $\lambda_Y^L \Lambda = \pi\, c\, (\varphi_0^{\rm ret})^k$ — the nine Λ are hexagonal-resolvent outputs (not free numbers); the *charged-lepton* ones are exact, the *quark* ones are reduced to the ($U_{\rm wall}$) selector. The $\varphi_0^{\rm ret}$-order ladder is $t(0) > \{b, c, \tau\}(2) > \{s, \mu\}(3) > \{u, d\}(4) > e(5)$. [E] (leptons exact, quark ratios closed v49/v71) / [O] (absolute quark scale)
2. **H2 reduced via the Mehta–Seshadri framework.** Mehta–Seshadri supplies the equivalence (stable parabolic $\leftrightarrow$ irreducible unitary); the TFPT-specific step — "stable = transport-minimal" and the geodesic C_6 word length — is conditional on the unitarity (U) of $\nabla_F^\star$ (document 3) and the parabolic-degree$\to C_6$ dictionary. H2 is the sharp realisation statement: the D_4-equivariant stable structure on $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ realises the branch with $\det R = h(D_5) = 8$, principal 2-minors $(2, 3, 5)$. [E] (given (U))
3. **Strong CP / dynamics via the explicit gap.** $\theta_{\rm eff} = 0$ needs only γ_5-Hermiticity + polar mass + sheet involution + reflection positivity (no mass gap). The full dynamics closes on the physical sector because the C_6 transport transfer operator has the *explicit* spectrum $\{(k/3)^6\}$, giving the mass gap $\Delta = 6 \log \frac{3}{2} = 2.43$ in closed form $\Rightarrow$ OS reconstruction $\Rightarrow$ measure on the admissible sector. [E] (phys. sector)
4. **$E_6 \times A_2$ flavor shadow.** The residue matrix sits inside an E_8 secondary branching: $248 = \|R\|_F^2 + \det R + 2(\mathbf{1}^\top Ra) N_{\rm fam} = 78 + 8 + 2 \cdot 27 \cdot 3$, with $\|R\|_F^2 = 78 = \dim E_6$, $\det R = 8 = \dim A_2$. The seven E_8 slices form an audit raster (atlas note). [O]
5. **SM completeness.** The discrete structure and dimensionless core are closed; the dimensionful EW/QCD masses $(m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s)$ are schema-level via $\mathcal{R}_{\rm SM}$, with m_H resting on the closed UV quartic $\lambda_\Phi = 1/(16\pi^2)$ (near-criticality). The *exact* solar angle θ_{12} (leading $1/N_{\rm fam} = \frac{1}{3}$) is *conditionally derived*, pending the seam-misalignment lemma — the sharpest remaining SM tension point. [C]

16 Status summary and open audit points

Machine checks for this section: [verification/v22_open_gates_audit.py]

Statement	Status	Note
Dual-root dictionary (Y, cusps, $\gamma_{\rm car}$, $\varphi_{\rm base}$)	[E]	exact, from $\{1/2, -1/3\}$
$10b_1 = 5^2 + 4^2 = \Omega_{\rm adm}\gamma + 1$	[E]	Pythagorean carrier identity
$I_1^{\rm EW} = 41/48 = {\rm Var}(Y)\, b_1$	[E]	exact
$\delta_2 = 4! \cdot 5!\, c_3^8$, ${\rm Tr}\, X^2 = 5!$	[E]	factorial traces
$5/4 = g_{\rm car}/(g_{\rm car} - 1)$ universal compression	[E]	exact
$u = \frac{4}{3}c_3 + 48c_3^4$; one-decoder identities	[E]	exact
$\xi = \frac{3}{4}/(1 + 9/128\pi^3)$	[E]	corrected tree value
Ladder $A_{\rm EW} : \mathcal{A}_H : \mathcal{A}_\Lambda = 1 : g_{\rm car} : 2g_{\rm car}$	[E]/[C]	Hubble rung is a corollary
Ladder $= \binom{g_{\rm car}}{0} : \binom{g_{\rm car}}{1} : \binom{g_{\rm car}}{2}$; $2g_{\rm car} = \binom{g_{\rm car}}{2} \Leftrightarrow g_{\rm car} = 5$	[E]	new overdetermination of $g_{\rm car} = 5$
2D action lattice $S_x = a(\alpha_\star^{-1}/g_{\rm car}) + bL_0$	[E]/[C]	two integer labels per scale
Decuple bridge $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = \frac{3}{4\pi^2}e^{2\delta_{\rm ph}}[v/(5\beta_{\rm rad}^2\bar{M}_{\rm Pl})]^{10}$	[E]	exact, $10 = 2g_{\rm car} = \dim \Lambda^2 E$
Family = ladder = Pascal row $\{1, 5, 10\}$; $16 = 1{+}5{+}10$	[E]	central new identification
$g_{\rm car} = 5$ via $2^{g_{\rm car}-1} = \binom{g_{\rm car}}{0}{+}\binom{g_{\rm car}}{1}{+}\binom{g_{\rm car}}{2}$	[E]	Pascal closure (unique)
${\rm Tr}_{S^+}\, X^2 = 5!$ generates $40, 41, 240, \delta_2$	[E]	master moment
$\lvert R(E_8)\rvert = 16 \cdot 5 \cdot 3$; $\dim E_8 = 240 + 8$	[E]	E_8 compression
Even-flip atlas $240 = 160 + 80$	[E]/[O]	metrization open ($\neq 112 + 128$); Coxeter Stage A closed on the character space (v531)
$\delta_{\rm CKM} - \pi/3 = 3u(1 - u)$	[E]	CP = triple seed variance
Observable quadrilateral ($u = \Omega_b + \beta_{\rm rad}$)	[E]	closed UV-shadow cycle
Power tower $v{=}x^1$, $H{=}x^5R_H$, $\Lambda{=}x^{10}R_\Lambda$	[E]	K_5 vertices/edges
$496 = 16 \cdot 31 = \dim(E_8{\times}E_8) = \dim SO(32)$	[E]/[C]	heterotic-dimension count
$A_{\rm EW} = \alpha_\star^{-1}/\sqrt{\Delta_Y}$, $\Delta_Y = (b{+}s)^2 = 25$	[E]	discriminant action theorem
$\hat{m}_e\hat{m}_\mu\hat{m}_\tau \propto \frac{32}{9}u^{10}$; $(5, 3, 2) = (g_{\rm car}, b, s)$	[E]	lepton flavor decuple
$M_{\rm scal}/\bar{M}_{\rm Pl} = c_3^{7/2}$	[E]	scalaron seam power
$\delta_{\rm CKM} - \pi/3 = N_{\rm fam}\, u(1 - u)$	[E]	weak CP = $N_{\rm fam}\times$ seed variance
$L = R + W$: $\sum R = 22$, $\sum W = 18$	[E]	first generation = residue store
$\lvert R(E_8)\rvert/6 = \sum L = 40$ ($\mathbb{Z}_6$ slice)	[E]	one $\mathbb{Z}_6$ slice = flavor budget
$h_{E_8} = \lvert R\rvert/\,{\rm rank} = 30 = N_{\rm fam}\binom{5}{2} = N_{\rm fam}A_\Lambda$	[E]/[C]	carrier Coxeter theorem
$\lvert R^+(E_8)\rvert = {\rm Tr}_{S^+}\, X^2 = 120 = 5!$	[E]	positive-root moment
common decuple $\binom{5}{2}{=}A_\Lambda{=}u^{5+3+2}{=}h_{E_8}/N_{\rm fam}{=}10$	[E]/[C]	one carrier pair-decuple
$\lambda_C^6/\delta_{\rm top} = 1.061$ (winding $\approx$ seam defect)	[O]	flagged proximity (not exact)
$y_f \simeq \lambda_C^{L_f}$, top $L_t = 0 \Rightarrow y_t \simeq 1$	[E]/[C]	Froggatt–Nielsen ladder
$w = -1$ (true Λ); $N_{\rm eff} \simeq 3$	[E]	dark energy / neutrino species
3-generator lattice $\{\alpha_\star^{-1}, L_0, L_c\}$; $M_{\rm scal} = e^{-\frac{7}{2}L_c}$	[E]	seam axis $L_c = \ln(1/c_3)$
$\dim S^+ = 2\,{\rm rank}(E_8)$, $h_{E_8} = 2(\dim S^+{-}1) = 2^g{-}2$	[E]	rank–Coxeter coupling (ν^c closes it)
$\gcd(2h, 6r) = \dim \mathfrak{g}_{\rm SM} = 12$, ${\rm lcm} = 240$	[E]/[C]	SM algebra = gcd; E_8 count = lcm
$I_1^{\rm EW} = h_{E_8}/36 + 1/(6r) = 41/48$	[E]/[C]	Coxeter part + rank singlet
$M_{\rm scal}^2/\bar{M}_{\rm Pl}^2 = c_3^{r-1} = c_3^7$ exact; $A_s = N_\star^2c_3^7/24\pi^2$	[E]/[C]/[O]	seam power exact; A_s conditional (Starobinsky)
$h_{E_8} = 2{\cdot}3{\cdot}5 = 30$, $r = \varphi(30) = 8$	[E]/[C]	$2, 3, 5$ Coxeter cyclotomic compiler
${\rm Exp}(E_8) = (\mathbb{Z}/30)^\times$, $\sum = 120$, $\chi_C = \Phi_{30}$	[E]/[C]	exponents = primitive residues
lepton $(5, 3, 2)$: $\sum{=}10{=}A_\Lambda$, $\prod{=}30{=}h_{E_8}$	[E]/[C]	sum=action, product=Coxeter
Hubble $H_0/\bar{M}_{\rm Pl} = e^{-\alpha_\star^{-1}}/(2\pi\sqrt{\Omega_\Lambda})$	[E]	exact corollary

Λ-anchored closure $G_N = \frac{3c^3}{32\pi^3\hbar\Lambda_{\rm geom}}e^{-2\alpha_\star^{-1}}$	[E]/[C]	G_N to -0.35%
$g_{\rm car} = 5$ overdetermined	[C]	monotone $\alpha^{-1}(g_{\rm car})$
$\lvert R(E_8)\rvert = 240 = 16\cdot 15$, $248 = 8\cdot 31$	[E]	counting identities
$E_8 \leftrightarrow$ carrier-transition map	[C]	needs structure map
$\delta_{\rm CP} = 4\pi/3$, $\theta_{12} = \frac{1}{3} - \frac{u}{2}$	[C]	PMNS readouts
$\lvert V_{td}\rvert, \lvert V_{ts}\rvert, J_{\rm CKM}, J^{\nu}_{\rm CP}$; n_s, r	[C]	derived readouts

Open audit points: (1) *discharged* — the EM closure $F_{U(1)}(\alpha) = 0$ is now stated explicitly (Grammar III), with existence/uniqueness and root $\alpha_\star^{-1} = 137.0359992168$ (2.9×10^{-10}, 1.9σ vs CODATA-2022); (2) keep $\gamma_{\rm car} = 5/6$ vs γ_E distinct; (3) type $M_{\rm Pl}$ vs $\bar M_{\rm Pl}$ and "pure" ($\mathcal{A}_\Lambda = 274.07$) vs "total" ($-\ln(\delta_{\rm top}e^{-2\alpha_\star^{-1}}) = 283.10$, $-\ln(\rho_\Lambda/\bar M_{\rm Pl}^4) = 276.65$) actions consistently; (4) version the Λ prefactor branch; (5) the scalaron-mass→ Starobinsky-M identification is the *only* condition for $A_s = N_\star^2 c_3^7/(24\pi^2)$ (the seam power $M_{\rm scal}^2/\bar M_{\rm Pl}^2 = c_3^7$ itself is exact); (6) the E_8 *lattice* is now built explicitly as $(D_5 \oplus A_3) + \mu_4$-glue (Section 3), so the former "blind 240-metrisation" is closed. The flavor status (companion parabolic-weight note) now splits cleanly: the *anchor multiset* $\{1, 1, 2\}$ is derived (Schur–Horn + stability) [E]; the *sector assignment* $(1, 1, 2)$ follows from the D_5 cusp ordering [E]/[C]; the *first-generation* $+6$ *winding* is the hierarchy convention [C]; the six *non-anchor residues* are then forced (residue sets $= D_6$ orbit of $\{0, 1, 3\}$, minus anchor, monotone) [E]. The remaining geometric input at the time of writing was (H2), the specific D_6 cusp labelling on $\mathbb{P}^1 \setminus \mu_4$ — sharply characterised by the residue-matrix spectral selector ($\det R = h(D_5)$, principal minors $(2, 3, 5)$), with a naive cusp-position shortcut explicitly ruled out [C]. The explicit Riemann–Hilbert construction confirms the rigid $SU(3)$ system and its A_3 Coxeter rotation. *Status update:* the H2 dictionary has since been made *exact* (hexagon = sign-twisted family spectrum, addresses pinned, margins forced, v118–v122), the deck is *derived* (v141), and the residue is the R4′ selector establishment — two line pairings (v139/v142); what stays geometric is only the parabolic realisation premise (`GATE.QGEO`, research contracts).

Honest negatives (reported to keep the discipline visible). The TFPT lepton ladder gives a Koide ratio $Q = (\sum m_\ell)/(\sum\sqrt{m_\ell})^2 = 0.6645$, close to but *not* the empirical $2/3 = 0.66667$ (-0.33%); TFPT is Koide-*like* but does not reproduce the precise relation. The baryon asymmetry $\eta_B \approx 6\times 10^{-10}$ and the proton/electron ratio $m_p/m_e = 1836$ do *not* fall on a clean action charge or L_0 ladder; they remain outside the present grammar. These are stated so the strong items are not read as a uniform success.

17 Consolidated picture

The three nested grammars.

$$\begin{aligned}
\textbf{Carrier:}\quad & 3+2 \Rightarrow \{y_-, y_+\} = \{-\tfrac{1}{3}, \tfrac{1}{2}\} \Rightarrow Y,\ \dim S^+ = 16,\ N_{\rm fam} = 3,\ \Omega_{\rm adm} = 48,\ b_1 = \tfrac{41}{10},\\
\textbf{Seed:}\quad & u = \tfrac{4}{3}c_3 + 48c_3^4 \Rightarrow \{\lambda_C,\ \beta_{\rm rad},\ \Omega_b,\ \sin^2\theta_{13}\},\\
\textbf{Action:}\quad & \alpha_\star^{-1} \Rightarrow A_{\rm EW} = \tfrac{\alpha_\star^{-1}}{g_{\rm car}},\ \mathcal{A}_\Lambda = 2\alpha_\star^{-1},\ \mathcal{A}_H = \alpha_\star^{-1} \Rightarrow \frac{\rho_\Lambda}{\bar M_{\rm Pl}^4} = \frac{3}{4\pi^2}e^{2\delta_{\rm ph}}\Big[\frac{v_{\rm geo}}{5\beta_{\rm rad}^2\bar M_{\rm Pl}}\Big]^{10}.
\end{aligned}$$

A small grammar generates many sectors: the Standard-Model packet, flavor, the electroweak hierarchy, the cosmological constant, the metrology, and the inflation/CMB interface. The strongest new statements are exact: the decuple bridge ties Λ to the tenth power of the stripped electroweak scale with $10 = 2g_{\rm car}$; the Hubble scale is the square root of Λ; and the Einstein normaliser is the corrected tree value $\frac{3}{4}/(1 + 9/128\pi^3)$. None of this is claimed as a proof of a final theory; it is offered as a tightening of the existing architecture for your assessment.

The compiler in one line: TFPT = even carrier code + seed decoder + 2·3·5 Coxeter compiler. From the three atoms $(2, 3, g_{\text{car}}{=}5)$:

$$h_{E_8} = 2{\cdot}3{\cdot}5 = 30, \qquad r_{E_8} = \varphi(30) = 8 = g_{\text{car}}{+}N_{\text{fam}},$$

$$|R(E_8)| = rh = 240, \qquad \dim E_8 = r(h{+}1) = 8{\cdot}31 = 248,$$

$$\operatorname{Exp}(E_8) = (\mathbb{Z}/30)^{\times}, \quad \textstyle\sum \operatorname{Exp} = 120 = \operatorname{Tr}_{S^+} X^2 = |R^+|, \quad \chi_C = \Phi_{30},$$

$$\dim S^+ = 1 + 10 + 5 = 16 = 2r, \quad A_{\text{EW}}{:}\mathcal{A}_H{:}\mathcal{A}_\Lambda = 1{:}5{:}10, \quad \mathcal{A}_H = \tfrac{h}{6},\ \mathcal{A}_\Lambda = \tfrac{h}{3},\ N_{\text{fam}} = \tfrac{h}{A_\Lambda} = 3,$$

$$\textstyle\sum L = \frac{|R^+|}{3} = 40, \quad 10b_1 = 1 + \frac{|R^+|}{3} = 41 = 1 + h\frac{\dim S^+}{\dim \mathfrak{g}_{\text{SM}}},$$

$$\det M_\ell \propto u^{10}, \qquad \frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} \propto \left[\frac{v_{\text{geo}}}{5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}}}\right]^{10}, \qquad A_{\text{EW}} = \frac{\alpha_\star^{-1}}{\sqrt{\Delta_Y}}, \quad \sqrt{\Delta_Y} = g_{\text{car}} = 5.$$

The common decuple (= 10): $\binom{g_{\text{car}}}{2}$ (carrier pairs) = A_Λ (cosmological action) = u^{5+3+2} (lepton determinant) = h_{E_8}/N_{fam} (Coxeter per family). Λ, the charged-lepton determinant, the E_8 Coxeter structure and the action ladder all read the *same* carrier pair-decuple. The open core is one question: can the even-flip atlas be metrised as a genuine E_8 root system (the 160+80 vs 112+128 obstruction)? Its Coxeter half (Stage A) is closed on the primitive-character space by the tensor clock $T_{30} = \operatorname{Comp}(\Phi_5) \otimes \operatorname{Comp}(\Phi_6)$ (`v531`); the atlas intertwiner and the metrization (Stage B) remain.

Part II
The Coxeter–cyclotomic compiler and the explicit E_8 construction

Abstract

This short companion compresses the TFPT bridge layer to a single discrete object. The claim is that the Standard-Model family, the cosmological action ladder, the E_8 counting data, the flavor budget, inflation and the cosmological constant are *not* independent coincidences but projections of one structure: the even-weight code C^+ on $g_{\text{car}} = 5$ slots, together with its $\mathbb{Z}_{30}$ Coxeter spectrum and the single seam seed $c_3 = 1/(8\pi)$. The decisive new result is that C^+ is precisely the half-spin weight orbit of $D_5 = \mathfrak{so}_{10}$ ($\text{Aut}(C^+) = W(D_5)$, order 1920), that the family geometry $\mathbb{P}^1 \setminus \mu_4$ is an $A_3 = \mathfrak{su}_4$, and that E_8 is then the standard *closure algebra* of $D_5 \times A_3$ — not an input. The fusion is exact: $\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4$, so the four family corners μ_4 are precisely the glue code that closes $D_5 \oplus A_3$ into the unimodular E_8 lattice. The glue is metric: the discriminant-form norms $q(D_5) = \frac{5}{4}$ and $q(A_3) = \frac{3}{4}$ sum to the E_8 root norm 2 — and these are exactly the pre-existing TFPT constants $\delta_2/\delta_{\text{top}}^2 = \frac{5}{4}$ and $\xi_{\text{tree}} = \frac{3}{4}$. The A_3 side is explicit: the residue pairing on $H^1(\mathbb{P}^1 \setminus \mu_4)$ is the A_3 Cartan form, and the carrier corner rotation $z \mapsto iz$ acts as the A_3 Coxeter element (order 4, spectrum $\{i, -1, -i\}$). The lift is no longer a count: the 240 E_8 roots are *explicitly* built from $D_5 \oplus A_3$ plus the spinor (μ_4) glue, all of norm 2, Cartan determinant 1. Seven statements carry it; every lattice identity below is checked exactly, and the analytic metric is computed and characterised (a D_4-equivariant deformation), so no *lattice* structural step is left open in the $D_5 \times A_3$ glue route (the upstream boundary kernel, EM closure and flavor Hecke saturation remain separate open problems). Master statement: *five carrier slots give D_5, four family corners give A_3, and μ_4 glues $D_5 \times A_3$ into the E_8 atlas.*

The compression.

$$\text{TFPT} = \underbrace{D_5 \text{ half-spin carrier}}_{C^+} + \underbrace{A_3 \text{ four-point family}}_{\mathbb{P}^1 \setminus \mu_4} + c_3 \text{ seam seed},$$

$$E_8 = \text{closure of } D_5 \times A_3$$

with $c_3 = \dfrac{1}{8\pi}$, $g_{\text{car}} = 5$, $h := 2 \cdot 3 \cdot g_{\text{car}} = 30$, $r := \varphi(h) = \varphi(30) = 8$, $N_{\text{fam}} = 3$, $N_\Phi = 1$, $\dim \mathfrak{g}_{\text{SM}} = 12$. Equivalently $C^+ + \mathbb{Z}_{30}$ Coxeter compiler $+\ c_3$ seed; the $D_5 \times A_3$ reading is the structural lift of the same object.

Theorem 1 — the even carrier code

Let E be the rank-$g_{\text{car}} = 5$ carrier, split $E = E_3 \oplus E_2$ (color $b = 3$, weak $s = 2$). The one-generation spinor packet is the even exterior algebra, equivalently the even-weight binary code

$$S^+ = \Lambda^0 E \oplus \Lambda^2 E \oplus \Lambda^4 E \;\cong\; C^+ = \{x \in \mathbb{F}_2^5 : |x| \text{ even}\}, \qquad \dim S^+ = |C^+| = 2^{g_{\text{car}}-1} = 16. \quad (69)$$

The two carrier roots $y_+ = \frac{1}{2}$, $y_- = -\frac{1}{3}$ generate all hypercharges ($Y(Q_L) = y_+ + y_- = \frac{1}{6}$, $Y(u^c) = 2y_-$, $Y(e^c) = 2y_+$, $Y(\nu^c) = 0$) and the transport cusps $\{1, \frac{2}{3}, \frac{1}{3}\} = \{2y_+, -2y_-, 2(y_+ + y_-)\}$, with $\gamma_{\text{car}} = \text{Tr}_E\, Y^2 = 3y_-^2 + 2y_+^2 = \frac{5}{6}$. **[E]**

Gauge-rank factorisation (`[verification/v79_review_identities.py]` **).** The carrier polynomial factors into the two gauge ranks directly:

$$\boxed{6Y^2 - Y - 1 = (2Y - 1)(3Y + 1)}, \qquad Y = \tfrac{1}{2}\ (s{=}2,\ SU(2)), \quad Y = -\tfrac{1}{3}\ (b{=}3,\ SU(3)),$$

i.e. $(sY - 1)(bY + 1)$ — the polynomial splits into one linear representative of each Standard-Model non-abelian factor; the 5-slot carrier *forces* the split into $SU(3) \times SU(2)$.

Hypercharge Lucas–Binet resonances [E] (mode reading [C], [verification/v79_review_identities.py])

The integer carrier charges $(q_+, q_-) = (3, -2)$ (the primitive trace-free pair, $X^2 - X - 6 = (X-3)(X+2)$, Lean `unique_carrier_pair`) generate the Lucas U-sequence

$$D_n = \frac{3^n - (-2)^n}{5} : \qquad D_1, \ldots, D_6 = 1,\ 1,\ 7,\ 13,\ 55,\ 133,$$

which is, *in order*, $(N_\Phi,\ \cdot\ ,$ scalaron exponent 7, $\Delta_Q{=}13$, quark numerator 55, $\dim E_7{=}133)$. In particular the quark ratio reads $c_u/c_d = D_5/(N_{\text{fam}}^2 D_4) = \frac{55}{117}$ — a single recurrence of the hypercharge generator, not a free integer. [E] for the arithmetic; the per-mode physical identification is a [C] cross-domain reading (only $n \le 6$, the carrier-slot range, is meaningful).

Theorem 2 — Exterior–Action Duality

On C^+ the flip $F = x \oplus y$ is even, so $|F| \in \{0, 2, 4\}$. The three relations R_0, R_2, R_4 form a symmetric association scheme (the even Hamming scheme on five bits) with

$$\boxed{\begin{array}{l}\text{valencies } (1,10,5) = (\dim \Lambda^0 E, \dim \Lambda^2 E, \dim \Lambda^4 E),\\ \text{spectral multiplicities } (1,5,10) = A_{\text{EW}}{:}\mathcal{A}_H{:}\mathcal{A}_\Lambda\end{array}} \qquad [\mathbf{E}]. \tag{70}$$

The eigenvalues (computed directly) are $A_2 : \{10^{(1)}, 2^{(5)}, (-2)^{(10)}\}$ and $A_4 : \{5^{(1)}, (-3)^{(5)}, 1^{(10)}\}$, with $A_2 + A_4 = K_{16} - \mathbf{1}$. The two graphs are named objects:

$$\begin{gathered} R_2(C^+) = \text{folded/complement, SRG}(16, 10, 6, 6) \text{ (the Clebsch complement);} \\ \boxed{R_4(C^+) = \text{Clebsch graph, SRG}(16, 5, 0, 2)} \qquad [\mathbf{E}]. \end{gathered} \tag{71}$$

Remark. The Standard-Model family is the *valency* side $(1, 10, 5)$ of one scheme; the cosmological action ladder is its *spectral* side $(1, 5, 10)$. The action ladder is therefore not merely "also binomial" — it is the spectral dual of the even exterior family. [C]

This duality is the MacWilliams transform ([verification/v79_review_identities.py]). C^+ is the even-weight code $[5, 4, 2]$; its dual $(C^+)^\perp$ is the repetition code $[5, 1, 5] = \{00000, 11111\}$ (the uniform/vacuum word). The MacWilliams identity sends the weight enumerator of the dual to that of C^+, namely $(1, 10, 5)$ — so the valency↔spectral duality *is* the coding-theoretic UV/IR duality, and the carrier grading $\Lambda^{\text{even}}(5) = 1{+}10{+}5$ is the MacWilliams image of the repetition vacuum. [E] (code identity) / [C] (holographic reading).
The full Bose–Mesner algebra (machine-checkable) fixes the duality beyond eigenvalues:

$$A_2^2 = 10\,\mathbf{1} + 6A_2 + 6A_4, \quad A_2 A_4 = 3A_2 + 4A_4, \quad A_4^2 = 5\,\mathbf{1} + 2A_2, \tag{72}$$

with eigenmatrix and multiplicity orthogonality

$$P = \begin{pmatrix} 1 & 10 & 5 \\ 1 & 2 & -3 \\ 1 & -2 & 1 \end{pmatrix}, \quad \det P = -64, \qquad \boxed{P^\top \operatorname{diag}(1, 5, 10)\, P = \operatorname{diag}(16, 160, 80)} \qquad [\mathbf{E}], \tag{73}$$

so the flip split $240 = 160{+}80$ ($|F| = 2, 4$) appears directly in the scheme's eigenmatrix orthogonality.
Carrier $= D_5$ (verified). The 16 vertices of C^+ are exactly the weights of the $D_5 = \mathfrak{so}_{10}$ half-spinor $\frac{1}{2}(\pm 1, \ldots, \pm 1)$ with an even number of minus signs. The flip-2 and flip-4 graphs are translation- and permutation-invariant, so the *affine* group $C^+ \rtimes S_5$ (translations $\rtimes$ coordinate permutations) acts by graph/scheme automorphisms; the group closure gives the automorphism group of the even Hamming scheme (equivalently the Clebsch graph),

$$\boxed{\mathrm{Aut}_{\text{aff}}(C^+) = C^+ \rtimes S_5, \qquad |\mathrm{Aut}_{\text{aff}}(C^+)| = 2^{g_{\text{car}}-1}\, g_{\text{car}}! = 16 \cdot 120 = 1920 = |W(D_5)|} \qquad [\mathbf{E}], \tag{74}$$

and $1920 = r\,|R(E_8)| = 8 \cdot 240$. (The *linear* code automorphism group alone is only the coordinate group S_5 of order 120; the factor $2^{g_{\rm car}-1} = 16$ is the translation/scheme part. It is the affine/scheme group that equals $W(D_5)$.) Edges of the Clebsch graph: $|E(R_4)| = \frac{16\cdot 5}{2} = 40 = |R(D_5)| = \sum_{f,j} L_{f,j}$ (the entire flavor transport budget). So the five-slot even code is the half-spin weight structure of $\mathfrak{so}_{10}$, the classical home of one SM family with ν^c.

Theorem 3 — the Coxeter closure equation selects $g_{\rm car} = 5$

Requiring the even family to equal the action truncation through pair degree, $\dim S^+ = 2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$, i.e.

$$\boxed{2^g - 2 = g(g+1) \iff g = 5} \qquad [\mathbf{E}], \tag{75}$$

the unique positive-integer solution. At $g = 5$ the two sides equal 30, so *carrier exhaustion* $2^g - 2$ and the *Coxeter period* $g(g+1)$ coincide only at five slots: $2^{g_{\rm car}} - 2 = 30 = h$ and $g_{\rm car}(g_{\rm car}+1) = 30$.

Theorem 4 — the $\mathbb{Z}_{30}$ cyclotomic Coxeter compiler

From the three atoms $(2, 3, g_{\rm car})$, with $h = 2 \cdot 3 \cdot g_{\rm car} = 30$ and $r = \varphi(h) = 8$:

$$\boxed{|R(E_8)| = rh = 240, \qquad \dim E_8 = r(h+1) = 8 \cdot 31 = 248, \qquad h + 1 = 2^{g_{\rm car}} - 1 = 31} \qquad [\mathbf{E}]. \tag{76}$$

The E_8 exponents are the primitive residues modulo 30,

$$\mathrm{Exp}(E_8) = (\mathbb{Z}/30\mathbb{Z})^\times = \{1, 7, 11, 13, 17, 19, 23, 29\}, \qquad \chi_C(t) = \Phi_{30}(t) = t^8 + t^7 - t^5 - t^4 - t^3 + t + 1, \tag{77}$$

and $|(\mathbb{Z}/30)^\times| = \varphi(30) = 8 = r$. The local SM algebra and the root atlas are the two arithmetic projections of $(r, h) = (8, 30)$:

$$\Omega_{\rm adm} = 6r = 48, \quad D_{\rm start} = 2h = 60, \quad \gcd(6r, 2h) = \dim \mathfrak{g}_{\rm SM} = 12, \quad \mathrm{lcm}(6r, 2h) = rh = 240. \tag{78}$$

Theorem 5 — the root transition atlas and the positive-edge moment

The directed even-flips of C^+ are the complete graph K_{16}:

$$\boxed{\begin{gathered} |R^+(E_8)| = \binom{\dim S^+}{2} = \binom{16}{2} = 120 = \mathrm{Tr}_{S^+} X^2 = 5! = \sum_{m \in (\mathbb{Z}/30)^\times} m, \\ |R(E_8)| = 2\binom{16}{2} = 240 \end{gathered}} \qquad [\mathbf{E}]. \tag{79}$$

$(X = 6Y.)$ So the positive E_8 roots are the unordered transitions of one family; the second hypercharge moment, the positive-root count and the sum of the E_8 exponents coincide at 120. The CP pairing $m \leftrightarrow 30 - m$ of the exponents gives four pairs, each of sum $h = 30$:

$$\boxed{\mathrm{Tr}_{S^+} X^2 = (N_{\rm fam} + N_\Phi)\, h = 4 \cdot 30 = 120 = \dim \mathfrak{so}_{16} = \underbrace{45}_{D_5} + \underbrace{15}_{A_3} + \underbrace{60}_{(10,6)}} \qquad [\mathbf{E}], \tag{80}$$

so the hypercharge moment is simultaneously four CP-paired Coxeter periods *and* the adjoint half $\mathfrak{so}_{16} = (45, 1) \oplus (1, 15) \oplus (10, 6)$ of the E_8 closure.

How the root *origin* is now obtained — see Theorem 7. A naive flip-size-preserving map $R_{\rm flip} \to \mathbb{R}^8$ cannot be E_8: the flip split $240 = 160 + 80$ ($|F| = 2, 4$) differs from the D_8

split $112 + 128$. The correct route is *not* a direct 240-metrization but the standard branching $E_8 \supset D_5 \times A_3$ (Theorem 7), which builds the root spaces from known D_5 and A_3 data. Until the A_3 lattice is derived from the four punctures (Theorem 7, [O]), E_8 remains a Coxeter atlas with a concrete, falsifiable lift.

Theorem 6 — downstream readouts from (h, r, c_3)

All discrete budgets and the seam seed are functions of $(h, r, c_3) = (30, 8, 1/(8\pi))$:

$$\dim S^+ = 2r = 16, \qquad h = 2(\dim S^+ - 1) = 30, \tag{81}$$

$$10b_1 = 5r + 1 = 41 = 1 + h\tfrac{\dim S^+}{\dim \mathfrak{g}_{\mathrm{SM}}}, \qquad I_1^{\mathrm{EW}} = \frac{5r+1}{6r} = \frac{41}{48}, \tag{82}$$

$$u = \frac{r}{6}c_3 + 6r\,c_3^4 = \frac{4}{3}c_3 + 48c_3^4, \qquad \delta_2 = 4!\tbinom{16}{2}c_3^8 = 2880\,c_3^8, \tag{83}$$

$$\xi_{\mathrm{tree}} = \frac{6}{r} = \frac{\dim \mathfrak{g}_{\mathrm{SM}}}{\dim S^+} = \frac{3}{4}, \qquad \frac{M_{\mathrm{scal}}^2}{\bar{M}_{\mathrm{Pl}}^2} = c_3^{\Omega_{\mathrm{adm}} - 10b_1} = c_3^{r-1} = c_3^7. \tag{84}$$

Two-graph product table. With the carrier-pair graph K_5 ($|E(K_5)| = \binom{5}{2} = 10 = \mathcal{A}_\Lambda$) and the family-corner graph K_4 on μ_4 ($|E(K_4)| = \binom{4}{2} = 6$, $|R(A_3)| = 2|E(K_4)| = 12$), every large budget is a product of the two graphs:

$$\boxed{40 = 10 \cdot 4, \quad 60 = 10 \cdot 6, \quad 120 = 10 \cdot 12, \quad 240 = 10 \cdot 24} \qquad [\mathrm{E}], \tag{85}$$

i.e. $(\sum L,\ D_{\mathrm{start}},\ |R^+(E_8)|,\ |R(E_8)|) = 10 \cdot (|\mu_4|, |E(K_4)|, |R(A_3)|, |S_4|)$. Two occupancy identities follow,

$$\begin{gathered} \boxed{10b_1 = 1 + |\mu_4|\,|E(K_5)| = 1 + 4 \cdot 10 = 41}, \\ \boxed{\Omega_{\mathrm{adm}} = N_{\mathrm{fam}} \dim S^+ = (N_{\mathrm{fam}} + N_\Phi)|R(A_3)| = 3 \cdot 16 = 4 \cdot 12 = 48} \qquad [\mathrm{E}], \end{gathered} \tag{86}$$

so the Higgs singleton is the fourth (μ_4) direction that closes $N_{\mathrm{fam}} = 3$ to **4** of A_3. **Occupancy deficit ladder.** The admissible occupancy splits cleanly,

$$\boxed{\begin{gathered} \Omega_{\mathrm{adm}} = \underbrace{40}_{\sum L\text{ (flavor)}} + \underbrace{1}_{\text{Higgs}} + \underbrace{7}_{\text{scalaron}} = 48, \qquad \Omega_{\mathrm{adm}} - \sum L = 8 = h(D_5) = r(E_8), \\ \Omega_{\mathrm{adm}} - 10b_1 = 7 = h(D_5) - 1 \end{gathered}} \qquad [\mathrm{E}], \tag{87}$$

so the scalaron exponent $r - 1 = 7$ is exactly the post-flavor, post-Higgs occupancy remainder, $M_{\mathrm{scal}}^2/\bar{M}_{\mathrm{Pl}}^2 = c_3^{\Omega_{\mathrm{adm}} - 10b_1} = c_3^7$. With $h(D_5) = 8$ and $|R^+(A_3)| = 6$ the seed itself is a D_5/A_3 ratio–product, $u = \frac{h(D_5)}{|R^+(A_3)|}c_3 + h(D_5)|R^+(A_3)|c_3^4$, and $\xi = \frac{|R^+(A_3)|}{h(D_5)}\,(1 + |R^+(A_3)|^2 c_3^3)^{-1} = \frac{3}{4}(1 + 36c_3^3)^{-1}$. The seed readouts and the two decuples are

$$\lambda_C^2 = u(1-u), \quad \beta_{\mathrm{rad}} = 2c_3 u, \quad \sin^2\theta_{13} = e^{-5/6}u, \quad \boxed{\omega_b = \frac{\lambda_C}{\mathcal{A}_\Lambda} = \frac{\lambda_C}{10} = 0.02244}\ (\text{Planck } 0.02237), \tag{88}$$

$$\boxed{\begin{aligned} &\det M_\ell \propto u^{h/N_{\mathrm{fam}}} = u^{10}, \\ &\frac{\rho_\Lambda}{\bar{M}_{\mathrm{Pl}}^4} = \frac{3}{4\pi^2}e^{2\delta_{\mathrm{ph}}}\left[\frac{v_{\mathrm{geo}}}{5\beta_{\mathrm{rad}}^2 \bar{M}_{\mathrm{Pl}}}\right]^{h/N_{\mathrm{fam}}} \end{aligned}}, \tag{89}$$

with lepton powers $(5, 3, 2) = (g_{\mathrm{car}}, b, s)$: $\sum = 10 = \mathcal{A}_\Lambda$ (the Λ decuple), $\prod = 30 = h$ (the Coxeter period). The first-generation transport sums read the non-primitive $\mathbb{Z}_{30}$ load,

$$(G_1, G_2, G_3) = (22, 13, 5) = (h - r,\ \dim \mathfrak{g}_{\mathrm{SM}} + 1,\ g_{\mathrm{car}}), \qquad h - r = 30 - \varphi(30) = 22, \tag{90}$$

Flavor transport is a $D_5\times A_3$ object. The transport packets $L_u=(7,3,0), L_d=(7,5,2), L_e=(8,5,3)$ split into a D_5 budget and an A_3 geometry:

$$\boxed{\sum_{f,j} L_{f,j} = 40 = |R(D_5)| \text{ (Clebsch edges)}, \qquad L_{f,j} \bmod 6 \text{ lives on } C_6 = Z_6 = |R^+(A_3)|} \quad \textbf{[E]}, \tag{91}$$

the residues being the D_6-orbit of the perfect set $\{0,1,3\} \subset C_6$ (cyclic distances $\{1,2,3\}$): $u\equiv\{0,1,3\}$, $d\equiv2-\{0,1,3\}$, $e\equiv2+\{0,1,3\}$, with the universal first-generation $+6$ winding. So the carrier D_5 sets the total budget and the family A_3 (its 6-element positive-root hexagon) sets the per-entry distances. The scalaron mass is a pure seam power **[E]**; the amplitude is a *conditional* readout, kept **[C]**/**[O]** until the scalaron variable is formally identified with the Starobinsky M parameter. That identification is now *heat-kernel grounded* (G2): the spectral-action Seeley–DeWitt coefficient $a_4|_{R^2}\sim R^2/72$ delivers exactly the Starobinsky R^2 term, with $M^2/\bar{M}_{\rm Pl}^2 = 6(4\pi)^2/f_0$ and the TFPT closure c_3^7 fixing the cutoff moment f_0 ([`verification/v36_spectral_action_g2.py`]):

$$\underbrace{\frac{M_{\rm scal}^2}{\bar{M}_{\rm Pl}^2} = c_3^{r-1} = c_3^7,}_{\textbf{[E]}} \qquad \underbrace{A_s = \frac{N_\star^2}{24\pi^2}\frac{M_{\rm scal}^2}{\bar{M}_{\rm Pl}^2} = \frac{N_\star^2 c_3^7}{24\pi^2} = 2.17\times10^{-9}}_{\textbf{[C]}/\textbf{[O]} \text{ (if } M_{\rm scal}=M_{\rm Star})}. \tag{92}$$

Theorem 7 — the $D_5\times A_3$ lift of the E_8 atlas

Since $A_3 = \mathfrak{su}_4 \cong \mathfrak{so}_6 = D_3$, one has the standard chain $E_8 \supset \mathfrak{so}_{16} \supset \mathfrak{so}_{10}\oplus\mathfrak{so}_6 = D_5\times A_3$, with the exact branching (verified dimensions and root/weight counts):

$$\begin{gathered}\boxed{\mathfrak{e}_8 = (45,1)\oplus(1,15)\oplus(10,6)\oplus(16,4)\oplus(\overline{16},\bar{4})},\\ \underbrace{45{+}15{+}60{+}64{+}64}_{=248} = \dim E_8, \qquad \underbrace{40{+}12{+}60{+}64{+}64}_{=240} = |R(E_8)|.\end{gathered} \tag{93}$$

The adjoint **120** of $\mathfrak{so}_{16}$ is $(45,1)\oplus(1,15)\oplus(10,6)$ and its spinor **128** is $(16,4)\oplus(\overline{16},\bar{4})$. The TFPT dictionary makes each summand a known object:

piece	value	TFPT reading
$(45,1)$	$45=40+5$	D_5 carrier adjoint: Clebsch edges $40=\|R(D_5)\|$ plus carrier rank $g_{\rm car}=5$
$(1,15)$	$15=12+3$	A_3 family+Higgs adjoint: $\dim\mathfrak{g}_{\rm SM}=12$ plus family rank $N_{\rm fam}=3$
$(10,6)$	$10\cdot6$	Λ decuple $\mathcal{A}_\Lambda=\binom{5}{2}=10$ times Z_6 transport $\binom{4}{2}=6$
$(16,4)\oplus(\overline{16},\bar{4})$	$2\cdot64$	matter half-spinor S^+ times the four corners μ_4, plus conjugate sheet

Proof route (replaces the old blind 240-metrization). *Stage 1 (D_5,* ***[E]****): C^+ is the D_5 half-spin orbit,* $R_4=$ Clebsch, Aut $=W(D_5)$, $|E(R_4)|=40=|R(D_5)|$ — Theorem 2. *Stage 2 (A_3,* ***[E]****):* the carrier geometry fixes $X_f^\circ=\mathbb{P}^1\setminus\mu_4$ with $H_1(X_f^\circ,\mathbb{Z})=\mathbb{Z}^3$ and a rigid $D_4\subset PGL_2$ generated by $z\mapsto iz$ and $z\mapsto1/z$. The logarithmic forms $\omega_i=dz/(z-p_i)$ regular at infinity span H^1 by their differences, dual under the residue pairing to the puncture loops γ_i with $\operatorname{Res}(\omega_i,\gamma_j)=\delta_{ij}$; hence the canonical (residue) metric on the puncture classes is I_4, and the twelve puncture-difference cycles $\gamma_i-\gamma_j$ are exactly the A_3 roots:

$$\boxed{\begin{aligned}&\{\gamma_i-\gamma_j\}_{i\neq j}:\ \#=12=|R(A_3)|=\dim\mathfrak{g}_{\rm SM},\quad \text{rank}=3=N_{\rm fam},\\ &\text{Gram}=\text{Cartan}(A_3),\ \det=4\end{aligned}} \quad \textbf{[E]}, \tag{94}$$

with $\dim A_3=15=\dim S^+-1$, and the A_3 positive roots equal the exponent sum $1{+}2{+}3=6=Z_6=|E(K_4)|$. *The corner rotation realises the A_3 Coxeter element.* The carrier generator $z\mapsto iz$ permutes μ_4 as the 4-cycle $(p_1p_2p_3p_4)\in S_4=W(A_3)$ and acts on $H^1=\mathbb{C}^3$ as an isometry of the Cartan form with

$$\boxed{\text{ord}=4=h(A_3)=|\mu_4|,\quad \text{spec}=\{i,-1,-i\},\quad \chi(t)=\tfrac{t^4-1}{t-1}=\Phi_4(t)\Phi_2(t)} \quad \textbf{[E]}, \tag{95}$$

i.e. the eigenvalues are the A_3 exponents $\{1, 2, 3\}$ as fourth roots of unity — the exact analogue of the E_8 Coxeter element reading Φ_{30}. The full $D_4 = \langle z \mapsto iz, z \mapsto 1/z\rangle$ embeds in $W(A_3) = S_4$ (order 8, index 3) by isometries. *Stage 3 (standard, [E]):* assemble E_8 from the $D_5 \times A_3$ branching — no 240-map required.

The μ_4 glue theorem (why the four corners appear everywhere). The two factors have equal discriminant groups, and E_8 is unimodular:

$$\boxed{\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4, \qquad [\,E_8 : D_5 \oplus A_3\,] = \sqrt{\tfrac{\det D_5 \cdot \det A_3}{\det E_8}} = \sqrt{\tfrac{4\cdot 4}{1}} = 4 = |\mu_4|} \quad [\mathbf{E}]. \tag{96}$$

So $E_8 = (D_5 \oplus A_3) + \mu_4$-*glue*: the four family corners $\mu_4 = \mathbb{Z}_4$ are precisely the glue code that fuses the carrier and the family into the unimodular E_8 lattice. The glue pairs the D_5 spinor class ($= S^+$, the family) with the A_3 class — exactly the $(16, 4) \oplus (\overline{16}, \overline{4})$ spinor block. Hence μ_4 plays three roles at once: family corners, A_3 origin, and E_8 glue.

Discriminant-norm glue lemma (and the TFPT $5/4, 3/4$ identification). The two glue generators carry the discriminant quadratic form $q \in \mathbb{Q}/2\mathbb{Z}$:

$$\boxed{\begin{array}{ll} q(D_5) = \tfrac{n}{4}\big|_{n=5} = \tfrac{5}{4} \text{ (spinor class)}, & q(A_3) = \tfrac{n}{n+1}\big|_{n=3} = \tfrac{3}{4} \text{ (corner class)}, \\ q(D_5) + q(A_3) = 2 \equiv 0 \pmod 2, & q(D_5) - q(A_3) = \tfrac{1}{2} \end{array}} \quad [\mathbf{E}]. \tag{97}$$

The diagonal $\langle (s, g)\rangle \cong \mathbb{Z}_4$ is isotropic ($k^2 \cdot 2 \equiv 0$ for all k), so the glue closes to an *even unimodular* lattice — this is E_8.

The two glue norms are just $(g_{\rm car}, N_{\rm fam})$ over the glue index. Because rank $D_5 = g_{\rm car} = 5$ (so $q(D_5) = \frac{n}{4}|_{n=5} = g_{\rm car}/|\mu_4|$) and rank $A_3 = N_{\rm fam} = 3$ (so $q(A_3) = \frac{n}{n+1}|_{n=3} = N_{\rm fam}/|\mu_4|$, with $n+1 = 4 = |\mu_4|$),

$$\boxed{q(D_5) = \frac{g_{\rm car}}{|\mu_4|} = \tfrac{5}{4}, \qquad q(A_3) = \frac{N_{\rm fam}}{|\mu_4|} = \tfrac{3}{4}}$$

and *all four arithmetic operations* on the pair are then forced compiler atoms:

$$\underbrace{q(D_5){+}q(A_3) = \tfrac{g_{\rm car}+N_{\rm fam}}{|\mu_4|} = 2}_{E_8 \text{ root norm}}\,, \qquad \underbrace{q(D_5){-}q(A_3) = \tfrac{|\mathbb{Z}_2|}{|\mu_4|} = \tfrac{1}{2}}_{\delta \text{ (lepton transport)}}\,,$$

$$\underbrace{q(D_5)q(A_3) = \tfrac{g_{\rm car}N_{\rm fam}}{|\mu_4|^2} = \tfrac{15}{16}}_{\dim \mathfrak{su}(4)/\dim S^+}\,, \qquad \underbrace{\tfrac{q(D_5)}{q(A_3)} = \tfrac{g_{\rm car}}{N_{\rm fam}} = \tfrac{5}{3}}\,.$$

So the distinguished lepton transport value $\delta = \frac{1}{2} = (g_{\rm car} - N_{\rm fam})/|\mu_4| = |\mathbb{Z}_2|/|\mu_4|$ is the carrier-minus-family count over the glue index — a fully atomic origin, not a chosen rational (it also equals the harmonic-frame holonomy diagonal modulus, [`verification/v41_leg_assignment.py`]); and (sum, ratio) invert uniquely back to $(\frac{5}{4}, \frac{3}{4})$. [`verification/v51_boundary_half_step.py`] The sum $\frac{5}{4}+\frac{3}{4} = 2$ is literally the E_8 spinor-root norm $\frac{1}{2}(\pm)^8$ split across $D_5 \times D_3$: $5 \cdot \frac{1}{4} + 3 \cdot \frac{1}{4} = 2$. Strikingly, both norms are pre-existing TFPT constants:

$$\boxed{\frac{\delta_2}{\delta_{\rm top}^2} = \frac{2880\, c_3^8}{(48\, c_3^4)^2} = \frac{5}{4} = q(D_5), \qquad \xi_{\rm tree} = \frac{\dim \mathfrak{g}_{\rm SM}}{\dim S^+} = \frac{12}{16} = \frac{3}{4} = q(A_3)} \quad [\mathbf{E}], \tag{98}$$

so $\delta_2/\delta_{\rm top}^2 + \xi_{\rm tree} = 2$: the two-loop/defect ratio is the D_5 spinor norm and the gravitational tree normalization is the A_3 corner norm, and their sum is the E_8 root norm.

The whole integer skeleton from one pair $(g_{\rm car}, N_{\rm fam}) = (5,3)$ [E] ([verification/v53_compiler_core.py])

Every load-bearing integer of the compiler flows from the single boundary pair $(g_{\rm car}, N_{\rm fam}) = (5,3)$ by sum, difference and squares:

$$\operatorname{rank} E_8 = g_{\rm car}{+}N_{\rm fam} = 8, \qquad |\mathbb{Z}_2| = g_{\rm car}{-}N_{\rm fam} = 2, \qquad |\mu_4| = \tfrac{g_{\rm car}+N_{\rm fam}}{2} = 4,$$

and $(N_{\rm fam}, |\mu_4|, g_{\rm car}) = (3,4,5)$ is *the* Pythagorean triple. This is not decorative: the grand-mass-volume exponent $\Delta_Y = g_{\rm car}^2 = 25$ ($\det M_{\rm SM} \sim (\varphi_0^{\rm ret})^{25}$; document 3) splits over the sectors as a *difference of squares*,

$$\boxed{\Delta_Y = g_{\rm car}^2 = \underbrace{N_{\rm fam}^2}_{\text{down}=9} + \underbrace{(g_{\rm car}{-}N_{\rm fam})(g_{\rm car}{+}N_{\rm fam})}_{=|\mathbb{Z}_2|\cdot\operatorname{rank} E_8} = 9 + 16 = \dim S^+}$$

i.e. the down sector carries $N_{\rm fam}^2$ and up+lepton carries $|\mathbb{Z}_2|\cdot\operatorname{rank} E_8 = \dim S^+$ (the K-row sums $(6,9,10)$). **The anchor is the Dirac square root of this skeleton:** $a = (1,1,2)$ is the *unique* 3-multiset whose elementary symmetric functions are $(|\mu_4|, g_{\rm car}, |\mathbb{Z}_2|) = (4,5,2)$, equivalently its characteristic polynomial is $\chi_a(t) = (t{-}1)^2(t{-}2) = t^3 - |\mu_4|\, t^2 + g_{\rm car}\, t - |\mathbb{Z}_2|$ — the compiler atoms *are* its char-poly coefficients. Sharper still ([verification/v491_p2_partition_corollary.py]): the partition needs only $e_1 = 4$ — three *positive integers* summing to 4 already force $\{1,1,2\}$, so $g_{\rm car} = e_2 = 5$ and $|\mathbb{Z}_2| = e_3 = 2$ are *corollaries* of the four marks given the weight typing (integrality = Birkhoff–Grothendieck splitting exponents, positivity = $h^0{=}0$, sum = par-deg-0 "given (U)"); every dropped assumption is negative-controlled, and `AX.P2.01` stays a declared axiom — the residual is the typing, not the integer 5. That typing is itself now hardened ([verification/v499_p2_weights_deligne_bg.py], `P2.TYPING.01`): given the four marks, rank 3, the cusp class and (U), Deligne + Birkhoff–Grothendieck + Mehta–Seshadri stability force $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ outright ($h^0{=}0$ is the positivity), leaving only the module identification and (U) as [C] residuals — and the module-identification rest is now *one* residual with the QGEO modulus rest, both reducing to the same order-4 clock = Coxeter/cusp-class carrier datum (v503, `QGEO.EMERGE.LIGHT.01`; markers unchanged); the clock-rigidity theorems then reduce that common residual to one alignment bit, with the clock's order fermionically forced, $U^2 = (-1)^F$ exactly ([verification/v506_seam_clock_rigidity.py], `SEAM.CLOCK.RIGIDITY.01`); and the bit-origin theorems derive the deck's *class* (a half-period translation of the seam double) — only its position stays the [C] carrier input ([verification/v507_seam_bit_origin.py], `SEAM.BIT.ORIGIN.01`); the freedom theorems then type that position half as *topology* (the covering deck is free on the seam circle, the edge class is excluded), so the residual is the square-modulus datum $\tau = i$ alone ([verification/v510_seam_bit_freedom.py], `SEAM.BIT.FREEDOM.01`); and the flag-transitivity equivalence web restates that residual in discrete form — flag transitivity of the four marks ($V_4 \to D_4$) $\iff \tau = i$, with bare mark-transitivity automatic for every configuration and no local jet seeing the side bit ([verification/v512_seam_tau_flag.py], `SEAM.TAU.FLAG.01`) — a residual that has since survived ten side-blind derivation attacks (v521/v525/v529) and carries a physical definition as the twist-class choice with a gauge-robust order parameter, in principle interferometrically readable (v528; it remains formal input) — and, since v534, the first *dynamical* selection datum: reflection positivity on the interacting mark family keeps exactly the $\delta = \pi/2$ member alive (toy-fenced). So the theory's only transcendental is π (through $c_3 = \frac{1}{8\pi}$, $8 = \operatorname{rank} E_8$, and the seed $\frac{1}{6\pi}$, $6 = |R^+(A_3)|$); everything else is this $(5,3)$-generated integer skeleton.

Boundary Carrier Selection Theorem (Theorem A) [E]/[C]

E_8 is not assumed: it is *selected* as the unique even-unimodular audit hull of the admissible boundary-carrier data. **Boundary side:** $\mu_4 = \{1, i, -1, -i\}$ gives $X_f = \mathbb{P}^1 \setminus \mu_4$ with $H_1 = \mathbb{Z}^3$, hence A_3 and $N_{\rm fam} = 3$. **Carrier side:** a minimal D-type carrier with a 16-dim half-spinor forces D_5 (only $n = 5$ gives $2^{n-1} = 16$; $S^+ = \Lambda^{\rm even}\mathbb{C}^5 = 1{+}10{+}5$). **Glue:** $\operatorname{disc}(D_5) = \operatorname{disc}(A_3) = \mathbb{Z}_4$ (isomorphic, cyclic, anti-isometric), $q(D_5) + q(A_3) = 2$ (the E_8 root norm), glue index $4 = |\mu_4|$, and E_8 is the unique even unimodular rank-8 lattice. **Exclusion:** the four conditions

$$\text{(i)}\ \dim S^+{=}16, \quad \text{(ii)}\ N_{\rm fam}{=}3, \quad \text{(iii) cyclic } \mathbb{Z}_4 \text{ glue}, \quad \text{(iv)}\ q_D{+}q_A{=}2$$

are met *only* by $D_5 \oplus A_3$; D_8 (non-cyclic $\mathbb{Z}_2{\times}\mathbb{Z}_2$, no family geometry), $E_7{\oplus}A_1$ (1 family, no 16-spinor), $E_6{\oplus}A_2$ (2 families, $\mathbb{Z}_3$), $A_4{\oplus}A_4$ (no D-spinor, $\mathbb{Z}_5$) and A_8 (single factor) each fail at least one. **Typing:** the lattice/classification facts are [E]; the physical selection (reflection-positive seam $c_3 = 1/8\pi$ +

minimal chiral carrier ⇒ exactly these inputs) is the [C]/[O] selection axiom (P1 analytic interface + P2 carrier interface), not a zero-input derivation. [`verification/v47_selection_theorem.py`]

Group-level closure (μ_4 as the common centre). The lattice glue lifts to a clean group statement, because both factors share the *same* centre and μ_4 *is* the diagonal:

$$\boxed{Z(\mathrm{Spin}(10)) = \mathbb{Z}_4, \quad Z(SU(4)) = \mathbb{Z}_4, \quad |\mu_4| = 4, \quad G_{\text{closure}} = \frac{\mathrm{Spin}(10) \times SU(4)}{\Delta\mathbb{Z}_4} \subset E_8} \qquad \text{[E]}, \tag{99}$$

with $\mathrm{rank} = 5 + 3 = 8 = \mathrm{rank}\, E_8$. Every adjoint block is diagonal-$\mathbb{Z}_4$ neutral (generator $\omega = i$; charges mod 4): $(45, 1), (1, 15) : 0$; $(10, 6) : 2{+}2{=}0$; $(16, 4) : 1{+}3{=}0$; $(\overline{16}, \overline{4}) : 3{+}1{=}0$, so $45{+}15{+}60{+}64{+}64 = 248$ are genuine reps of the quotient. This promotes μ_4 from a lattice glue index to the *common centre quotient* — the physically cleanest language for the gluing: the D_5 half-spinor 16 pairs with the $SU(4)$ fundamental 4 so the spinor charges ± 1 cancel the fundamental charges ∓ 1.

Root-length completion. Every E_8 block reaches norm 2:

block	norm split	reading
$(45, 1), (1, 15)$	$2,\ 2$	D_5, A_3 root adjoints
$(10, 6)$	$1 + 1 = 2$	$SO(10)$ vector $\oplus$ $SO(6)$ vector
$(16, 4) \oplus (\overline{16}, \overline{4})$	$\frac{5}{4} + \frac{3}{4} = 2$	D_5 spinor norm + A_3 corner norm

The old question "can the 240 flip atlas be metrized to E_8?" is now the sharp question "are the TFPT D_5 and A_3 metrics in standard normalization?" — and the $5/4, 3/4$ match says yes.

Explicit construction (the lift is now built, not counted). Embedding $D_5 \oplus A_3 = D_5 \oplus D_3 \subset D_8$ and adjoining the even spinor coset (the μ_4 glue), the 240 roots are produced explicitly and verified:

$$\boxed{\{\pm e_i \pm e_j\}_{i<j}^{8}\ (112)\ \cup\ \{\tfrac{1}{2}(\pm)^8 :\ \text{even}\}\ (128)\ =\ 240 \text{ roots, all norm 2, Cartan } \det = 1} \qquad \text{[E]}. \tag{100}$$

Splitting each root into its D_5 (first five) and A_3 (last three) parts gives exactly the branching blocks — $(2, 0){:}40$, $(0, 2){:}12$, $(1, 1){:}60$, and $(\frac{5}{4}, \frac{3}{4}){:}128$ — and the 128 spinor roots realise $\frac{5}{4} + \frac{3}{4} = 2$ on the nose. All 240 vectors are integer combinations of one set of 8 simple roots, so the assembled lattice *is* E_8. The blind 240-metrization is therefore eliminated: E_8 is explicitly $(D_5 \oplus A_3) + \mu_4$-glue.

Remark (the metric, computed)**.** This reframes E_8 from origin to closure: E_8 is the closure algebra of the D_5 carrier and the A_3 family geometry, glued by μ_4. The root system, Cartan form, $W(A_3) = S_4$ and the corner-rotation $= A_3$ Coxeter element are exact (residue pairing + explicit carrier symmetry). The transcendental geometric metric $G_{ij} = -\log |p_i - p_j|$ on μ_4 has been computed: it is only D_4-symmetric, with a regularisation-independent anisotropy $-\log 2$ on the root space, hence *not* proportional to $\mathrm{Cartan}(A_3)$. It need not be: the E_8-relevant A_3 is the canonical *residue* pairing, and the geometric metric is a D_4-equivariant deformation that preserves the root system, Weyl group and Coxeter element. So the lattice data is exact [E]; the analytic metric is a *characterised* deformation, not an open gap.

Uniform Coxeter law. All three factors obey $|R| = h \cdot \mathrm{rank}$, and the two TFPT Coxeter numbers are themselves carrier data:

$$\boxed{\begin{aligned} &h(D_5) = 8 = r(E_8) = \varphi(30), \quad h(A_3) = 4 = |\mu_4|, \\ &h(D_5)\, h(A_3) = 2^{g_{\text{car}}} = 32, \quad 2^{g_{\text{car}}} - 2 = 30 = h(E_8) \end{aligned}} \qquad \text{[E]}, \tag{101}$$

so the Coxeter closure of Theorem 3 reappears as $h(E_8) = h(D_5)h(A_3) - 2$, and $h(D_5) + h(A_3) = 12 = \dim \mathfrak{g}_{\mathrm{SM}} = |R(A_3)|$.

Theorem 8 — the $(1,1,2)$ anchor microcode and the φ_0-ladder

The parabolic flavor bundle $E_\bullet = \mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$ on $\mathbb{P}^1\setminus\mu_4$ has splitting type $a=(1,1,2)$, whose power sums $p_n(a)=2+2^n$ generate the whole compiler budget:

The anchor power compiler.

$$(p_1,\dots,p_5)=(4,6,10,18,34)=(|\mu_4|,\ |R^+(A_3)|,\ \mathcal{A}_\Lambda,\ N_{\rm fam}|R^+(A_3)|,\ \mathbb{Z}_6\ \text{lift}),$$

$$q(A_3)=\tfrac{p_2}{2p_1}=\tfrac{3}{4},\quad q(D_5)=\tfrac{p_3}{2p_1}=\tfrac{5}{4},\quad \textstyle\sum L=p_1p_3=40,$$

$$\Omega_{\rm adm}=2p_1p_2=48,\qquad D_{\rm start}=p_2p_3=60,\qquad |R(E_8)|=4p_2p_3=240.$$

The elementary symmetric data $\chi_a(t)=(t-1)^2(t-2)$ give $(e_1,e_2,e_3)=(4,5,2)=(|\mu_4|,g_{\rm car},|\mathbb{Z}_2|)$, so the anchor reconstructs the carrier rank, and $10b_1=e_1^2+e_2^2=4^2+5^2=41$. The seed itself is the two-term anchor form $\varphi_0^{\rm ret}=\frac{2p_1}{p_2}c_3+2p_1p_2\,c_3^4=\frac{4}{3}c_3+48c_3^4$.

The φ_0-ladder. With $\lambda_Y=\sqrt{\varphi_0^{\rm ret}(1-\varphi_0^{\rm ret})}$, every fermion mass is the single closed form $\hat m_{f,j}=\frac{v_{\rm geo}}{\sqrt2}\lambda_Y^{L_{f,j}}\Lambda_{f,j}$ with $\lambda_Y^L\Lambda=\pi\,c\,(\varphi_0^{\rm ret})^k$, so all nine holonomies are resolvent outputs and the masses form the $\varphi_0^{\rm ret}$-order ladder $t(0)>\{b,c,\tau\}(2)>\{s,\mu\}(3)>\{u,d\}(4)>e(5)$.

Secondary E_8 atlas. Beyond the main $D_5\times A_3$ split, the residue matrix appears in the $E_6\times A_2$ branching, $248=\|R\|_F^2+\det R+2(\mathbf{1}^\top Ra)N_{\rm fam}=78+8+2\cdot27\cdot3$, and the scalaron exponent in $E_7\times A_1$, $112=7\cdot16$. The seven E_8 slices are charted in the companion note `e8_secondary_branching_atlas` as a falsifiable audit raster.

The $E_6\times A_2$ block is the $\tau{=}\omega$ face ([`verification/v407_dn_pairings_omega.py`], `SEAM.EQUIV.02`). A_2 is the Eisenstein lattice of the family CM point $\tau{=}\omega$ (Cartan det $=3=N_{\rm fam}$), and the (d,n) selector's three pairings $\{n\cdot\mathbf{1},\ n\cdot a,\ n\cdot\sigma\}=\{2,8,121\}$ are exactly its family-slice atoms $\{|Z_2|,\ \dim A_2{=}\det R{=}\operatorname{rank}E_8,\ \|Pl(K)\|_1^2\}$, so deriving (d,n) (the R4′ residue, v139) folds into the dual keystone (v405), leaving only "why $11{=}\|Pl(K)\|_1$".

Master statement and status

> Five carrier slots give D_5, four family corners give A_3, and $D_5\times A_3$ gives the E_8 atlas.

Equivalently: $\mathcal{A}_\Lambda=\frac{h}{N_{\rm fam}}=10=\binom{5}{2}$, $h=30=2\cdot3\cdot5$, $(5,3,2):\ \sum=10,\ \prod=30$; Λ, $\det M_\ell$, the E_8 Coxeter structure and the action ladder read the same five-slot pair-decuple. The Standard model, flavor, Λ, inflation and E_8 are projections of the even carrier code C^+ ($=D_5$ half-spinor) on five slots together with the four-point family A_3.

Statement	Status	Note
Even code C^+, $\dim S^+ = 16$, dual-root Y	[E]	Theorem 1
Exterior–Action duality $(1,10,5) \leftrightarrow (1,5,10)$	[E]	scheme eigenvalues verified
R_4 = Clebsch $(16,5,0,2)$, $R_2 = (16,10,6,6)$; $\mathrm{Aut}_{\mathrm{aff}} = W(D_5) = 1920$	[E]	carrier = D_5 half-spinor
$2^g - 2 = g(g+1) \Leftrightarrow g = 5$	[E]	Coxeter closure (unique)
h=30=2·3·5, r=$\varphi(30)$=8, χ_C=Φ_{30}	[E]/[C]	cyclotomic compiler
$\lvert R^+\rvert = \binom{16}{2} = \mathrm{Tr}\, X^2 = 4h = 120$	[E]	positive-edge moment
$40, 60, 120, 240 = 10 \cdot (4, 6, 12, 24)$; 41=1+4·10	[E]	$K_5 \times K_4$ product table
$10b_1$=$5r$+1, u totient, M^2_{scal}=c_3^{r-1}, ω_b=$\lambda_C/10$	[E]/[C]	downstream readouts
$A_s = N_\star^2 c_3^7/24\pi^2$	[C]/[O]	if $M_{\mathrm{scal}} = M_{\mathrm{Star}}$
$\mathfrak{e}_8 = (45,1) \oplus (1,15) \oplus (10,6) \oplus (16,4) \oplus (\overline{16},\overline{4})$	[E]/[C]	$D_5 \times A_3$ lift
240 roots built from D_5⊕A_3+μ_4 glue, det 1	[E]	explicit E_8 construction
flavor $\sum L$=40=$\lvert R(D_5)\rvert$, entries on C_6=$\lvert R^+(A_3)\rvert$	[E]	transport = D_5×A_3
A_3 from $\mathbb{P}^1 \setminus \mu_4$: residue pairing = Cartan, det 4	[E]	Stage 2 (Theorem 7)
corner $z \mapsto iz = A_3$ Coxeter el.: $\{i, -1, -i\}$, $\Phi_4\Phi_2$	[E]	order 4 = $h(A_3) = \lvert\mu_4\rvert$
$\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4$, glue = μ_4	[E]	$E_8 = (D_5 \oplus A_3) + \mu_4$
$q(D_5)$=$\frac{5}{4}$=$\delta_2/\delta^2_{\mathrm{top}}$, $q(A_3)$=$\frac{3}{4}$=ξ_{tree}, sum 2	[E]	discriminant-norm glue
120= $\dim \mathfrak{so}_{16}$; Ω_{adm}=40+1+7	[E]	adjoint half; deficit ladder
$h(D_5)$=8=$r(E_8)$, $h(A_3)$=4=$\lvert\mu_4\rvert$, $\lvert R\rvert$=$h \cdot r$	[E]	uniform Coxeter law
$\mathrm{Exp}(E_8)$ as 41/48 budget deficits	[O]	exponent ledger (fingerprint)
geometric metric D_4-deform (anisotropy $-\log 2$), residue $= A_3$	[E]	metric computed/characterised

Scope. This note compresses; it does not re-derive the upstream boundary kernel (the P1 hardening chain) or the carrier rigidity theorem (Lean), and it makes no claim of a final theory. The arithmetic and the D_5/Clebsch/$W(D_5)$, A_3/residue-Cartan/Coxeter, and μ_4-glue identifications are exact, and the E_8 lattice is now built explicitly from $D_5 \oplus A_3$ plus the μ_4 glue (240 roots, det 1). The only residual [C] is the analytic identification of the carrier geometry's L^2 cusp Hodge metric with the canonical residue metric up to scale, which does not affect the explicit root/lattice/Weyl construction. Honest non-fits (Koide 0.6645 $\neq 2/3$, η_B, m_p/m_e) stay outside the compiler.

Appendix: machine-checkable $E_8 = (D_5 \oplus A_3) + \mu_4$ construction

The lift is reproducible by a short script (all entries below verified exactly).

Cartan matrices and discriminant groups.

$$C(D_5) = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & -1 \\ 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 0 & 2 \end{pmatrix}, \quad C(A_3) = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}; \quad \det C(D_5) = \det C(A_3) = 4.$$

Smith normal forms give $\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4$; the glue index is $[\,E_8 : D_5 \oplus A_3\,] = \sqrt{4 \cdot 4/1} = 4 = \lvert\mu_4\rvert$, with discriminant-form norms $q(D_5) = \frac{5}{4}$, $q(A_3) = \frac{3}{4}$, $q(D_5)+q(A_3) = 2 \equiv 0 \pmod 2$ (even-glue). Writing $\mu := \lvert\mu_4\rvert = 4$ the norms are $q(D_5) = \frac{\mu+1}{\mu}$, $q(A_3) = \frac{\mu-1}{\mu}$, so $q(D_5)q(A_3) = 1 - \frac{1}{\mu^2} = \frac{15}{16}$ and

$$\boxed{\dim S^+ (1 - q(D_5)q(A_3)) = 16 \cdot \tfrac{1}{16} = 1} \qquad \text{[E]}, \tag{102}$$

i.e. the glue product fills all $15 = \dim S^+ - 1$ non-trivial half-spin states and the defect is exactly the singleton (the ν^c/Higgs closure). And E_8 has a tidy μ_4-cardinal form, $\dim E_8 = \binom{\lvert\mu_4\rvert^2}{2} + 2\lvert\mu_4\rvert^3 = \binom{16}{2} + 2 \cdot 64 = 120 + 128 = 248$ [C].

Explicit roots. Split $\mathbb{R}^8 = \mathbb{R}^5_{(D_5)} \oplus \mathbb{R}^3_{(A_3)}$. The 240 roots are

$$\{\pm e_i \pm e_j : 1 \le i < j \le 8\}\ (112)\ \cup\ \{\tfrac{1}{2}(\pm 1)^8 :\ \text{even } \#\text{ of } -\}\ (128),$$

all of norm 2. Splitting each by $(\|\cdot\|^2_{D_5}, \|\cdot\|^2_{A_3})$ gives the branching blocks

$$(2,0)\colon 40\ [D_5],\quad (0,2)\colon 12\ [A_3],\quad (1,1)\colon 60\ [(10,6)],\quad (\tfrac{5}{4},\tfrac{3}{4})\colon 128\ [(16,4)\oplus(\overline{16},\overline{4})],$$

so $\mathfrak{e}_8 = (45,1) \oplus (1,15) \oplus (10,6) \oplus (16,4) \oplus (\overline{16},\overline{4})$, $248 = 45{+}15{+}60{+}64{+}64$, $240 = 40{+}12{+}60{+}64{+}64$.

Simple-root Gram (unimodular check). With the Bourbaki simple roots $\alpha_1 = \frac{1}{2}(e_1{-}e_2{-}\ldots{-}e_7{+}e_8)$, $\alpha_2 = e_1{+}e_2$, and $\alpha_k = e_{k-1}{-}e_{k-2}$ for $k = 3,\ldots,8$ (so $\alpha_3 = e_2{-}e_1,\ \alpha_4 = e_3{-}e_2,\ \ldots,\ \alpha_8 = e_7{-}e_6$), the Gram matrix $G_{ij} = \langle \alpha_i, \alpha_j\rangle$ is the E_8 Cartan matrix with $\det G = 1$ and all $G_{ii} = 2$; every one of the 240 roots is an integer combination of the α_i. Hence the assembled lattice is E_8, not merely a 240-count. [E] (The shifted index $e_{k-2}{-}e_{k-3}$ would duplicate α_3 and make G singular — use $e_{k-1}{-}e_{k-2}$.)

Part III

Appendix: ablation of the EM fixed point

The single sharpest reviewer test is: *was the closure $F_{U(1)}(\alpha) = 0$ engineered so that α comes out?* The honest answer is an ablation — vary each input and see which actually move the root. Three one-parameter scans (all others held at their compiler values) separate the *load-bearing* integers from the *fine* last-digit selector.

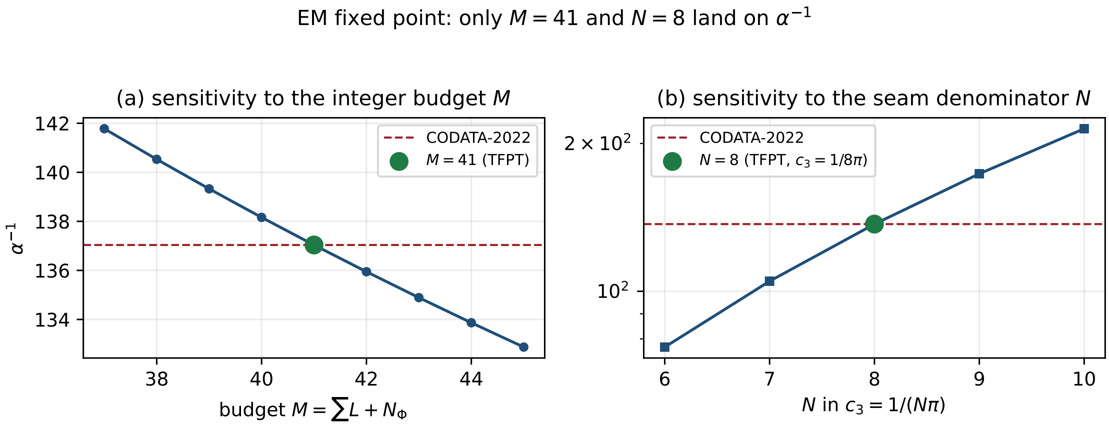


Figure 7: EM-closure ablation (data plot, `verification/make_figures.py`). **(a)** $\alpha_\star^{-1}$ vs the integer budget M: only $M = 41$ lands on CODATA-2022. **(b)** $\alpha_\star^{-1}$ vs the seam denominator N in $c_3 = 1/(N\pi)$: only $N = 8$ lands. Both integers are fixed elsewhere (flavor budget; rank/$h(D_5)$), so the equation is not freely tunable to 137.036.

(A1) The transport budget $M = \sum L + N_\Phi$ (exponent held at $-\frac{5}{4}$):

M	38	39	40	**41**	42	43	44
$\alpha_\star^{-1}$	140.530	139.326	138.162	**137.0359992**	135.946	134.890	133.866

Each unit of M shifts $\alpha_\star^{-1}$ by ≈ 1.09, so *only* $M = 41$ lands on 137.036. This is a hard lock — and $41 = 10b_1 = \sum L + N_\Phi = 40 + 1$ is fixed by the compiler (flavor budget 40 + one Higgs). **Load-bearing.**

(A2) The seam "8" in $c_3 = 1/(N\pi)$ (here $\delta_{\text{top}} = 48c_3^4$ tracks c_3):

N	6	7	**8**	9	10
$\alpha_\star^{-1}$	76.97	104.85	**137.036**	173.53	214.34

Only $N = 8$ — i.e. $c_3 = 1/(8\pi) = 1/(\operatorname{rank} E_8 \cdot \pi)$ — reproduces 137.036. The "8" is the same one fixed by the bootstrap ($h(D_5) = \operatorname{rank} E_8 = \varphi(30)$) and the Gauss–Bonnet hardening. **Load-bearing.**

(A3) The carrier discriminant exponent (at $M = 41$, $c_3 = 1/(8\pi)$):

exponent	$-\frac{1}{4}$	$-\frac{3}{4}$	$-\frac{\mathbf{5}}{\mathbf{4}}$	$-\frac{7}{4}$	$-\frac{9}{4}$
$\alpha_\star^{-1}$	137.035995	137.035997	**137.0359992**	137.036001	137.036003

The exponent moves only the *seventh* digit; all candidates agree to six. Here $-\frac{5}{4} = q(D_5)$ is selected, but this is a *fine* selector, not a hard lock. **Fine (last-digit).**

Ablation verdict — honest

The EM fixed point is a **hard test of two integers** — the budget $M = 41$ and the seam denominator $8 = \operatorname{rank} E_8$, each decisive (a single-unit change misses by ~ 1 in $\alpha_\star^{-1}$) — and a **fine test of one norm**, the carrier discriminant exponent $-\frac{5}{4} = q(D_5)$ (seventh digit). The equation is therefore *not* freely tunable to 137.036: the two integers that hit it are exactly the compiler/bootstrap numbers, fixed elsewhere. What it does *not* do is pin the carrier norm strongly — that is honest, and it is the right level of claim. [E]

Appendix: computational verification

The load-bearing identities of this document (marked [E], [E], [E]) are re-derived from $\{c_3, g_{\rm car}\}$ alone and machine-checked by the suite in the repository folder `verification/`; the inline tags `[verification/...]` point to the exact script. Relevant here:

- `v1_e8_glue.py` — the μ_4 glue ($q(D_5)+q(A_3)=2$, disc $=\mathbb{Z}_4$, $240, 248$).
- `v2_carrier_pascal.py` — $g_{\rm car}=5$, 16=1+5+10, $N_{\rm fam}, \Omega_{\rm adm}, b_1$.
- `v3_em_alpha.py` — the EM fixed point $\alpha_\star^{-1}$=137.0359992168 (existence/uniqueness, ablation).
- `v4_flavor_matrix.py` — R with det =8, minors $\{2, 3, 5\}$, $\sum L$=40.
- `v7_gravity_cosmo.py` — scalaron c_3^7, Λ/A_s readouts.

Run `python verification/run_all.py` (needs `mpmath, numpy, sympy`); it exits 0 iff all checks pass. Full map: `verification/README.md`.

External works built upon

The TFPT discrete core (the two axioms, the E_8 glue, the integer skeleton) is self-contained and machine-checked. Its *bridges* to continuum/AQFT physics, to gravity and to the lattice/singularity picture build on established mathematics, listed here by topic; the body refers to these by author name.

Modular theory and algebraic quantum field theory.

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- J. J. Bisognano, E. H. Wichmann, *On the duality condition for a Hermitian scalar field / for quantum fields*, J. Math. Phys. **16** (1975) 985–1007; **17** (1976) 303–321.
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